

Chapter 7

Design of the Hydraulic Components



As per the state of the art, the hydraulic design of the various types of impellers, diffusers, volutes and inlet casing is based on empirical methods such as described in this chapter. The first design created that way is then subject to analysis and optimization by CFD. Even then the accuracy of performance prediction is not always satisfactory. This, because the 3D-flow through the pump depends on the complex shapes of the flow paths given by the inlet casing, impeller and collector. To compound the issue, the interaction of the main flow with the flow in the impeller side rooms can have an unexpectedly large influence on the Q-H-curve and efficiency (an example can be found in Chap. 9.1). Always remember that it is the *combination of all parameters and shapes* of the hydraulic channels which determines the flow patterns - hence performance. In order to reduce the uncertainties of performance prediction, a systematic approach to hydraulic design is advocated. To this end, Chap. 7.14 introduces a novel concept for a fully analytical description of the impeller geometry. Prior to starting the hydraulic design all requirements the pump has to fulfill and the boundary conditions imposed should be thoroughly reviewed and documented (see “hydraulic specification” in Chap. 17).

7.1 Methods and Boundary Conditions

7.1.1 *Methods for the Development of Hydraulic Components*

This chapter deals with one-dimensional calculation procedures and design methods for impellers, volute casings, diffusers and inlet casings. For developing these components, the main dimensions and blade angles are calculated in a first

step. Subsequently, the hydraulic contours are designed based on certain rules and methods. Many pump manufacturers employ computer programs for this work, and the drawings are generated on 2D-CAD systems. However, these methods are also increasingly replaced by 3D-CAD systems with which fully three-dimensional geometry models of a component can be created, Figs. 2.3 and 7.45. The complex hydraulic channels can be evaluated better with such models than with the conventional two-dimensional representations in various sections and views. Even more importantly, 3D-CAD systems provide the capability to directly manufacture the hydraulic components (or the casting patterns) by NC milling, stereo lithography or other fast-prototyping processes, [16]. The advantages of such processes are evident in terms of geometrical accuracy and lead times (not in the least also for model tests with milled or stereo-lithographed components). Since manual designs on the drawing board are rather the exception, the subsequent discussion of the design methods emphasizes the fundamental aspects of the design processes rather than a very detailed description of geometrical operations.

The development of hydraulic components often comprises the following steps:

1. Calculation of the main dimensions and blade angles by one-dimensional methods based on empirical correlations for slip factors and hydraulic efficiencies (based on databases and experience), Chap. 3.
2. Generating an initial design.
3. Optionally, quasi-3D methods can be applied to impellers for a pre-optimization of the design (“Q-3D”, see Chap. 8.2).
4. Verification or optimization using 3D-Navier-Stokes programs where the geometry is changed, if required, until a satisfactory result is obtained.
5. Works or model tests for verification (and further optimization if necessary).

Frequently step (3) is skipped since Q-3D methods operate with velocity distributions that may considerably deviate from the reality (Chap. 8.2).

Efficiency, head, partload characteristics and cavitation depend on the local velocity and pressure distribution which cannot be captured in simple calculation and design processes. For this reason, all one-dimensional design methods are affected by uncertainties - even if derived from a large number of tests. Ultimately, it is the combination of the various geometry parameters and the shaping of the channels and blades that produce good hydraulics; these effects cannot be forced into unequivocal design rules. Experience still plays a crucial role in the hydraulic design, since empirical correlations, statistical data or “design methodologies” form the basis of the majority of pumps available on the market. Published design recommendations differ considerably; hydraulic components designed by different methods can nevertheless produce equivalent results. For instance, impellers can be designed for a specific pumping task with very different geometric parameters (e.g. blade angle, blade number, outlet width, blade twist etc.) and may achieve essentially the same efficiency and the same head coefficient.

7.1.2 The Hydraulic Specification

Depending on the pump type and application, the hydraulic design must meet a number of requirements and boundary conditions. Prior to starting the actual design, a hydraulic specification should be established thoroughly (obviously it is not necessary to specifically consider the following points in every application). A more comprehensive format for the hydraulic specification is proposed in Chap. 17.7.

1. The calculation is invariably based on the best efficiency point, defined by: Q_{opt} , H_{opt} at a specified speed n . If the planned main operating point, or the rated flow Q_R , in the plant does not coincide with the best efficiency point, the design flow rate Q_{opt} should be selected so that the condition $0.8 < Q_R/Q_{opt} < 1.1$ is satisfied.
2. The maximum flow rate Q_{max} should be defined to check the cavitation characteristics. To do so, the planned mode of operation must be known (important, for instance, in parallel operation, Chap. 11.1).
3. The suction conditions ($NPSH_A$) have a large impact on the design in many cases, refer also to Chap. 15.
4. In most applications a “stable” Q-H-curve is demanded where the head rises continuously when the flow rate is reduced.
5. The shut-off head H_o is often limited because of its impact on pipeline design pressure and costs. At $n_q < 40$, a head rise of $1.2 < H_o/H_{opt} < 1.25$ is desired (as close as possible to, but not lower than, 1.2).
6. With semi-axial and axial pumps $n_q > 100$, both shut-off head H_o and power P_o must be kept as low as possible because P_o determines the size of the motor when $P_o > P_{opt}$.
7. The cavitation characteristics must satisfy three criteria: (1) the pump must be able to operate in the planned range; (2) cavitation noise and vibrations must be limited; (3) no cavitation damage, Chap. 6.7.
8. To reduce pump size and costs, the head coefficient is frequently selected rather near the upper limit - except in cases where a steep Q-H-curve is explicitly demanded.
9. With respect to energy costs, and for ecological reasons, the highest possible efficiency must be aimed at (the energy costs are often capitalized and rated at 2000 to 5000 \$ per kW of pumping power).
10. The hydraulic configuration is - sometimes profoundly - affected by the requirements of the mechanical design such as: the pump type, the longitudinal section of the pump, the required shaft diameter (criteria: torque, sag due to gravity or radial thrust, critical speed), impeller mounting (slide-fit, shrink-fit, shaft protection sleeves), annular seal clearances, axial thrust balancing (leakage) and the manufacturing costs.
11. The manufacturing process envisaged for the hydraulic components and/or the patterns may have an influence on the hydraulic design as well, e.g. on the draft to be applied on the blades for cast components or if sheet metal is used for

making the impellers and diffusers. Likewise, the materials must be considered e.g. metal versus plastics or ceramics.

12. The hydraulic excitation forces and pressure pulsations must be limited to an allowable magnitude.

Consequently, a pump must by not be optimized only for the best efficiency point but the entire operating range of $Q = 0$ to $Q_{\max}$ should be implicitly considered in the design. Since there is no comprehensive theory satisfying all of the listed requirements, usually the hydraulic design is based explicitly or implicitly on experience, statistical data, empirical approaches or proven similar hydraulics. Even if numerical methods are employed for further optimization, it is expedient to perform the initial design on such bases, since partload and cavitation characteristics cannot be controlled sufficiently in a wide operating range utilizing numerical methods.

7.1.3 Calculation Models

The main dimensions and blade angles of impeller and collector are selected applying one-dimensional methods so that the requirements listed above are satisfied as much as possible. During this calculation, impeller and diffuser can either be idealized as “cascades” or as “channels”. Both models have some justification and are therefore briefly outlined:

Cascade model: Impeller and diffuser are considered as cascades. Consequently, the calculation focuses on matching the (geometrical) vane angles to the flow angles. Accordingly, the design is based on the velocity vectors, incidences and the slip factor. At partload and overload, losses due to excessive incidence are considered to play a critical role. Calculating cavitation according to Eq. (6.10) is also based on the idea of a flow around a profile. C. Pfleiderer can be considered as a proponent of the cascade concept, [B.1].

Channel model: Impeller and diffuser consist of channels whose cross sections change along the flow path. Deceleration or acceleration conditions are therefore the relevant physical quantities (instead of the angles of incidence and deviation). The channel model does not exclusively utilize velocity vectors but rather considers the velocities calculated from the continuity equation as $w = Q/A$ with the local cross section of the channel. A complete calculation method according to the channel model is not known - it would also be hardly sensible to ignore the incidence at the impeller inlet - but H. H. Andersons “area ratio method” goes in that direction, [B.4, B.11]. Instead of determining head and impeller outlet angle via slip coefficients, the design is based on the area ratio A_v , which relates the cross section $A_{2q} = a_2 \times b_2$ between the blades at the impeller outlet to the throat area of the collector $A_{3q} = a_3 \times b_3$. The following relationships apply:

- The area ratio is defined by: $A_v = z_{La} \times A_{2q}/(z_{Le} \times A_{3q})$
- Cross section at the impeller outlet: $z_{La} \times a_2 \times b_2 \approx 0.95 \times \pi \times d_2 \times b_2 \times \sin \beta_{2B}$
- Cross section at diffuser inlet: $z_{Le} \times a_3 \times b_3$

The main dimensions are then determined in the following way:

1. Select a value for A_v based on statistical data of executed pumps from the correlation: $A_v = (42 \text{ to } 53)/n_q$.
2. Obtain ψ_{th} and c_{3q}/u_2 from empirical functions that can be approximated by: $\psi_{th} = A_v^{0.23}$ and $c_{3q}/u_2 = 0.31 \times A_v^{0.45}$.
3. Estimate the hydraulic efficiency (Table 3.9), determine $\psi = \eta_h \times \psi_{th}$, calculate d_2 from Eq. (T7.1.3). Calculate u_2 and $c_{3q} = u_2 \times (c_{3q}/u_2)$.
4. Select the number of diffuser vanes or volute channels and determine the throat area of the collector from $A_{3q} = Q_{opt}/(z_{Le} \times c_{3q})$.
5. Select b_2^* from Fig. 7.2 and the number of impeller blades. Calculate b_2 and A_{2q} from $A_{2q} = A_v \times z_{Le} \times A_{3q}/z_{La}$.
6. The distance between the impeller blades a_2 is determined from $a_2 = A_{2q}/b_2$. The outlet angle β_{2B} is finally determined so that the required a_2 is obtained.

Both models are incomplete idealizations of the complex flow processes in a pump. Determining the impeller outlet angle - as outlined above - according to the channel model appears neither expedient nor does it afford any physical insight since the approach is purely empirical. However, the channel model can provide a meaningful addition to the cascade model since it better explains some experimental facts and allows checks of the design according to additional criteria:

- The Q-H-curve falls steeply when the fluid at $q^* \gg 1$ is accelerated from the impeller outlet to the collector throat area. Then some of the static pressure generated in the impeller is re-converted into kinetic energy which is only recuperated downstream of the throat area with increased losses, Fig. 5.18. In the case of a diffuser, the “shock losses” generated by increased incidence could be considered as the cause of the steeper gradient of the Q-H-curve. However, in the case of a volute or annular collector the effects of incidence are negligible.
- Reducing the throat area of the collector increases the head at partload since c_{3q}/c_2 increases and the flow is better able to follow the imposed deceleration. That way the onset of stall is retarded and zones of stalled fluid are reduced, Fig. 5.18. This phenomenon is hardly affected by the cutwater or diffuser vane inlet angle and poorly explained by the cascade model.
- The NPSH₃ curve increases steeply if the flow vector w_1 at the impeller inlet is greatly accelerated towards the throat area, Fig. 5.13 ($w_{1q} \gg w_1$).
- As explained in Chaps. 4.1.3, 5.2.4 and 5.3.1, the flow in the impeller and collector separates mainly when an allowable deceleration has been exceeded and less because of excessive incidence.

- If angles and channel cross sections are not well matched it is necessary to use the blade distances a_1 and a_2 (or A_{1q}) as design criteria in addition to the blade angles.

Since blade angles *and* channel cross sections have to match the intended flow, cascade and channel models are jointly used in the design procedures discussed in this chapter.

Airfoil model: If the blades are widely-spaced (low L/t), it is also possible to apply the airfoil theory. The lift coefficients of the individual airfoil may then be corrected to account for the influence of neighboring blades using suitable corrections, (Chap. 7.6).

7.2 Radial Impellers

The methods for impeller design discussed in this chapter are essentially generic. Based on these methods, radial impellers for pure fluids can be developed without additional information. Semi-axial impellers are treated in a similar way; however, some particular aspects must be considered which are explained in Chap. 7.5. Methods for axial impellers are presented in Chap. 7.6.

The procedure is the same for open and closed impellers; in the case of large clearances the performance loss must be considered as per Chap. 3.6.5.

7.2.1 Determination of Main Dimensions

Prior to actually drawing an impeller, the main dimensions and blade angles must be determined. The relevant calculation is explained step by step in the following; the required equations are listed in Table 7.1.

1. Given quantities: n , Q_{opt} , H_{opt} and the boundary conditions as per Chap. 7.1.2. As a result, the specific speed n_q can be calculated without which a meaningful design would scarcely be possible. Additionally, the approach flow conditions must be defined. In many applications the approach flow angle is $\alpha_1 = 90^\circ$, and the c_m -distribution over the approach flow cross section is assumed constant.

2. Efficiencies: To calculate the head, the hydraulic efficiency has to be assumed; the equations in Table 3.9, or Figs. 3.30 to 3.32, can be used for this purpose. In general the hydraulic efficiency should be assumed conservatively in order to avoid the risk of a deficit in head.

The volumetric flow through the impeller exceeds the useful flow rate by the annular seal leakage losses and the balance water flow, Eq. (T7.1.1). These volumetric losses must initially be estimated for the impeller calculation, Chap. 3.6, Eqs. (T3.5.9 and 3.5.10). The balance water flow can be calculated when the axial thrust balance device (if any) has been dimensioned.

Table 7.1 Impeller calculation		
Given quantities	$n, Q_{\text{opt}}, H_{\text{opt}}, \alpha_1$; hence the specific speed is known	Eq.
Impeller flow rate	$Q_{L,a} = Q_{\text{opt}} + Q_{\text{sp}} + Q_E = Q_{\text{opt}}/\eta_v$	7.1.1
Shaft diameter	$d_w = \left(\frac{16 P_{\text{max}}}{\pi \omega \tau_{\text{al}}} \right)^{\frac{1}{3}} = 3.65 \left(\frac{P_{\text{max}}}{n \tau_{\text{al}}} \right)^{\frac{1}{3}}$	$\frac{P_{\text{max}} \text{ in W}}{n \text{ in rpm}}$ $\tau_{\text{al}} \text{ in N/m}^2$
Impeller outlet diameter	$d_2 = \frac{60}{\pi n} \sqrt{\frac{2g H_{\text{opt}}}{\Psi_{\text{opt}}}} = \frac{84.6}{n} \sqrt{\frac{H_{\text{opt}}}{\Psi_{\text{opt}}}}$	Ψ_{opt} from Eq. (3.26) or Table 7.2
Impeller inlet diameter for minimum relative velocity	$d_{1*} = f_{d1} \sqrt{d_n^{*2} + 1.48 \times 10^{-3} \Psi_{\text{opt}} \frac{n_q^{1.33}}{(\eta_v \delta_r)^{0.67}}}$	$\delta_r = 1 - \frac{c_{1m}}{u_{1m} \tan \alpha_1}$
	Application	f_{d1}
	Normal impeller	1.15 to 1.05 falling from $n_q = 15$ to 40
	Suction impeller	1.25 to 1.15
Impeller inlet diameter for selected $\beta_1 = \beta_{1a}$	$d_1 = 2.9 \sqrt[3]{\frac{Q_{L,a}}{f_q n k_n \tan \beta_1} \left(1 + \frac{\tan \beta_1}{\tan \alpha_1} \right)}$	7.1.5
Impeller inlet diameter for selected coefficients λ_c, λ_w	$d_{1,\text{opt}} = \sqrt{d_n^2 + 10.6 \left(\frac{Q_{L,a}}{f_q n} \right)^{\frac{2}{3}} \left(\frac{\lambda_c + \lambda_w}{\lambda_w} \right)^{\frac{1}{3}}}$	7.1.6
Blade angle for selected incidence $i_1 = 0$ to 4°	$\beta_{1B} = \beta'_1 + i_1 = \arctan \frac{c_{1m} \tau_1}{u_1 - c_{1u}} + i_1$	With constant c_{1m} und τ_1 : $\tan \beta_{1B}(r) = \frac{r_a}{r} \tan \beta_{1B,a}$
Slip factor	$\epsilon_{\text{im}} = \exp \left\{ -\frac{8.16 \sin \beta_{2B}}{Z_{L,a}} \right\}$	$k_w = 1 - \left(\frac{d_{1m}^* - \epsilon_{\text{Lim}}}{1 - \epsilon_{\text{Lim}}} \right)^3$
	$\gamma = f_1 \left(1 - \frac{\sqrt{\sin \beta_{2B}}}{Z_{L,a}^{0.7}} \right) k_w$	$k_w = 1$ for $d_{1m}^* \leq \epsilon_{\text{Lim}}$ Radial: $f_1 = 0.98$ Semi-axial: $f_1 = 1.02 + 1.2 \times 10^{-3} (n_q - 50)$
Head ($A_2 = \pi d_{2b} b_2$)	$H = \frac{\eta_h u_2^2}{g} \left\{ \gamma - \frac{Q_{L,a}}{f_q A_2 u_2 \tan \beta_{2B}} \left[\tau_2 + \frac{A_2 d_{1m}^* \tan \beta_{2B}}{A_1 \tan \alpha_1} \right] \right\}$	7.1.9
Effective blade loading, according to [70]	$\xi_{\text{eff}} = \frac{2\pi \Psi_{\text{opt}}}{\eta_h Z_{L,a} L_{\text{sch}}^* (w_1^* + w_2^*)}$	7.1.10
Allowable blade loading [70]	$\xi_{\text{al}} = \left(\frac{n_{q,\text{Ref}}}{n_q} \right)^{0.77}$ $n_{q,\text{Ref}} = 40$ Band width $\pm 15\%$	7.1.11
Lift coefficient according to [26]	$\zeta_a = \frac{\pi \Psi_{\text{th}}}{Z_{L,a} L_{\text{sch}}^* \sqrt{\varphi_2^2 + \left(1 - \frac{\Psi_{\text{th}}}{4} \left\{ 1 - \frac{\sin \epsilon_{\text{MS}}}{Z_{L,a}} \right\} \right)^2}}$	allowable $\zeta_a < 0.9$

3. Shaft diameter d_w : Apart from the criteria mentioned in item 7.1.2.10, the torque has to be transmitted. When the material and the allowable sheer stress τ_{al} have been selected, the minimum required shaft diameter can be calculated from Eq. (T7.1.2). To satisfy all criteria given in item 7.1.2.10, it is frequently necessary to increase the shaft diameter above the minimum that has been calculated based on the torque only.

4. Impeller outer diameter d_2 : The head coefficient is selected from Eq. (3.26), Fig. 3.24 or from Table 7.2 for sewage pumps. Subsequently, the impeller outer diameter d_2 is calculated from Eq. (T7.1.3). In order to achieve a stable Q-H-curve, good partload behavior and acceptably low hydraulic excitation forces, the head coefficient ψ_{opt} must be limited. The range of typical head coefficients is shown in Fig. 3.24 as a function of the specific speed. If ψ_{opt} is selected in the upper range of the band, flatter Q-H-curves and smaller impeller diameters are obtained, but the risk of a Q-H-curve instability increases. Head coefficients near the lower limit of the band lead to steeper Q-H-curves with less risk of instability. If extremely steep Q-H-curves are demanded, it is obviously possible to select ψ_{opt} below the range in Fig. 3.24 - but the pump becomes correspondingly larger and more expensive.

With low specific speeds (below $n_q = 20$ to 25) one tends to obtain better efficiencies with high head coefficients since the impeller disk friction loss falls with the 5th power of the impeller diameter, Eq. (T3.6.2). The smaller n_q , the greater is this influence. With $n_q > 50$ the impeller disk friction effects are insignificant and turbulent dissipation due to uneven velocity distributions largely determines the efficiency; in this case a high head coefficient tends to be less favorable.

5. Impeller blade number z_{La} : The selection of the blade number z_{La} depends on various criteria:

A. To reduce pressure pulsations and hydraulic excitation forces, impeller and diffuser blade/vane numbers must be matched according to Chap. 10.7.1: z_{La} and z_{Le} must be selected so that the values $m = 0$ and $m = 1$ for the parameter $m = \left| v_2 \times z_{La} - v_3 \times z_{Le} \right|$ are avoided up to $v_2 = 3$ and $v_3 = 3$.

B. The hydrodynamic blade loading should be in an optimum range: If the loading is too low, unnecessarily high friction losses must be expected. If the loading is too high, the turbulent dissipation losses increase due to uneven flow distribution. The blade loading can only be verified after completing the impeller design. This can be done using empirical values for the blade loading in step 14.

C. Eight and more blades are not normally recommended out of consideration of the stability of the Q-H-curve.

D. Less than 5 blades are unfavorable for high heads per stage since the impeller outlet flow becomes very non-uniform over the circumference due to the large blade spacing. The consequence would be unnecessarily high pressure pulsations, noise and vibrations. Therefore avoid $z_{La} < 5$ for $H_{st} > 100$ m.

Based on the above criteria most radial and semi-axial impellers are designed in a range of approximately $10 < n_q < 120$ with 5 to 7 blades. If no stable Q-H-curve is required since the plant only demands a narrow operating range near the best efficiency point, 9 blades are also designed (impellers with straight radial blades are discussed in Chap. 7.3.3).

At high specific speeds the ratio of blade length to blade pitch (“solidity”) gains significance as a design criterion. The criteria A to D are sufficient for selecting blade numbers if used together with correlations for suitable hydrodynamic blade loading. Empirical formulae for blade number selection (as sometimes published) are not helpful because they are likely to violate the various criteria discussed above.

E. For suction impellers with high n_{ss} and flat curves $NPSH_3 = f(Q)$ one should select $z_{La} = 5$ (or possibly 6); since it would be difficult to get flat NPSH-curves with 7 blades.

6. Impeller inlet diameter d_1 : The choice of the inlet diameter depends on the requirements made on the cavitation characteristics. Three cases are distinguished:

A. Dimensioning for minimum relative velocity at the impeller inlet: This design tends to minimize leakage, friction and shock losses. It is recommended when the available $NPSH_A$ is sufficiently high so that cavitation is avoided. This is true, for example, for the second (and following) stages of multistage pumps (“series stages”). The minimum of the relative velocity w_1 is obtained from $w_1^2 = c_{1m}^2 + (u_1 - c_{1m}/\tan\alpha_1)^2$, with $c_{1m} = 4 \times Q / \{\pi(d_1^2 - d_n^2)\}$ and $u_1 = \pi \times d_1 \times n/60$, by differentiation of $\partial w_1 / \partial d_1$. Setting the expression obtained to zero, one solves for the inlet diameter $d_{1,min}$ and obtains Eq. (T7.1.4), where pre-rotation (if any) is covered iteratively by the swirl number δ_r . The inlet diameter is selected several percent above the minimum thus calculated to allow for secondary effects such as boundary layer blockage and uneven velocity distributions. Thus $d_1 = f_{d1} \times d_{1,min}$ is obtained with recommendations for the factor f_{d1} given in Table 7.1.

B. Design for a selected suction specific speed: For suction impellers a suction specific speed n_{ss} suitable for the intended application is chosen from Table D6.1. With the selected n_{ss} the approach flow angle β_{1a} is read from Fig. 6.21 and the inlet diameter is calculated from Eq. (T7.1.5). The inlet diameter can also be determined somewhat summarily from Eq. (T7.1.4) with a correspondingly larger factor f_{d1} (Table 7.1).

C. Design for selected λ -coefficients according to Chap. 6.3.2: Equation (T7.1.6) offers an alternative route to calculate d_1 when the coefficients λ_c and λ_w are chosen. The recommendations given in Table D6.1 should also be considered in this context: If the circumferential speed u_1 is high, the inlet diameter and the suction specific speed should not be selected too large.

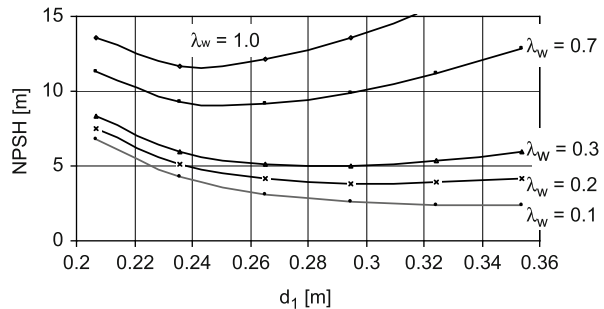
According to Chap. 6.3.2, Eq. (T7.1.6) yields the inlet diameter at which the required NPSH_R reaches its minimum for specified values of flow rate and rotor speed with *given* λ -coefficients. As discussed in Chap. 6.3.2, these coefficients are not well known since they depend on the impeller geometry and the approach flow. Consequently, the optimum inlet diameter calculated from Eq. (T7.1.6) is uncertain as well. In order to be able to evaluate the sensitivity of the required NPSH_R to changes of the λ -coefficients and the impeller inlet diameter, it is convenient to calculate $\text{NPSH}_R = f(d_1)$ for various values of λ_w from Eq. (6.10) and to plot it in a graph. Figure 7.1 shows an example of such a calculation. The smaller the value of λ_w , the greater is the inlet diameter at which NPSH_R reaches its minimum. At the same time, this minimum becomes flatter; hence the inlet diameter can be selected somewhat below the mathematical minimum without significant loss in suction capacity.

D. Dimensioning for minimum NPSH_i : Suction impellers of high-pressure pumps with $u_1 > 50$ m/s must be designed for short cavity lengths to avoid cavitation damage. Above $u_1 \approx 75$ m/s the impeller must be operated virtually without any cavitation. The inlet diameter can be calculated from Eq. (T7.1.6), into which the coefficient for cavitation inception $\lambda_{w,i}$ must be entered, Fig. 6.2 and Table D6.3. Since these values depend sensitively on blade profile, incidence and non-uniformities of the approach flow, CFD-calculations of the pressure distribution are useful to verify the design, Chap. 8.

As is evident from Figs. 6.19 and 6.20 and discussed in Chap. 6.3, the λ -coefficients and suction specific speeds are subject to a large scatter since many influence parameters are ignored in this simple approach. Sizing the impeller inlet on the basis of above concepts therefore implies some uncertainty. The risk of cavitation problems (vibrations, noise and erosion) should be evaluated according to Chap. 6 and Tables 6.1 and 6.2 when determining the impeller inlet geometry.

7. Blade inlet diameter at the inner streamline d_{1i} : The lower limit for d_{1i} is usually given by the method of mounting the impeller on the shaft. For between-bearing impellers and $n_q < 25$ to 30, the diameter d_{1i} should be selected as close as possible to the lower limit in order to improve the stability of the Q-H-curve. With overhung impellers and high specific speeds, d_{1i} should not be

Fig. 7.1 Optimization of the impeller inlet diameter d_1



selected too small since very large blade inlet angles would be obtained. These cause excessive flow separations near the hub during partload operation.

8. Impeller blade inlet angles: Once the impeller inlet diameter has been determined from step (6), it is possible to calculate all quantities of the inlet velocity triangle according to Table 3.1. The blade inlet angle is obtained by adding the incidence i_1 to the flow angle β'_1 (with blade blockage) from Eq. (T7.1.7): $\beta_{1B} = \beta'_1 + i_1$. If cavitation is not an issue (e.g. series stages of multistage pumps or low-head pumps with ample $NPSH_A$), the incidence should be selected between 0 and 4°. The flow rate of shockless entry according to Eq. (T.3.1.10) is then close to, or slightly above, the best efficiency point. In contrast, suction impellers should be designed with low blade loadings - i.e. flat pressure distributions - at the impeller inlet. With increasing tip speed u_1 at the impeller inlet, lower incidences should be chosen in order to reduce the thickness of the cavities and hence the risk of cavitation damage, see also [10]. The blades can then be designed with very low (even strongly negative) incidence. When this is done, care must be taken that the minimum throat area A_{1q} is sufficiently large (see item 15 below).

Instead of using the incidence, it is also possible to apply an “angle exaggeration”. This constitutes the ratio of the blade angle tangent to the tangent of the flow angle with blockage and is defined as $a_{ex} = \tan\beta_{1B}/\tan\beta'_1$. For radial impellers $a_{ex} = 1.1$ to 1.2 may be assumed and for semi-axial impellers $a_{ex} = 1.05$ to 1.1. If the impeller is designed for low $NPSH_i$, $a_{ex} = 0.95$ to 1.05 can be selected.

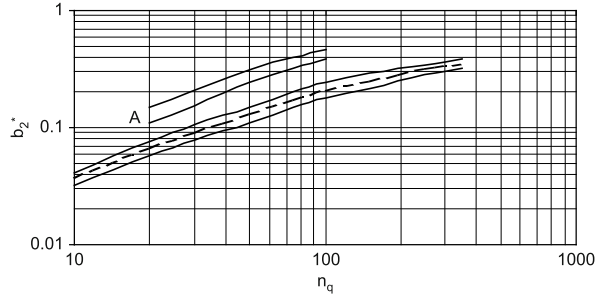
The above calculation of the flow and blade angles is performed for the outer, mean and inner streamlines (with high n_q use 5 or more streamlines). The meridional velocity c_m *upstream* of the leading edge is mostly assumed constant over the inflow cross section. Depending on the local hub blockage, a different value of c_m should be used for the calculation of each streamline if applicable, Table 3.1.

If the pump has to be operated far to the right of the best efficiency point ($Q_{max} \gg Q_{opt}$), inlet diameter and incidence must be enlarged accordingly. However, this causes the cavitation characteristics in the best efficiency point and at partload to deteriorate; therefore, the design options and risks should be carefully evaluated in such cases.

9. Outlet width b_2 : Blade number, blade outlet angle and outlet width cannot be selected independently of one another. They must be matched so that the demanded head coefficient is achieved with a stable Q-H-curve. As shown by the outlet velocity triangle in Table 3.2, a large outlet width reduces the meridional velocity c_{2m} thereby increasing c_{2u} (and the head) - *unless the flow separates*. With given values of outlet angle and blade number the head consequently increases with the outlet width; the Q-H-curve becomes correspondingly flatter. As discussed in Chap. 5, the outlet recirculation intensifies with increasing b_2/d_2 and, consequently, shut-off pressure and power consumption rise as well.

A sufficiently large outlet width is required to achieve a stable Q-H-curve (in particular at low n_q). Conversely, the non-uniformity of the flow at the impeller

Fig. 7.2 Impeller outlet width. Radial and semi-axial impellers: $b_2^* = b_2/d_{2a}$; axial impellers: $b_2^* = 1/2(1 - \nu)$ Curves of range A: sewage pumps with $z_{L,a} \geq 2$



outlet grows with the width of the impeller, increasing the turbulent dissipation losses in the collector as well as pressure pulsations and excitation forces. Finally, it should be ensured to maintain $b_2 < b_1$.

Since the above effects cannot be calculated theoretically, the relative outlet width $b_2^* = b_2/d_{2a}$ is commonly selected from empirical data. To render the impeller discharge flow near the best efficiency point as uniform as possible and to avoid unnecessary turbulent dissipation losses, it should be attempted to select b_2^* as low as is permissible with respect to the stability of the Q-H-curve. Figure 7.2 gives a band for $b_2^* = f(n_q)$ which approximately covers the data provided in [B.8, B.18, 25]. Set $b_2^* = 1/2(1 - \nu)$ for axial impellers. The mean curve for the outlet width in Fig. 7.2 can be approximated by Eq. (7.1). For sewage pump impellers with 2 or 3 blades Eq. (7.1a) can be used to determine the impeller outlet width.

$$b_2^* = 0.017 + 0.262 \frac{n_q}{n_{q,Ref}} - 0.08 \left(\frac{n_q}{n_{q,Ref}} \right)^2 + 0.0093 \left(\frac{n_q}{n_{q,Ref}} \right)^3 \quad (7.1)$$

$$b_2^* = 0.02 + 0.5 \frac{n_q}{n_{q,Ref}} - 0.03 \left(\frac{n_q}{n_{q,Ref}} \right)^2 - 0.04 \left(\frac{n_q}{n_{q,Ref}} \right)^3 \quad n_{q,Ref} = 100 \quad (7.1a)$$

10. Outlet angle β_{2B} : The blade outlet angle must be determined so that the specified head, or the head coefficient selected in step (4), is achieved with the values of d_2 , $z_{L,a}$ and b_2 determined previously. The outlet angle β_{2B} which fulfills this requirement can be determined by trial and error when calculating the respective slip factor and the head from Eq. (T7.1.8 and T7.1.9), see also Tables 3.2 and 3.3. The hydraulic efficiency must be assumed according to Chap. 3.9.

Outlet angles of radial impellers with 5 to 7 blades are commonly in the range of 15 to 45°; in many cases angles are selected between 20 and 27°. Matching outlet angles and outlet width is an optimization task during which the requirements in terms of efficiency and stability of the Q-H-curve (shut-off pressure and saddle) have to be considered. A possible criterion is to minimize the deviation angle $\delta_2 = \beta_{2B} - \beta_2$ which should not exceed 10 to 14° in order to limit the turbulent dissipation losses caused by non-uniform flow distributions.

A specified head can be achieved according to Eq. (T7.1.9) with very different combinations of b_2 , z_{La} and β_{2B} . Therefore no strict rules can be given for selecting the outlet angle. The blade length, obtained from the design, depends on the inlet and outlet angles; consequently this applies also to the resulting hydraulic blade loading. All these parameters must be collectively optimized and cannot be determined independently of one another (see also step 4).

With specific speeds above $n_q \approx 40$ to 60 the blade outlet angle should not be made constant over the impeller outlet width. The outer streamline is unloaded so that $\beta_{2B,a} < \beta_{2B,m} < \beta_{2B,i}$. The head is calculated on the basis of the mean angle $\beta_{2B,m}$. In this way, it is possible to match the lengths of all streamlines; in particular it can be ensured that the outer streamline is sufficiently long and, therefore, lowly loaded.

After the outlet angle has been determined, all parameters of the outlet triangle according to Table 3.2 can be calculated. To avoid premature stall and efficiency losses the deceleration ratio w_2/w_{1a} should not fall below the value of 0.7 (“de Haller criterion”).

11. Blade thickness e : Requirements in terms of castability and mechanical strength determine the blade thickness which depends on the head per stage or tip speed u_2 , the specific speed (impeller outlet width), the number of blades and the material and corrosion effects which limit the allowable alternating stress. Experience shows that these requirements are usually met when $e/d_2 = 0.016$ to 0.022 is selected for multi-vane impellers. The upper range applies to high-pressure impellers with more than 600 m of head per stage. The lower range applies to low heads and low specific speeds. With increasing blade width b_2 (i.e. also with growing specific speed) the blade stresses rise at a given head; thicker blades must be selected accordingly. A rational for checking the blade thickness with respect to the risk of impeller fatigue fractures is provided in Chap. 14.1. Equation (T14.3.1) in Table 14.3 allows to estimate the blade thickness as a function of the number of blades, the impeller tip speed and the specific speed; the correlation works even for single-channel impellers. The minimum thickness required to achieve good casting qualities must be observed in the case of cast impellers; it depends on the casting process and is 3 to 5 mm.

12. Blade leading edge profile (Fig. 7.3): Unfavorable leading edge profiling generates local excess velocities and a correspondingly intense low-pressure peak which impairs the cavitation behavior and may even affect the efficiency. Designing the leading edge as a semi-circle is very unfavorable in this regard and only acceptable for small pumps or applications with low requirements. Elliptical inlet profiles provide favorable pressure distributions. If the elliptical profiling extends only over a short distance, the blades react less sensitively to (excessive) incidence. In contrast, long wedge-like profiles perform well only at shockless entry ($i_1 = 0$) since high excess velocities are generated in incident flow. Long, thin (wedge-like) profiles are also unfavorable with respect to the casting quality (blade cracks caused by rapid cooling of thin profiles) and in terms of strength and risk of blade cracking.

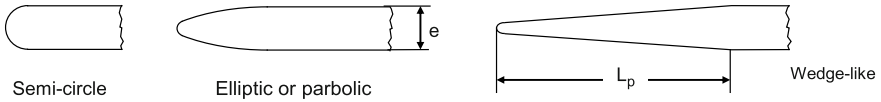


Fig. 7.3 Impeller blade leading edge profiles

According to calculations on individual profiles [65] quoted in [B.1], the minimum pressure coefficient $c_{p,\min,SF}$ at zero incidence (shockless entry) is:

$$\Delta p = c_{p,\min} \frac{\rho}{2} w_1^2 \quad \text{with} \quad c_{p,\min,SF} = 0.373 \frac{e}{L_p} \left(2 + 0.373 \frac{e}{L_p} \right) \quad (7.1a)$$

L_p is the length of the profile measured from the leading edge to the point of the transition to the full blade thickness e . Equation (7.1a) yields $c_{p,\min,SF} = 2.05$ for a semi-circle ($e/L_p = 2.0$), while an elliptical profile with $e/L_p = 0.2$ yields $c_{p,\min,SF} = 0.155$. As per Chap. 6.3.2 (Fig. 6.19), these results cannot be transferred directly to impellers since λ_w depends on the absolute value of the blade angle and other geometrical parameters of the impeller and the inlet. However, Eq. (7.1a) can be used for selecting and assessing a leading edge profile.

The effects of parabolic, elliptical, semi-circular and blunt leading edge profiles were investigated by testing and by CFD in [4], see also Chap. 8.7. The particular parabolic profile investigated was slightly better than the specific elliptical profile. This finding applied to all cavitation criteria (inception, 1 and 3% head drop and head breakdown). In this particular test the σ_3 -values at BEP were 0.18 for the parabolic profile, 5% higher for the elliptical, 17% higher for semi-circular and 25% higher for the blunt leading edge profile. Evidently, the semi-circular and the blunt leading edge perform less well than the parabolic and elliptical profiles, see also Eq. (7.1a). The suspected reason for the parabolic profile to be slightly superior to the elliptical is that the parabolic profile was thinner which results in lower excess velocities as long as the incidence is low. It should be noted that the thickness distribution of parabolic and elliptical profiles can be made nearly identical by a suitable selection of the profile parameters.

The profiling and shaping of the blades are discussed further in Chap. 7.2.4

13. Blade trailing edge profiles: Two design philosophies are used in practice:

A. The blades according to Fig. 3.6 are tapered towards the trailing edge to roughly half of the blade thickness according to $e_2 = 1/2 e$ in order to reduce the width of the wake, turbulent dissipation losses and pressure pulsations.

B. The full blade thickness is configured up to the trailing edge to preserve the possibility for under-filing in case of a deficit in head, Chap. 4.5.2. Outlet profiling is sometimes waived also for cost reasons.

After performing step 13, all parameters required for a first impeller design according to Chap. 7.2.2 are established. When the first design is available, blade

lengths and channel cross sections are known so that blade loading and deceleration ratios or cross sections can be verified.

14. Blade loading: Hydrodynamic blade loading can vary in a wide range without quantifiable adverse effects on efficiency and hydraulic excitation forces. Truly reliable criteria for the optimum blade length are hence not available. However, various limits for the hydrodynamic blade loading have been suggested. Using these blade-loading coefficients, it is possible to verify the selected blade length and blade number. Applying such criteria allows a more consistent approach to impeller design.

A. The evaluation of a number of proven radial pumps produced a correlation for the blade loading that can be used to verify the blade length, [70]. According to this, the blade number and length are selected appropriately if the effective blade load ξ_{eff} calculated from Eq. (T7.1.10) is roughly within the range of the allowable loading ξ_{al} given by Eq. (T7.1.11). In general, it is recommended to select the blade loading typically 10% below the value given by Eq. (T7.1.11) in order to reduce pressure pulsations and hydraulic excitation forces.

B. Another possibility of checking the blade loading was published in [26]: the lift coefficient ξ_a calculated from Eq. (T7.1.12) should not exceed the value 0.9. Whether this criterion has proved itself in practice is not known. For axial impellers with $\varepsilon_{\text{MS}} = 0$, Eq. (T7.1.12) leads to the relationship Eq. (T7.5.8b) valid for propeller pumps.

15. Throat area A_{1q} : As discussed in Chap. 7.1.3, blade angles and channel cross sections must be matched to the intended flow. In the design point the deceleration of the relative velocity vector w_1 to the mean flow velocity w_{1q} in the throat area must not be too high in order to avoid premature inlet recirculation. However, with respect to cavitation at high flows, excessive acceleration must not be allowed either. According to evaluations on operating pumps the throat area should be selected so that the ratio w_{1q}/w_{1m} in the best efficiency point with axial approach flow is in the range of 0.75 to 0.85 and with radial inlet casings (between-bearing pumps) between 0.65 and 0.75. At the maximum flow rate w_{1q}/w_{1m} should not appreciably exceed the value 1.0. The vector of the relative velocity w_{1m} is calculated on the mean streamline d_{1m} (not d_{1b}).

16. Blade distance a_2 at the outlet: the distance between the blades at the outlet a_2 must match the outlet angle β_{2B} in order to ensure that the slip factor (hence the head) are obtained. To this end, an angle $\beta_{a2} = \arcsin a_2/t_2$ can be defined and compared with β_{2B} . Experience shows that the ratio $\sin \beta_{a2}/\sin \beta_{2B}$ should be in the range of 0.7 to 0.9, Chap. 3.3 and Eq. (3.8).

Since the impeller design is to a large extent empirical, it is recommended to systematically apply consistent design methods and rules such as outlined above. This approach fosters a better predictability of hydraulic performance and allows one to faster build a data base and gain experience.

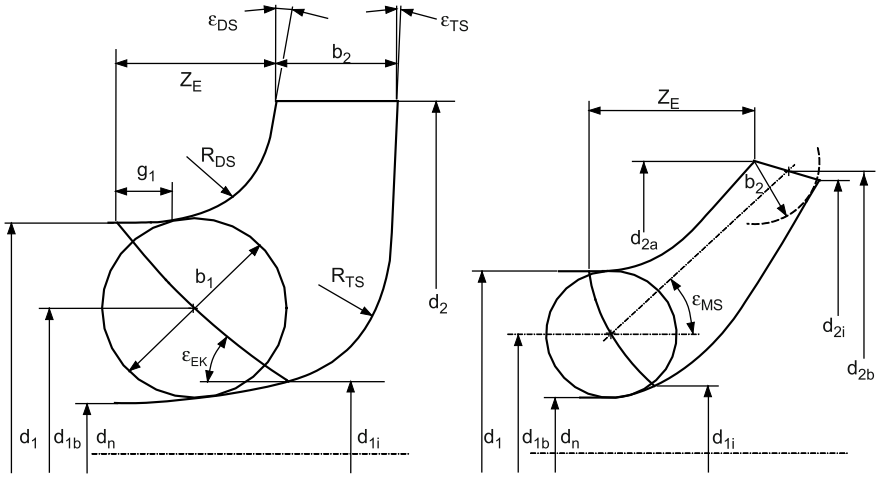


Fig. 7.4 Design parameters for the meridional section of the impeller. *Left:* Radial impeller, *right:* semi-axial impeller

7.2.2 Impeller Design

7.2.2.1 Design of the Meridional Section

A section through the impeller axis is called “meridional section”. In this presentation the blade leading and trailing edges are projected into the drawing plane through “circular projection”, Fig. 7.4.

To be able to design the meridional section, details for the position of the leading edge are required next to the dimensions (d_2 , b_2 , d_1 , d_{1i} , d_n) determined above. Recommendations for the axial extension z_E and the radius of curvature R_{DS} of the front shroud are given by Eq. (7.2):

$$z_E = a \frac{d_{1a} - d_n}{2} \left(\frac{n_q}{n_{q,Ref}} \right)^x \quad \begin{array}{l} R_{DS} = (0.6 \text{ to } 0.8)b_1 \\ b_1 = 1/2(d_1 - d_n) \\ n_{q,Ref} = 1 \end{array} \quad \begin{array}{l} \text{for example:} \\ a = 0.75 \\ x = -0.05 \\ \text{see also Table 7.8} \end{array} \quad (7.2)$$

Favorable flow conditions are achieved by moving the leading edge forward into the impeller eye. In this way low blade loadings and correspondingly moderate low-pressure peaks are obtained and cavitation is reduced. The leading edge at the outer streamline should not be located in the region of high curvature but preferably before the start of the front shroud curvature. Therefore, the radius R_{DS} should *not* be tangent to the point defined by z_E , but a short section $g_1 = (0.2 \text{ to } 0.3) \times b_1$ should be introduced with only a minor increase in radius in order to achieve flatter pressure distributions. For small pumps, or when a short axial extension of the

impeller is a priority, smaller values are selected for z_E and R_{Ds} than calculated from Eq. (7.2).

The angle ε_{Ds} which defines the shape of the front shroud near the impeller outlet is determined according to the specific speed. It can be used to *influence the velocity profile at the impeller outlet*. For $n_q < 20$, $\varepsilon_{Ds} = 0$ is often selected. For radial impellers with higher specific speeds ε_{Ds} increases to approximately 15 to 20°. Selecting $\varepsilon_{Ds} > 0$, increases the flow deceleration in the impeller for given b_2 and β_{2B} and thus the head produced by the impeller; this applies also for low n_q .

With d_2 , b_2 , z_E , d_1 , g_1 , ε_{Ds} and R_{Ds} it is now possible to draw the outer streamline which can be defined by a free curve or assembled from straight lines and circular arcs. When using a design program, Bezier functions are often used. Unfortunately, general rules are not known which would ensure favorable hydraulic features. Meridional section and blade development must rather be optimized together.

The angle ε_{Ts} which defines the shape of the rear shroud near the impeller outlet can be chosen positive or negative. At $n_q < 30$, $\varepsilon_{Ts} = 0$ is often designed (some manufacturers also use negative values). At high specific speeds ε_{Ts} is always positive with $\varepsilon_{Ts} < \varepsilon_{Ds}$.

The inner streamline must be shaped in a way that the cross sections $A = 2 \times \pi \times r \times b$ vary continuously along the mean streamline. To this end, a mathematical rule may be selected which describes the cross section A or the channel width b in the meridional section from b_1 to b_2 as a function of the developed length of the outer streamline, Fig. 7.5. In addition to a linear or parabolic function, a cubic law could also be considered to obtain smaller changes in cross section near the inlet and outlet of the impeller. After the desired rule $b = f(L)$ has been defined, the width is marked off with dividers at a number of checkpoints on the outer streamline. Subsequently, the inner streamline is drawn as an envelope curve. The same construction can also be performed in sub-operations with several streamlines. Finally, the streamlines can be constructed from the theory of potential flows.

The blade leading edge is initially obtained from the chosen parameters d_1 , z_E , d_{1i} and the designed inner streamline. The position of the leading edge has a major effect on the partload behavior according to Chap. 5. The following criteria serve for shaping the leading edge: (1) In multistage pumps with $n_q < 25$ the angle ε_{EK}

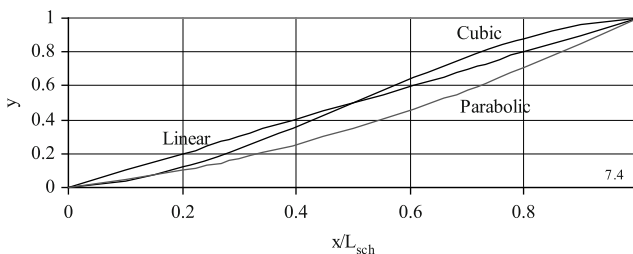


Fig. 7.5 Options for the development of the flow area in the impeller; L_{sch} = blade length

must be designed as large as possible to prevent the Q-H-curve from drooping towards $Q = 0$. (2) For impellers with an axial inlet $\varepsilon_{EK} = 30$ to 40° is about the optimum according to [25]. (3) If ε_{EK} is selected too small, problems with hub cavitation and instabilities of the Q-H-curve may be encountered. (4) Choosing ε_{EK} too large can result in too low values of d_{li} and too large blade angles $\beta_{1B,i}$ on the hub.

As discussed in Chap. 5, the meridional section largely determines the flow distribution over the blade width and, consequently, the turbulent dissipation losses, the partload behavior and the stability of the Q-H-curve.

Criteria for the meridional section to be checked:

- The leading edge should join the outer streamline with the largest possible angle to reduce the local blockage in the case of twisted blades.
- The curvatures of the outer and inner streamlines and the cross sections $A = 2 \times \pi \times r \times b$ should vary continuously along the mean streamline.

7.2.2.2 Blade Design

The blade design serves to define the shape of the blades along the outer, inner and mean streamlines so that the inlet and outlet angles established in Chap. 7.2.1 are obtained. The blade describes a three-dimensional curve along each surface of revolution; for instance, along the outer streamline defined by the meridional section (or front shroud). This curve can be described through its projections into the meridional section and the plan view. Every point of this curve is defined in space by the coordinates r , z and ε .

With reference to Fig. 7.7a consider an element of the outer streamline which is positioned between the points 5 and 6 on the radii r_5 and r_6 . In the meridional section it has the (approximate) length:

$$\Delta m = \sqrt{\Delta r^2 + \Delta z^2} \quad (7.3)$$

In the plan view, Fig. 7.7b, e, the position of the points 5 and 6 is defined by these radii and the angle $\Delta \varepsilon$. The distance between points 5 and 6 is:

$$\Delta g = \sqrt{\Delta r^2 + \Delta u^2} \quad (7.4)$$

The length of the streamline element in space is obtained from the length in the plan view Δg and the extension in axial direction Δz (normal to the plan view):

$$\Delta L = \sqrt{\Delta g^2 + \Delta z^2} \quad (7.5)$$

By means of Eqs. (7.3 and 7.4) the relationship between the true length ΔL of the streamline element and its projection Δm in the meridional section and the projection Δu in the plan view is obtained:

$$\Delta L = \sqrt{\Delta g^2 + \Delta z^2} = \sqrt{\Delta r^2 + \Delta u^2 + \Delta z^2} = \sqrt{\Delta m^2 + \Delta u^2} \quad (7.6)$$

The distances ΔL , Δu and Δm according to Fig. 7.7c constitute a right-angled triangle which includes the true angle β between the streamline element ΔL and the tangential direction Δu .

If a streamline is developed step by step into the drawing plane according to this procedure, its true length and the true angles relative to the circumferential direction are obtained. This procedure is used to design the blade coordinates in the plan view from a defined meridional section and a blade development to be specified. In the above method, the curvilinear triangles have been replaced by rectilinear ones. The smaller the elements are selected, the better this approximation becomes. Of the different methods for designing the blades, the Kaplan method is described [B.2]; it comprises the following steps:

1. Depending on the width of the impeller (i.e. n_q) one to five additional surfaces of revolution are drawn into the meridional section in addition to the outer and inner streamlines. These surfaces can be designed as streamlines of identical part flows. To this end, (estimated) normals to the streamlines are drawn. In doing so, the width between two streamlines is determined in a way that the same flow $\Delta Q = 2 \times \pi \times r \times \Delta b \times c_m$ travels through each partial channel (frequently with the assumption $c_m = \text{constant}$).
2. All surfaces of revolution or streamlines are sub-divided into n elements of identical length Δm (e.g. $\Delta m = 8 \text{ mm}$) according to Fig. 7.7a so that the points a_1 to a_n , b_1 to b_n etc. are obtained.
3. In the blade development according to Fig. 7.7d, n parallel straight lines are drawn at the distance Δm . Lengths $L_{a,m}$, $L_{b,m}$, $L_{c,m}$ etc. of the developed meridional section are obtained (which constitute the projection of the streamlines into the meridional section).
4. The length $L_{a,u}$ of the streamlines in circumferential direction is still unknown; within certain limits it can be freely selected. A possible approach consists in defining a development law according to Fig. 7.6. Starting with $\beta_{(j=0)} = \beta_{2B}$ at point A in Fig. 7.7d, the circumferential length Δu of the element Δm is calculated from Eq. (7.7) for each step j to obtain the position of the next point.

$$\Delta u_j = \frac{\Delta m_j}{\tan \beta_j} \quad (7.7)$$

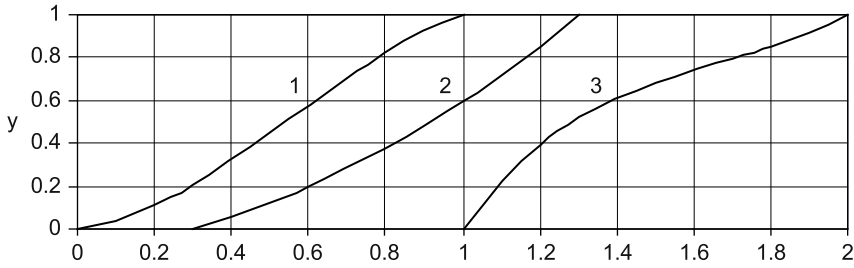


Fig. 7.6 Blade development options (the figures on the abscissa have no meaning)

The angles for each step are obtained from the selected law according to which the angle is to develop as a function of the blade length from β_{2B} to β_{1B} . In general it is possible to write:

$$\beta_j = \beta_{2B} - y(x)(\beta_{2B} - \beta_{1B}) \quad (7.8)$$

Any random function which satisfies the conditions $y(0) = 0$ and $y(1) = 1$ can be substituted for $y(x)$. If a linear angle development from β_{2B} to β_{1B} is desired, the function $y(x)$ has to be formulated as:

$$y(x) = \frac{1}{L_{a,m}} \sum_0^j \Delta m_j \quad (7.9)$$

This method can also be used in a way that the outlet angle is kept constant over n_a sections ($\beta_j = \beta_{2B}$ for $j < n_a$) and/or that the inlet angle is kept constant over n_e sections ($\beta_j = \beta_{1B}$ for $j > n - n_e$). Accordingly, Eq. (7.19) is only formulated for the middle part of the blade in which the angle changes from β_{2B} to β_{1B} .

5. An alternative way of determining the blade development is as follows: a line S1 is drawn from point A with an angle β_{2B} , Fig. 7.7d. A point E at distance $L_{a,m}$ is then selected such that a line S2 started with the angle β_{1B} intersects the line S1 in a point P. The ratio $L_p/L_{a,m}$ has an effect on the blade length and the channel cross sections. It must be selected so that the intended blade distances a_1 and a_2 as well as acceptable blade loadings according to the criteria in Eqs. (T7.1.10 to 7.1.13) are obtained. A smooth curve is then nestled into the geometry defined by the points A and E and the lines S1 and S2. This curve forms the chosen blade development.
6. The blade geometry must be represented in the plan view so as to allow the manufacturing of the impeller or pattern. The blade points 1 to 13 in the development (Fig. 7.7d) define triangles with the sides ΔL , Δu and Δm . These points are situated on the radii r_1 to r_{13} (see the meridional section in Fig. 7.7a). Starting on the outer radius the points are transferred to the radii r_1 , r_2 etc. with

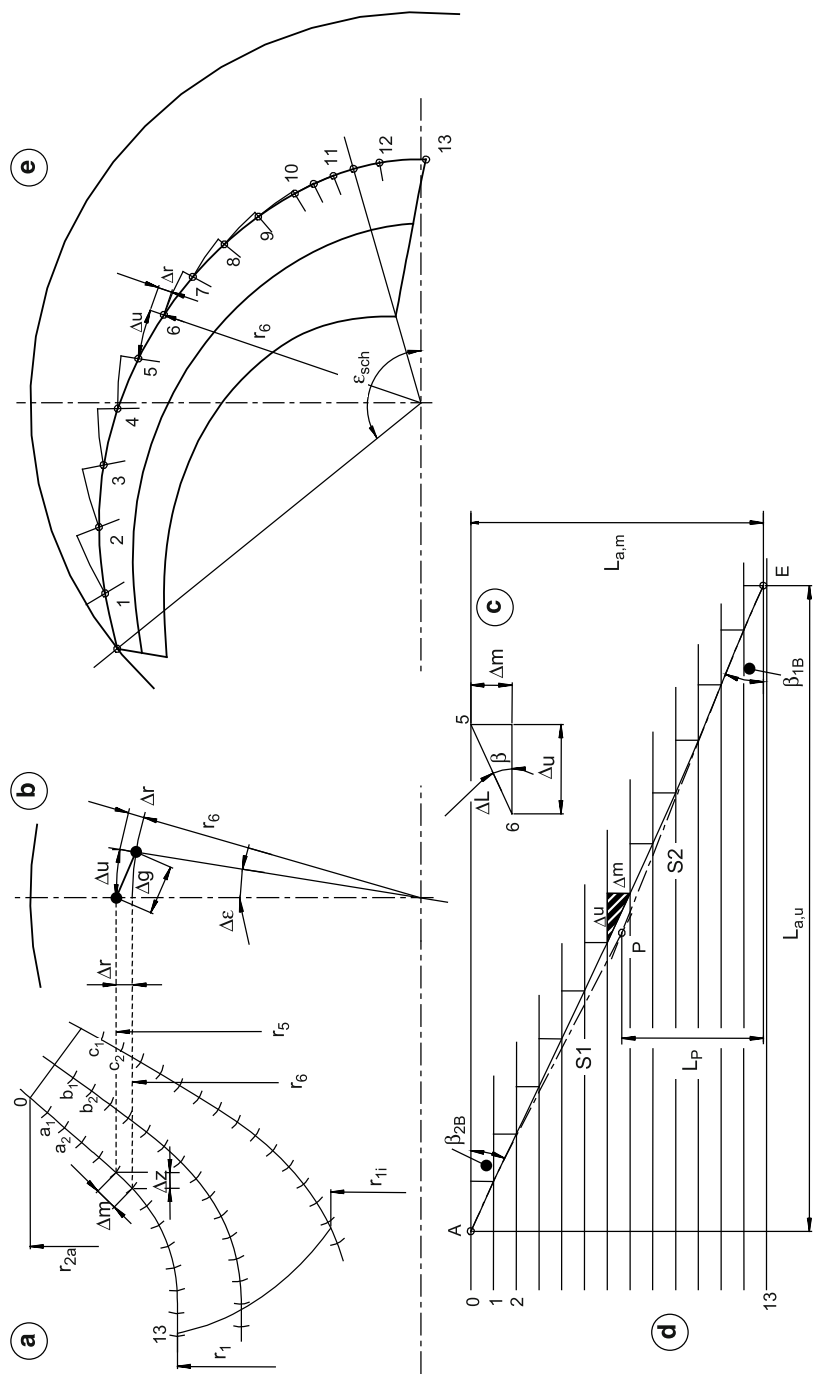


Fig. 7.7 Blade design according to Kaplan's method

- Δu and Δr according to Fig. 7.7e. This yields the points 1, 2, 3, etc. in the plan view. Joining these points produces the streamline projected into the plan view.
7. The wrap angle ε_{sch} , formed by the blade in the plan view depends on the blade angles β_{1B} and β_{2B} , the radii ratio d_1/d_2 (hence n_q) and the streamline length in the meridional section. Wrap angles of radial impellers usually are within the following ranges: with $z_{La} = 5$: $\varepsilon_{sch} = 130$ to 160° ; with $z_{La} = 6$: $\varepsilon_{sch} = 120$ to 140° and with $z_{La} = 7$: $\varepsilon_{sch} = 100$ to 130° .
 8. Steps 4 to 6 are performed for all surfaces of revolution (streamlines).
 9. To generate sufficient checkpoints for the manufacturing process, radial sections (A to Q in Fig. 7.8) are drawn in the plan view and their intersections with the surfaces of revolution or streamlines transferred into the meridional section (shown in Fig. 7.8 by way of Section K). The radial sections should yield smooth curves in the meridional section in order to avoid waviness in the blade surface. These radial sections need not always be curves; they can also be straight lines.
 10. For manufacturing it is also possible to use contour lines ("board lines"). These are equidistant sections in the meridional section normal to the axis, No. 0 to 12 in Fig. 7.8. Their intersections with the streamlines and radial sections are transferred to the plan view and joined to form the board lines. These must present smooth curves in the plan view in order to prevent wavy blade surfaces.
 11. The blade surface designed in this way constitutes the camber, the suction or the pressure surface - depending on the angles (camber, suction or pressure surface) that were used for the development. The selected blade profile can be drawn into the development as section normal to the blade surface in order to better appreciate its effect on the approach flow.
 12. The blade profile is represented as developed onto the camber line where it is dimensioned as thickness = $f(L)$ (dimensioning not shown in Fig. 7.8).

7.2.3 Criteria for Shaping the Blades

When constructing the blade development, the designer enjoys a certain freedom which he may utilize to obtain specific hydraulic features. Impellers designed for high $NPSH_A$, i.e. when cavitation is not an issue, will be primarily optimized for efficiency and the stability of the Q-H-curve. In order to minimize the losses, a smooth variation of geometric parameters should be targeted (for example by increasing the blade angles to a linear function from inlet to outlet, i.e. $\partial\beta/\partial L = \text{constant}$). On the outer streamline the outlet angle β_{2B} is usually several degrees larger than the inlet angle $\beta_{1B,a}$ and a development similar to curve 2 in Fig. 7.6 is obtained. On the inner streamline of an impeller with axial inlet (end suction pump) $\beta_{1B,i}$ is generally much larger than β_{2B} so that a development similar to curve 3 in Fig. 7.6 is obtained.

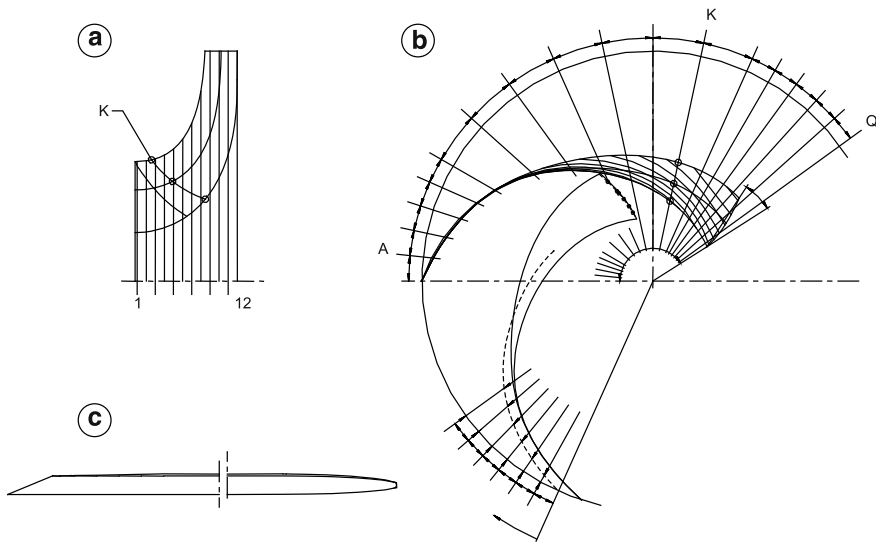


Fig. 7.8 Representation of the impeller coordinates by board lines 0 to 12 and radial sections A to Q. **a** Meridional section, **b** plan view, **c** blade profile (developed)

It is also possible to specify a law for $u \times c_u$ as a function of the blade length. For example, the design could be done with low blade loading at the inlet because of cavitation, with the strongest blade load in the middle part where the blades overlap and moderate blade loading towards the outlet with the object to achieve as uniform a flow as possible. It is thus attempted to minimize turbulent dissipation losses and pressure pulsations.

The developments of all streamlines must be matched to each other so that the desired leading edge is obtained. Since the inflow angles are usually small (between 12° and 18°), the approach flow of the blade leading edge can be assessed qualitatively by way of its position in the plan view. If the leading edge is radial (No. 1 in Fig. 7.9), it is hit by an almost perpendicular approach flow. If the leading edge is inclined by the angle ε against the radial direction, it meets the approach flow at an angle which tends to cause lower velocity peaks and losses (compare to a cylinder exposed to an inclined versus a perpendicular approach flow). A leading edge shape according to Nos. 2, 3 or 4 in Fig. 7.9 corresponds to a “sweep back” which is used in axial pumps, compressors and inducers. It tends to displace fluid away from the hub. Special shapes such as No. 5 in Fig. 7.9 (“sweep forward”) are sometimes used to extend and unload the outer streamline (see Chap. 5). As discussed in Chap. 5, the leading edge also influences the partload behavior - especially shapes such as 4 and 5 in Fig. 7.9.

The blade twist at the inlet is obtained from Eq. (T7.1.7) from the calculated flow angles and the selected incidences. This involves major angle variations over the blade height if $d_{1i} \ll d_{1a}$ and $\beta_{1B,i} \gg \beta_{1B,a}$. In contrast, the blade twist at the

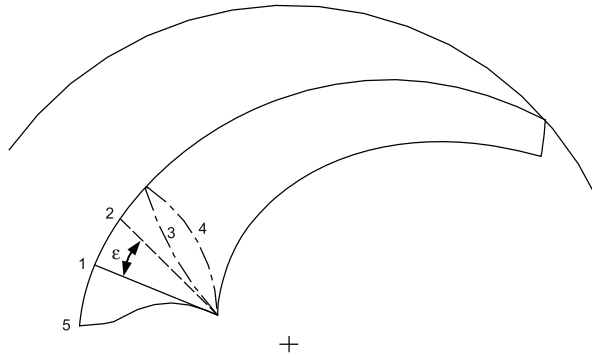


Fig. 7.9 Different shapes of the impeller blade leading edge

impeller outlet can be selected freely. The blades of radial impellers of low specific speeds are frequently not twisted at the outlet. The higher the specific speed, the more an outlet twist will be required to optimally match the streamline lengths. In order to reduce pressure pulsations, the blades can be twisted at the impeller outlet. The effectiveness of this measure increases with the ratio of the outlet width b_2 to the pitch of the blades t_2 and with the specific speed.

The blades can be arched deliberately in order to increase the throat area A_{1q} . The radial sections will then appear as pronounced curves in the meridional section. This design is used especially with radial impellers of high specific speeds (compare to the impellers of Francis turbines).

In general, all streamlines, blade developments, radial sections, board lines and channel cross sections should vary continuously along the flow path in order to prevent unnecessary accelerations, decelerations and velocity differences over the channel. Turbulent dissipation losses caused by uneven velocity distributions can thus be minimized.

7.2.4 Criteria for Suction Impeller Design

From the time when boiler feed pumps were built with heads up to 600 to 800 m per stage - or with u_1 up to 80 m/s - (so called “advanced class feed pumps” are operating even with 1200 m per stage), the design philosophy of the suction impellers has changed. In the past - with low circumferential speeds at the impeller inlet - large inlet angles and high incidences were selected and blades with long wedge-like inlet profiles were designed to obtain sufficiently low values of $NPSH_3$, concept “H” in Fig. 7.10c. Even when operating with low incidence, such impellers generate pronounced low-pressure peaks and large detached cavities. At $u_1 > 50$ m/s these can result in severe cavitation erosion.

In view of such damage, the manufacturers of high-pressure pumps were faced with the need to develop new blade shapes by means of stroboscopic observation of the cavities generated at the impeller inlet. These new designs feature: (1) Small incidences. (2) Relatively small absolute blade and approach flow angles which subsequently produce lower incidences at partload (example: with $\beta_{IB} = 20^\circ$ an incidence of 10° is obtained at $Q/Q_{SF} = 0.5$, while the incidence is only $i_1 = 5^\circ$ at $\beta_{IB} = 10^\circ$). (3) Thick, short profiles which are shaped to be less sensitive to incident flow. (4) Low blade loadings in the inlet region up to roughly the throat area, concept “L” in Fig. 7.10d.

In order to achieve flat pressure distributions, the blades can also be designed with an asymmetric leading edge profile where the pressure surface is arranged according to the calculated/selected blade angle $\beta_{IB,PS} = \beta_{IB}$ and the thickness distribution is applied to the pressure rather than on the camber surface, Fig. 7.10e. The maximum blade thickness can be selected equal to the nominal thickness e_{nom} (left sketch in Fig. 7.10e) or greater than the nominal thickness (right sketch in Fig. 7.10e). In [10] $e_{max}/e_{nom} = 1.37$ had been successfully applied to reduce cavitation damage. In doing so, the blade leading edge was swept forward near the hub (compare to shape 4 in Fig. 7.9); this blading is called “biased wedge design”, [57].

Based on these characteristics, flat pressure distributions are obtained. In addition the pressure increase over the first part of the blade should be kept small by limiting the increase in radius and blade angles at the outer streamline over a length of approximately $t_1 = \pi \times d_1/z_{La}$. This limits the increase in $u \times c_u$ in that region. This design achieves low coefficients σ_i for cavitation inception and thin translucent cavities with low vapor volume and correspondingly low hydraulic cavitation intensity. Figure 7.10a schematically shows how the specific work should develop over the blade length to obtain flat pressure distributions according to Fig. 7.10b. Despite the flat blade angles in the inlet region over a length of t_1 the throat area between the blades must be designed large enough so that the maximum specified flow rate can be achieved, Chap. 7.2.1, item 15. This can result in S-shaped developments of the camber line according to Fig. 7.6, curve 1.

Before designing suction impellers, the risk of cavitation damage should be assessed according to Chap. 6 and especially Chap. 6.7. In many applications the risk of damage is low and the aim is to achieve the lowest possible $NPSH_3$ and high n_{ss} . In contrast, the suction impellers of high-pressure pumps with considerable damage potential must be designed to achieve a low $NPSH_i$ and low cavity volumes. Typical parameters of these two extreme cases are compared in Table D7.1 (applications in-between can be interpolated). However, high suction specific speeds can also be achieved by designing impellers with flat pressure distributions - implying negative incidence and suction-side profiling. This is increasingly done and will be more so in the future. Some examples were discussed in [81]. Impellers with flat pressure distributions, designed according to the principles discussed above, have been tested in a single-stage volute pump with axial inlet, [80]. The data of interest are listed in Table D7.1-1. Three impellers with suction specific speeds have been designed with *negative incidence*. Interestingly, the onset of inlet recirculation $q^*_{RB} = Q_{RB}/Q_{opt}$ of all impellers was in a narrow range. Reason: with

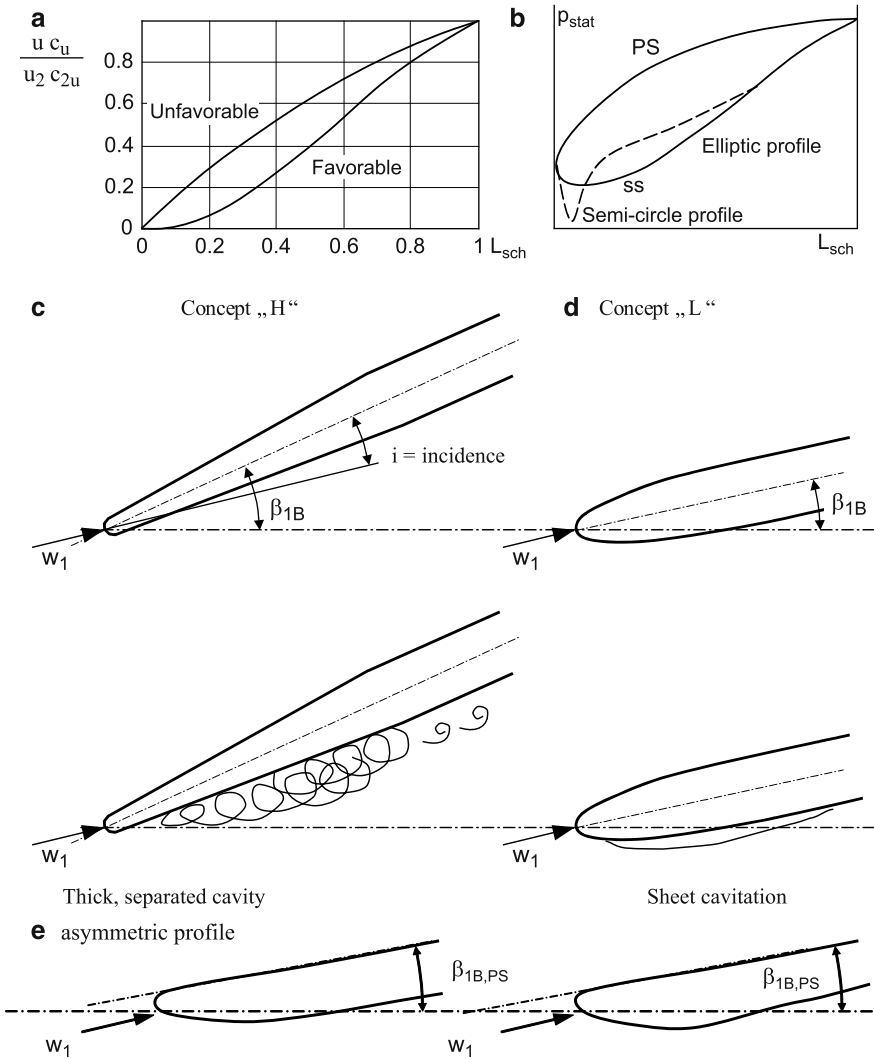


Fig. 7.10 Suction impeller design. **a** Work transfer; **b** pressure distribution; **c** design with high incidence and sharp profile; **d** design for a flat pressure distribution; **e** asymmetric profiling on the suction surface

increasing inlet diameter d_1^* , the onset of recirculation q_{RB}^* would be expected to rise, while q_{RB}^* drops as the incidence is reduced: in these tests, the effect of increasing the impeller inlet diameter apparently was compensated by reducing the incidence. Q_{RB} was predicted by CFD: incidentally, 2 cases were close to test, 2 cases were off (see Chap. 8.8). As in Fig. 6.21, the n_{ss} increases with decreasing approach flow angle β_{1a} ; the data in Table D7.1-1 are situated above the upper limit of the scatter in Fig. 6.21.

Table D7.1 Suction impeller design

Risk of erosion	Low	High
Impeller inlet tip speed	$u_1 < 35$ m/s	$u_1 > 50$ m/s
Fluid	Water above 220 °C and hydrocarbons	Water below 220 °C
Suction specific speed	$n_{ss} = 220$ to 260	$n_{ss} = 160$ to 180
Approach flow angle	$\beta_{1a} = 10$ to 15°	$\beta_{1a} = 14$ to 18°
Cavitation inception at $q^* = 1$	$\sigma_i = 1.2$ to 2	$\sigma_i = 0.5$ to 0.8
Incidence at $q^* = 1$	$i_1 = 3$ to 5° trend: smaller or negative	$i_1 = < 2^\circ$ or negative
Shockless entry at $q^* = 1$, but depending on Q_{max}	$Q_{SF}/Q_{opt} = 1.1$ to 1.2 trend: lower than 1.0	$Q_{SF}/Q_{opt} = 0.95$ to 1.05 or even lower
Resulting pressure distribution	Peaky	Flat

Figure 7.10.1 shows the cavitation coefficients $\sigma_3 = f(\phi_1)$ calculated from the test curves in [80]. Interestingly, the three curves of the impellers with $n_{ss} > 280$ coincide over their respective test range while the curve with $n_{ss} = 186$ seems shifted upwards. This observation may show the influence of the meridional section: the high n_{ss} -impellers had similar meridional sections with generous radii at the outer streamline (OSL). In contrast, the impeller with $n_{ss} = 186$ had a strong curvature at the OSL and the axial extension z_E of the blades is much shorter (z_E : rough estimates in Table D7.1-1, definition in Fig. 7.4). The sketches in Fig. 7.10.1 show, very approximately, the impeller shapes. As often observed, the plot against the flow coefficient looks more telling than the plot against ϕ_1/ϕ_{1SF} . This finding seems to emphasize the dominant impact of the stagger angle on cavitation phenomena. With 3D-leading edges it is also difficult to define a meaningful blade blockage. Figure 7.10.1 shows also the blade coefficients λ_{w3} calculated from Eq. (6.11).

Good cavitation characteristics of high-pressure pumps can only be achieved if the inlet casing is optimized so that no vortices (vortex ropes) are created. The inflow to the impeller must be as uniform as possible so that the incidence does not vary over the circumference by more than $\pm 1^\circ$. The partload behavior can be improved with a diffuser upstream of the impeller, [24].

The design criteria for suction impellers discussed above were predominantly developed empirically based on stroboscopic cavitation observations, partly supported by numerical calculations of inviscid flow. This follows from publications such as [11, 23–25, 34, 57, 58, 71]. Today efficient numerical methods can be employed for the prediction of cavitation inception and cavity length, [14, 27], provided the approach flow variations are kept under control.

Table D7.1-1. Suction impellers tested in [80], $n_q = 29$, type OH2												
d_1^*	β_{1a}	$i'_{1,a}$	n_{ss}	$\varphi_{1,opt}$	Q_{SF}/Q_{opt}	$\varphi_{1,SF}$	q_{RB}^*	$q_{RB,CFD}^*$	$\varphi_1/\varphi_{1,SF}$	b_2^*	β_{2B}	z_E/r_{1a}
	°	°	—	—	—	—	—	—	—	—	°	—
0.445	22.6	4.8	186	0.402	1.30	0.521	0.62	0.48	0.771	0.091	24.0	0.42
0.482	17.7	-7.0	287	0.308	0.67	0.207	0.65	0.63	1.490	0.082	26.3	0.83
0.500	15.3	-2.3	327	0.263	0.90	0.235	0.67	0.63	1.116	0.077	29.0	0.82
0.527	13.3	-3.4	346	0.227	0.80	0.182	0.64	0.74	1.248	0.086	27.5	0.60

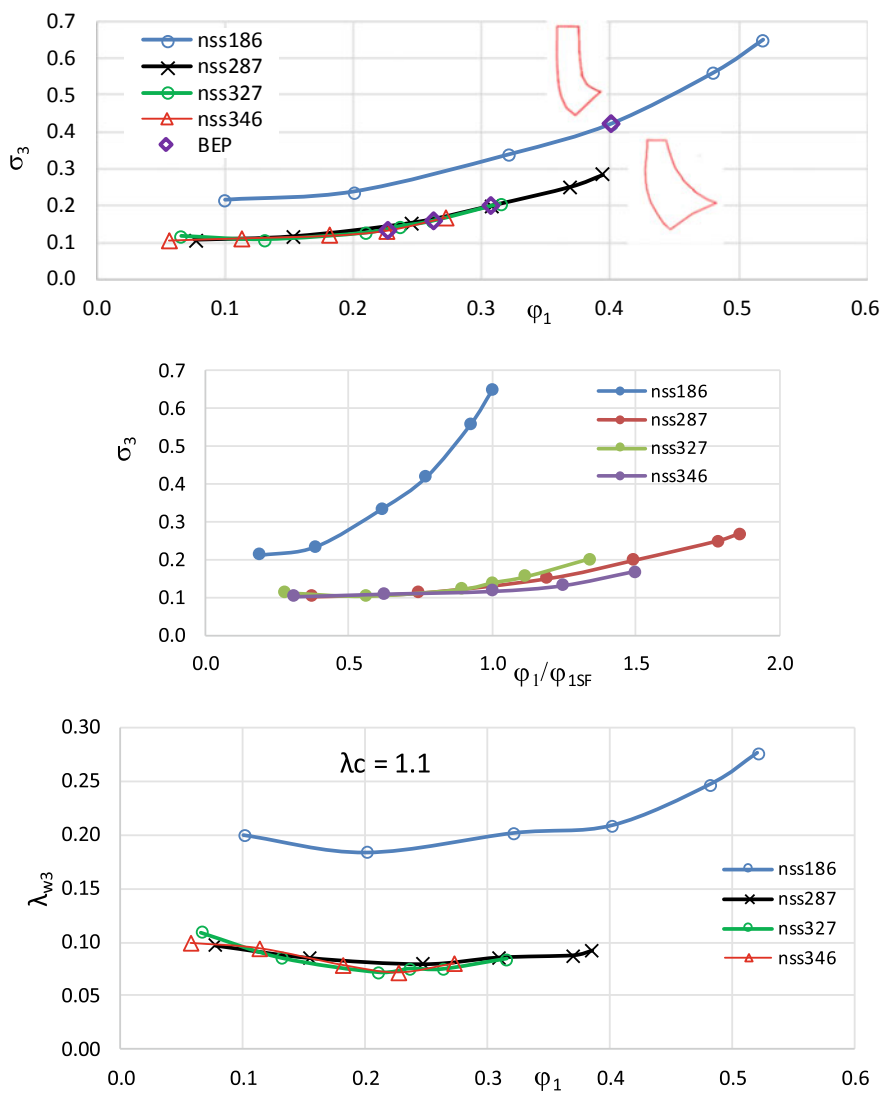


Fig. 7.10.1 Cavitation performance of impellers specified in Table D7.1-1; [80]

7.2.5 *Exploiting Three-Dimensional Effects in Design*

As explained in Chap. 5, the flow deflection (or the slip factor) and the losses are affected by secondary flows which, in the one-dimensional theory, can only be taken into account by empirical coefficients. The statistical data used do not do justice to the details of the geometry of a specific impeller. Applying numerical methods it is possible, however, to examine the impact of all geometrical characteristics on the flow. This opens up new possibilities for optimization by attempting to influence the flow by means of new kinds of geometrical effects and shapes. This does not only concern the minimization of the hydraulic losses in the design point but, especially, also the stability of the Q-H-curve and cavitation. 3D-effects increase with the blade height, i.e. with the specific speed. Examples for 3D-effects are:

- Sweep-forward: a leading edge pulled forward according to Fig. 7.9, shape 5, tends to improve the stability of the Q-H-curve of semi-axial pumps, Chaps. 5.5 and 5.6.
- Sweep-back reduces the extent of the cavitation zones in inducers and axial impellers.
- A leading edge pulled forward at the hub according to shape 4 in Fig. 7.9 helps to reduce the cavitation near the hub at partload according to [57].
- Axial blades can also be designed with a lean of their entire surface against a radial line. Numerical analyses of this effect were reported in [40].
- By means of blade twisting it is possible to affect the energy distribution over the blade height.
- The common practice to vary the impeller outlet angle over the blade height at high specific speeds can be employed more purposefully with the help of numerical calculations.
- The rear shroud need not necessarily be designed as a surface of revolution if the flow can be improved by a 3D-shape, [47]. A freely shaped rear shroud would not imply any major increase in expenditure for the pattern manufacture if NC-milling or stereo-lithography were employed for the purpose. However, no specific information is available on the benefit of such measures.

7.3 Radial Impellers for Small Specific Speeds

7.3.1 *Two-Dimensional Blades*

The outer and inner streamlines of cylindrical (untwisted) blades coincide in the plan view. At specific speeds below $n_q = 16$ to 18 untwisted blades can be designed, for example, as series stages of multistage pumps. Since the impeller losses at low specific speeds are small (see Fig. 3.33) and the shock losses at the

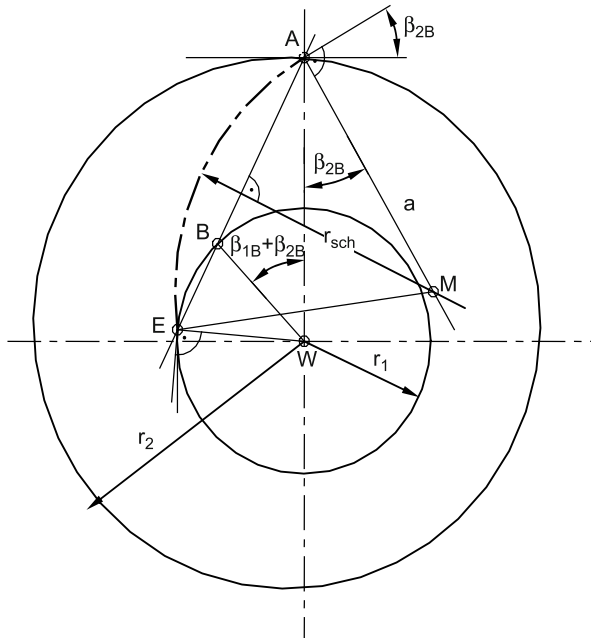


Fig. 7.11 Construction of circular blades

impeller inlet are insignificant because of the low ratio d_1/d_2 , the cylindrical blades imply virtually no loss in efficiency. However, if good cavitation characteristics are required, twisted blades must be employed.

Impellers with cylindrical blades are calculated and designed in a similar way as per Chap. 7.2.1. Impellers for undemanding services can be made with circular arc blades instead of designing the blade development according to Chap. 7.2.2.2. In the plan view the blades can be designed so that inlet and outlet angles are achieved by a single circular arc. The design is explained by means of Fig. 7.11: (1) Circles are drawn with the inlet and outlet radii of the impeller. (2) At any point A on the outer radius r_2 , the blade angle β_{2B} is applied and the normal $A-a$ is erected in A . (Note direction of rotation.) (3) From the shaft center W the angle $(\beta_{1B} + \beta_{2B})$ is applied in the direction of rotation on the radius $W-A$. The resulting line intersects the circle with r_1 in a point B . (4) A line through A and B produces a second intersection E with the circle r_1 , marking the blade leading edge. (5) The normal in the middle of the distance $A-E$ intersects the normal $A-a$ in a point M constituting the center of the circular arc. The latter is drawn with the radius $r_{sch} = MA = ME$. (6) The circular arc obtained constitutes the camber line of the blade. Suction and pressure surfaces of the blades are obtained from circular arcs with $r = r_{sch} \pm 0.5$ e around point M .

The radius r_{sch} of the circular arc blade can be calculated from Eq. (7.10):

$$r_{sch} = \frac{r_2^2 - r_1^2}{2(r_2 \cos \beta_{2B} - r_1 \cos \beta_{1B})} \quad (7.10)$$

Circular arc blades of this type result in an unfavorable development of the relative velocity which is decelerated to a value $w < w_2$ within the impeller channel. So as not to jeopardize the stability of the Q-H-curve, the blade leading edges must not be parallel to the rotor axis; they must indeed be brought forward to as small as possible a diameter d_{li} (which is limited by the hub diameter).

Radial impellers with cylindrical blades are mainly employed in the range $8 < n_q < 18$ for small pumps and for less demanding service up to about $n_q = 25$.

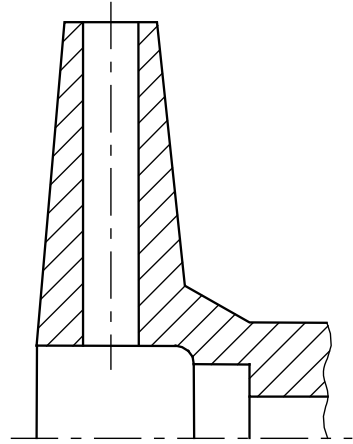
The small outlet width ratio b_2^* of impellers with very low specific speeds can cause casting problems because of the narrow channels. These difficulties are avoided by resorting to pumping disks or semi-open impellers with radial blades discussed in the following.

7.3.2 Pumping Disks with Channels of Circular Section

The work transfer in radial impellers of low specific speeds (i.e. with a small ratio d_1^*) is largely accomplished by centrifugal forces. This method of energy transfer implies very low losses in the impeller (Fig. 3.33). Hence the hydraulic efficiency of the pump is essentially determined by the losses in the collector. When centrifugal effects prevail, the impeller channels can be designed as very simple shapes without jeopardizing efficiency. A rotating disk with cylindrical holes according to Fig. 7.12 can be employed as an impeller, as described for instance in [46]. The holes can be drilled in radial direction with $\beta_{2B} = \beta_{1B} = 90^\circ$. Alternatively, the holes can be leaning back in circumferential direction with $\beta_{2B} = \beta_{1B} < 90^\circ$ to produce less flow deflection and correspondingly lower head coefficients but steeper Q-H-curves.

The collector is designed as a volute or an annular casing which is often preferred because of its simplicity. The fluid leaving the collector is decelerated further in a diffuser. The throat area of the diffuser mainly determines the best efficiency flow rate since the losses in the diffuser by far outweigh the impeller losses. When the fluid from the impeller outlet is accelerated into the diffuser throat (i.e. at $c_{3q} > c_2$), the diffuser losses increase. Consequently, head and efficiency drop. With sufficient acceleration, the flow cavitates in the throat area and head and efficiency drop almost vertically ("choking"). The design of the annular collector follows Chap. 7.12, the diffuser is calculated according to Chap. 1.6, and the best efficiency point flow rate is obtained from Chap. 4.2.

Fig. 7.12 Using a disk with drilled holes as an impeller



To obtain high head coefficients and low impeller losses, the velocity c_B in the holes must be selected low. It largely corresponds to the relative velocity in the holes with diameter d_B :

$$c_B = \frac{4Q}{\pi d_B^2 z_{La}} \quad \text{and} \quad \varphi_B = \frac{c_B}{u_2} \quad (7.11)$$

According to Chap. 5.2 the Coriolis acceleration is $b_c = 2 \times \omega \times w$. In radial holes b_c is largely responsible for the slip factor which increases with growing flow rate because of $w \approx c_B$. From measurements it can be deduced that the flow coefficient in the design point should not exceed $\varphi_B = 0.06$ to 0.08 if high head coefficients are desired. The impeller losses are then in the range of a few percent of the head. If high head coefficients are targeted, number and size of the holes must be selected such that flow coefficients of $\varphi_B < 0.08$ are obtained at the design point. It is then that head coefficients in the range of 1.1 to 1.25 are achieved. However, the Q-H-curves become flatter and the head drops slightly towards $Q = 0$. If a steeper characteristic is required, a higher flow coefficient must be selected for the design point; even backward leaning holes may be required. Conversely, the impeller losses increase and the efficiency drops.

Since the annular casing according to Chap. 7.12 has a larger cross section than corresponds to the impeller outlet velocity c_{2u} , the flow is decelerated abruptly (similar to a sudden expansion according to Chap. 1.2.3) and an intense secondary flow is generated. According to Chap. 5, the head can be increased if the exchange of momentum between the impeller and the flow in the collector is intensified (similar to a regenerative pump, Chap. 2). This increases the kinetic energy at the inlet to the diffuser whose pressure recovery rises accordingly. The exchange of momentum grows with decreasing φ_B ; it can be deliberately promoted by providing material recesses on the impeller outer diameter which accelerate the fluid

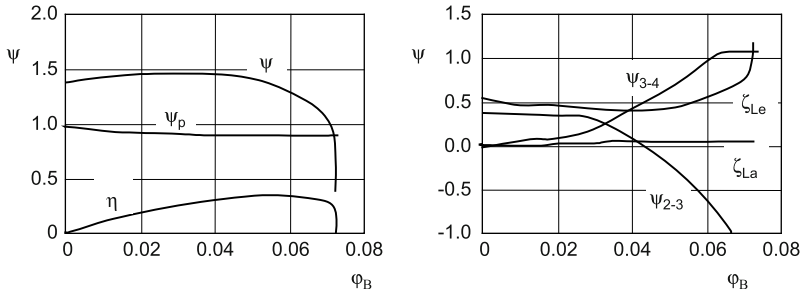


Fig. 7.13 Tests with a rotating disk with 12 drilled channels, Sulzer Pumps

circulating in the annular casing by exchange of momentum. In [46] for instance, blind holes were used for this purpose. In this way it is possible to considerably increase head coefficient and efficiency: at BEP $\psi_{\text{opt}} = 1.3$ to 1.4 is obtained while the maximum values at partload just reach $\psi_{\text{max}} = 1.5$. The Q-H-curve on the left of the best efficiency point becomes very flat and droops slightly towards shut-off.

Between impeller and casing, relatively large axial gaps (up to approximately $s_{\text{ax}} = 0.01 \times d_2$) can be permitted since the impeller sidewall operates like a friction pump. The head can even be improved slightly since the impeller sidewall boundary layer enters the collector with high circumferential speed - in this way increasing the kinetic energy of the fluid circulating there. However, a slight loss in efficiency is incurred since the increase of the power consumption exceeds that of the head.

Figure 7.13 shows the dimensionless characteristics of a rotating disk with 12 holes and recesses on the impeller outlet to increase the exchange of momentum between the annular casing and the impeller. The flow coefficient was calculated from Eq. (7.11). All head coefficients correspond to the definitions used in Chaps. 3 to 5. The Q-H-curve is very flat at flows below the best efficiency point; it drops vertically at $\phi_B \approx 0.073$, which is due to cavitation in the diffuser.

Noteworthy is also the flat curve of the static pressure rise ψ_p in the impeller which shows that slip factor and impeller losses change only moderately with increasing flow rate. At $Q = 0$, ψ_p almost reaches the theoretical value of the flow without losses. According to Chap. 4.1.4 this corresponds to a hydraulic impeller efficiency of 0.96. The pressure recovery in the diffuser ψ_{3-4} roughly increases with the square of the flow rate. The pressure increase ψ_{2-3} from the impeller outlet to the diffuser throat is significant below $\phi_B \approx 0.03$. Above $\phi_B \approx 0.044$ the static pressure is reduced since the flow is accelerated from the impeller outlet to the diffuser throat area (i.e. $c_{3q} > c_2$). The measured impeller losses are insignificant compared with the losses in the collector.

7.3.3 Impellers with Straight Radial Blades

Impellers with straight radial blades with $\beta_{2B} = \beta_{1B} = 90^\circ$ according to Fig. 7.14 are sometimes employed in small pumps with speeds up to 25,000 rpm¹. The range of specific speeds is about $n_q = 6$ to 12. This type of pump is often equipped with an annular casing from which the fluid is discharged through a diffuser. Alternatively, vaned diffusers or double volutes are in use. The impeller is semi-open or open to reduce disk friction and annular seal losses. Depending on the number of blades, this design demands tight axial clearances of $s_{ax}/b_2 = 0.01$ to 0.02 between blades and casing walls to obtain acceptable efficiencies.

The impeller outlet width is selected relatively large, similarly to the curve shown for sewage pumps in Fig. 7.2. The impeller usually has 16 to 24 (32) blades. Typically 3 to 6 blades have the full blade length from d_1 to d_2 while the rest are splitter blades - otherwise blade blockage at the inlet would become excessive. The large impeller outlet width and large number of blades produce slip factors close to 1.0 and foster a strong exchange of momentum with the flow in the annular collector. As with drilled disks, head coefficients of $\psi_{opt} = 1.2$ to 1.5 are achieved. The Q-H-curve becomes correspondingly flat and often slightly unstable, similar to that in Fig. 7.13. Blade numbers over 16 increase the head coefficient in the best efficiency point only to an insignificant degree but yield a higher shut-off head due to improved exchange of momentum, thereby increasing the stability of the Q-H-curve.

The hydraulic characteristics of radial bladed impellers are similar to those of the drilled disks discussed in Chap. 7.3.2. According to Fig. 3.26, efficiencies up to approximately 45% can be achieved—depending on specific speed, pump size and mechanical losses. High-speed pumps are often equipped with an inducer to lower NPSH₃; the suction specific speeds are typically between $n_{ss} = 350$ and 450.

Figure 7.15 shows the dimensionless characteristics of a radial bladed impeller measured with two different diffuser inserts the throat diameters of which amounted to 71 and 39% of the impeller outlet width. With the large diffuser insert a specific speed of $n_q = 11$, a head coefficient of $\psi_{opt} = 1.48$ and an efficiency of 43% were obtained. By reducing the diffuser throat area, the specific speed was reduced to $n_q = 7$ while the efficiency dropped to 23%. The power consumption was practically identical in both tests so that the best efficiency point shift was only achieved through throttling in the diffuser. The fact that cavitation in the diffuser plays a significant role is borne out by the curves for the cavitation coefficient which have changed drastically even though the impeller was identical.

¹These impellers are sometimes called “Barske impeller”, [5, 13, 43].

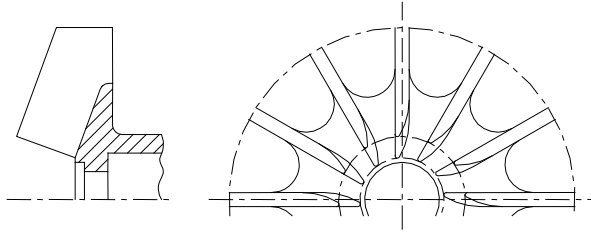


Fig. 7.14 Impeller with straight radial blades

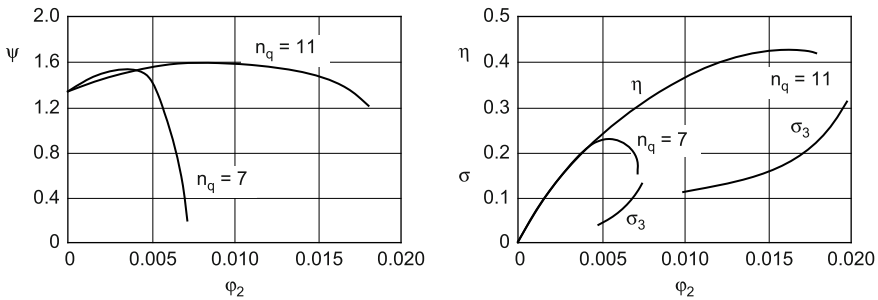


Fig. 7.15 Tests with an impeller with straight radial blades and two different diffusers, Sulzer Pumps

7.3.4 Double-Acting Impeller with Straight Radial Blades

A double-acting overhung impeller with straight radial blades is shown in Fig. 7.16 in a view from the inlet and in the meridional section. The impeller is used in a single-stage, single-entry end-suction pump of the type shown in Fig. 2.1. The heads achieved are well above 300 m, [12, 22]. Nearly identical blades are arranged on both sides of the center rib. Five large openings in the center rib near the hub allow half of the fluid to flow to the blades on the rear (that is why the impeller is labeled “double-acting”). On each side of the center rib there are 5 long and 20 short blades. The collector is built as a concentric casing followed by a partial volute which discharges into a diffuser.

In the present application the impeller is used for pumps in the range of roughly $1 < n_q < 10$. Figure 7.16c shows the dimensionless characteristics for $n_q = 2$, $n_q = 5$ and $n_q = 9$. While efficiencies above 50% are achieved with $n_q = 9$, the efficiency drops sharply for lower specific speeds.²

²In order to plot the curves for $n_q = 2$ to $n_q = 9$ on the same graph, a logarithmic scale was chosen for the flow. The shape of the efficiency curves therefore looks unusual.

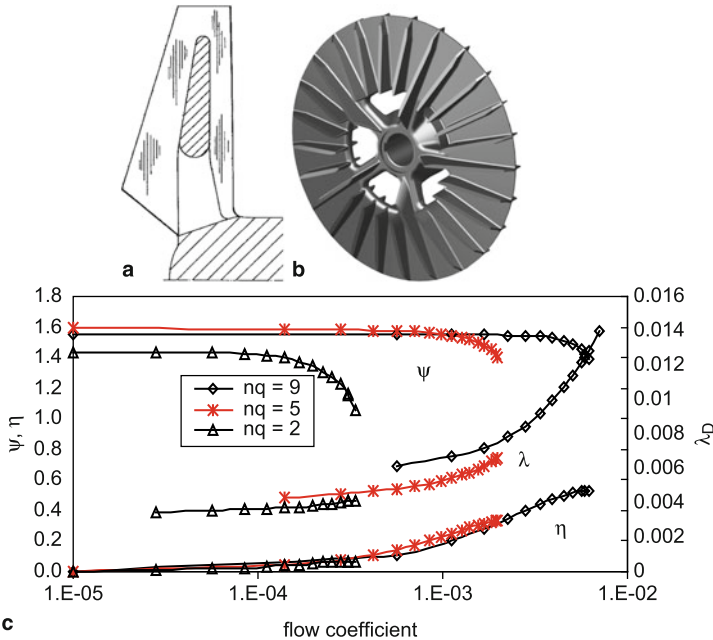


Fig. 7.16 End-suction pump with double-acting impeller; **a** meridional section; **b** front view; **c** performance characteristics, λ_D according to Eq. (T3.4.10)

The head coefficients at BEP are above 1.4; they drop at very low specific speeds due to increased losses in the collector. The Q-H-curves are stable but flat because of the high head coefficients. Since flow conditions on both sides of the impeller are very similar, the axial thrust is balanced to a large extent. There are no disk friction losses, which accounts for the relatively high efficiency. The gaps between the casing and the impeller blades can be made rather large without unduly impairing the efficiency. Operation on the right of the best efficiency point is limited by cavitation in the collector. Due to the low blade loading vibrations are very low.

“Double-acting” impellers with specific speeds of $n_q = 4, 8$ and 10 have been tested in [86]. In essence, they used grooves rather than blades and 6 large holes in the hub to allow flow to the rear side of the impeller; the concept is very similar to Fig. 7.16. Per side of the impeller, there were 180 grooves (long and short) with a depth of 3 mm (with $d_2 = 202$ mm). Groove width was 2 mm; the “land” between two grooves - which in fact corresponds to the blade thickness - was 1.5 mm. Performance was similar to Fig. 7.16, but the head coefficients were about 10% lower. Axial clearances as large as 5 mm (with groove depth 3 mm and total $b_2 = 12$ mm) had little effect on the head, while the efficiency suffered moderately (because of the scatter difficult to quantify). The suction specific speed was $n_{ss} = 194 \times (1 - 0.6n_q/n_{q,Ref})$ with $n_{q,Ref} = 10$ (tested between $n_q = 4$ and 9). As with the other pumps of low specific speed, the high head coefficients are achieved by an

intense exchange of momentum between the impeller channels and the casing near the impeller outlet. For this reason, performance is little affected by increased axial clearances between casing and impeller.

Another type of a “double-acting” impeller designed for $n_q = 6$ has been described in [87]. The meridional section of the *open* impeller with axial inflow resembles the shape of a conventional impeller for low n_q . The blades are arranged on both sides of a support plate which is attached to the hub by the blades themselves. Since the blade inlet angles can be selected to match the axial approach flow, the $NPSH_R$ is expected to be lower than with radial blades as applied in Fig. 7.16.

7.4 Radial Impellers for Non-clogging Pumps

In various applications coarse solids or fibrous matters are transported by the fluid pumped; this requires special designs to reduce the risk of clogging and damage. Typical applications are:

- Sewage pumps the operation of which is mainly threatened by clogging through textiles, plastic parts and foils.
- Dredge pumps where the excavated material can include coarse gravel and all kinds of waste.
- Turnip or fish pumps which are equipped with single-channel impellers in order to avoid damaging the goods pumped.
- Paper pulp pumps (Chap. 13.5).

Sewage pumps usually are single-stage pumps with radial, semi-axial or axial impellers. Radial and semi-axial pumps use volutes because diffusers would be prone to clogging. Axial pumps are built with diffusers with thick vane leading edges to reduce the risk of clogging. Two types of clogging should be distinguished for sewage pump design:

1. Clogging by large solids such as plastic parts or wood. To reduce the risk that foreign matter gets stuck in the impeller or volute, one specifies the diameter d_K of a ball which must be able to pass through the pump.
2. Clogging by fibrous or stringy matter such as textile waste or plastic bags. In essence, this issue is independent of the ball passage diameter and more difficult to handle than the passage of large solids. The reason is that fibrous matter can be caught at the impeller blade leading edge independently of the ball passage diameter or channel width. Fibrous matter can accumulate in the impeller eye and in the axial impeller sidewall clearances (gaps E and F). Clogging can also be caused by spinning fibrous matter into a long agglomeration in front of the impeller. Partload flow recirculation from the impeller could be instrumental in this process. In extreme cases, fibrous matter entering the impeller side rooms can lead to a blockage of the rotor. Expeller vanes can be applied and/or the impeller side rooms can be closed by introducing a gap A (Table 0.2) in order to reduce the ingress of fibrous matter into the side rooms. Various devices and

arrangements which are able to chop long fibers are applied to reduce the risk of clogging. In small pumps a chopper wheel with (for example) 3 knives can be installed in front of the impeller. In other designs cutting edges are integrated into the casing and impeller.

Figure 7.17 shows a selection of special impeller designs which were developed for pumping liquids containing solids with a risk of clogging or which are laden with gas.

Published design information on sewage pumps is scarce. The design data given in the following were derived from the sales curves of four major sewage pump manufacturers. The results of this study are presented in Table 7.2. Figure 7.18 shows the pressure coefficients thus derived for pumps with 1- to 3-bladed impellers and for vortex pumps. The curves shown represent in essence the upper limit, even though some individual data points can be found above these curves. However, many pumps offered on the market are designed for lower pressure coefficients - presumably with the object to increase the ball passage.

The efficiencies documented in the sales curves have been compared to those expected for normal water pumps from Table 3.9. To do so, for any given sewage pump with a BEP flow Q_{opt} and a specific speed n_q the reference efficiency $\eta_{\text{opt,R}}$ of a clear-water pump has been calculated from Eq. (T3.9.1). The ratio of the actual sewage pump efficiency to $\eta_{\text{opt,R}}$ yields an efficiency multiplier $f_\eta = \eta_{\text{opt,sewage}}/\eta_{\text{opt,R}}$. The multipliers resulting for the different pump types are given in Table 7.2. While the *absolute* efficiencies depend on the pump size and specific speed, no clear dependency of the multiplier f_η on the specific speed or the pump size could be established (with the exception of vortex pumps). Hence the concept of normalizing the sewage pump efficiency with Eq. (T3.9.1) provides a meaningful method for efficiency prediction and assessment of sewage pumps. Table 7.2 gives the average efficiency multipliers as well as the range in which most of the pumps analyzed were found.

The design of sewage pumps is largely determined by the size of foreign matter that must pass through the pump without clogging. The impeller inlet diameter d_1 , the impeller throat width a_1 or area A_{1q} , the impeller outlet width b_2 and the volute throat diameter d_{3q} or area A_{3q} all must be chosen to be larger than the ball passage diameter d_k . To select these parameters significantly larger than for clear-water pumps (which are optimized for efficiency) inevitably will impair the efficiency.

A sufficiently large blade distance is achieved by selecting the blade number between 1 and 4. The blade development is designed to yield the required a_1 as a minimum. The ball passage increases with the specific speed. Typical values for the ball passage are given in Table 7.2 and Fig. 7.19. In vortex pumps, the ball passage usually equals the diameter of the discharge nozzle. The same should be attempted for single-channel impellers, but in practice this goal is by far not always achieved as demonstrated by Fig. 7.20.

To obtain the necessary ball passage, the outlet width of impellers with low and medium specific speeds is designed considerably larger than for pumps intended for transporting pure liquids. The upper curve in Fig. 7.2 gives an idea of the relative

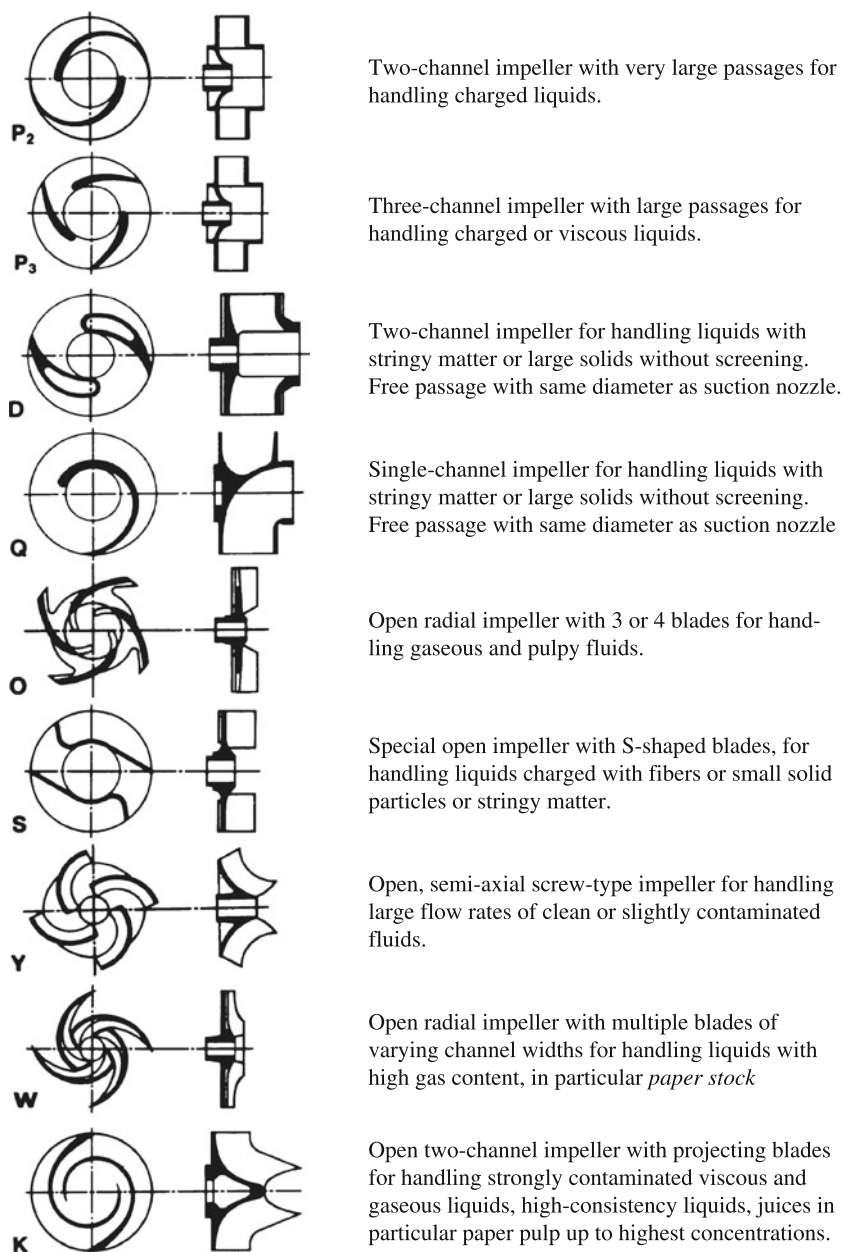


Fig. 7.17 Impellers for handling liquids charged with solids or gases, Sulzer Pumps

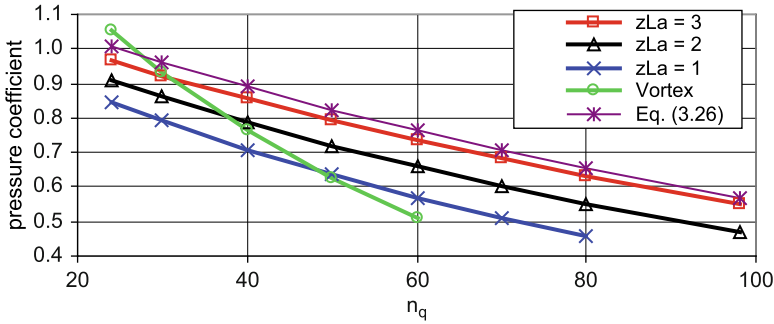


Fig. 7.18 Pressure coefficients for sewage pumps

outlet width of sewage impellers with 2 to 4 blades; single-channel impellers of small pump sizes are often larger. The data in Fig. 7.19 are representative of b_2/d_2 because of the condition $b_2 \geq d_k$. The large relative outlet width tends to yield high head coefficients, but because of the low blade numbers the slip factor γ drops, reducing the head (Chap. 3.3, Tables 3.2 and 3.3). Typical slip factors are given in Table 7.2; they depend on b_2/d_2 , d_1/d_2 , ε_{sch} and β_{2B} . The BEP head coefficients which can be achieved are shown in Fig. 7.18 and Table 7.2 which also provides head coefficients at shut-off.

The flow through pumps with single-channel impellers which are designed for very large ball passage is highly 3-dimensional and unsteady, see Chap. 9.3.9. Flow rate, pressure rise and torque are fluctuating strongly over one impeller revolution. Rough running and impairment in efficiency are the consequence. To appreciate the impact of the ball passage on efficiency, suppose that the ball passage is selected to the criterion $d_k = d_d$ (d_d = diameter of the discharge nozzle). As a consequence, the impeller inlet diameter d_1 , the outlet width b_2 and the volute or casing throat area d_{3q} must at least be equal to the discharge nozzle diameter: $d_{1a} = b_2 = d_{3q} = d_d$. The impact of selecting $d_k = d_d$ on the efficiency is threefold: (1) The large impeller outlet width causes a strongly non-uniform flow distribution and mixing losses in the casing; (2) The axial velocity at the impeller inlet $c_{ax} = 4Q/(\pi d_{1a}^2)$ is larger than the radial velocity at the blade leading edge $c_{rad} = Q/(\pi d_{1m} b_1)$. With $d_{1m} = (d_{1a} + d_{1i})/2$ the deceleration of the flow from c_{ax} to c_{rad} becomes: $c_{rad}/c_{ax} = d_{1a}/\{2b_1(1 + d_{1i}/d_{1a})\}$. With $d_{1a} = b_1 = b_2 = d_k = d_d$ we get $c_{rad}/c_{ax} = 0.5/(1 + d_{1i}/d_{1a})$ which yields $c_{rad}/c_{ax} = 0.25$ for $d_{1i}/d_{1a} = 1$ or $c_{rad}/c_{ax} = 0.33$ for $d_{1i}/d_{1a} = 0.5$. This strong deceleration at the impeller inlet, together with the sudden flow turn from the axial to the radial direction leads to flow separation and losses. (3) The requirement $d_{3q} = d_d$ implies that there is no flow deceleration in the diffuser; the associated sudden deceleration from the impeller outlet velocity c_2 to the throat velocity $c_{3q} = c_d$ could cause additional losses depending on the casing design. On the other hand, the flow deceleration in the volute is quite efficient, but the BEP may be shifted to higher flows due to the large throat area.

Fig. 7.19 Ball passage diameter d_k referred to impeller diameter d_2 for single-channel impellers from three manufacturers

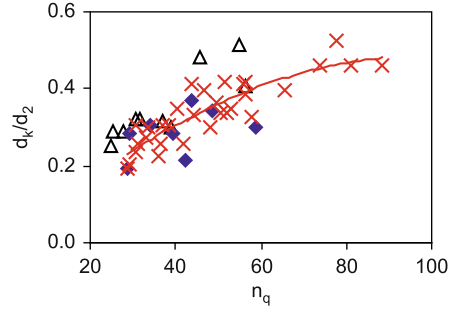
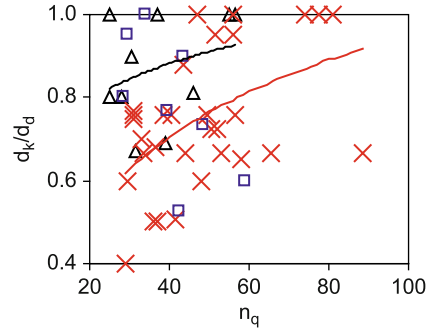


Fig. 7.20 Ball passage diameter d_k referred to discharge nozzle diameter d_d for pumps with single-channel impellers



Efficiency impairment with 2- and 3-bladed impellers is less severe than in single-channel impellers, see data in Table 7.2. Essentially it is due to the non-uniform flow in the impeller which is induced by the large impeller outlet width and the large pitch of the blades.

Studies on optimizing single-channel impellers were reported in [60, 64]. In [60] the impeller blade wrap angle was varied between 289° and 420° ; the optimum efficiency (64%) was found at 370° while it dropped to 50% at a wrap angle of 289° and was $\eta_{\text{opt}} = 61\%$ at 420° .

Open versus closed impellers: A parametric study of a pump with open single-channel impellers has been done in [8]; the main data were $n_q = 45$, $\beta_{1B} = 5^\circ$, $\beta_{2B} = 30^\circ$, $\varepsilon_{\text{sch}} = 325^\circ$. When increasing the axial gap width s between the casing and the blade of the open impeller from $s/b_2 = 0.005$ to 0.053, the BEP flow rate shifted to lower flow by about 30%, the head coefficient dropped by 12% and the efficiency diminished from 52 to 39%.

A closed impeller performed better than an open impeller with a gap width of $s/b_2 = 0.0057$: the BEP shifted slightly to higher flow; the shutoff head was not affected, but the difference in head increased linearly with the flow, attaining 35% at $q^* = 1.5$. The efficiency at BEP increased by 1%, but at $q^* = 1.5$ the efficiency of the closed impeller was 44% as opposed to the open impeller with only 35%.

According to a comparison of efficiency curves published in [49] closed and open impellers achieved the same efficiency at BEP, but the open impellers had

Table 7.2 Sewage pump design data			$n_q^* = n_q/n_{qRef}$ with $n_{qRef} = 1$	
Efficiency of water pumps according to Table 3.9 $Q_{Ref} = 1 \text{ m}^3/\text{s}$	$m = 0.1 a \left(\frac{Q_{Ref}}{Q} \right)^{0.15} \left(\frac{45}{n_q} \right)^{0.06}$		$Q \leq 1 \text{ m}^3/\text{s}$	$a = 1$
			$Q > 1 \text{ m}^3/\text{s}$	$a = 0.5$
	$\eta_{opt,R} = 1 - 0.095 \left(\frac{Q_{Ref}}{Q} \right)^m - 0.3 \left\{ 0.35 - \log \frac{n_q}{23} \right\}^2 \left(\frac{Q_{Ref}}{Q} \right)^{0.05}$			
Sewage pump efficiency		$\eta_{opt} = \eta_{opt,R} f_\eta$		
Pump type	Efficiency multiplier f_η		Blade wrap	Ball passage
	Average	Range	$\epsilon_{sch} [^\circ]$	d_k/d_2
$z_{La} = 1$	0.82	0.65 to 0.95	280 to 350	0.2 to 0.5 (Fig. 7.19)
$z_{La} = 2$	0.91	0.8 to 0.98	220 to 270	$d_k/d_2 = 0.13 \text{Ln}(n_q^*) - 0.26$
$z_{La} = 3$	0.94	0.9 to 0.99	170 to 220	$d_k/d_2 = 0.08 \text{Ln}(n_q^*) - 0.11$
Vortex	0.6	0.45 to 0.8	$f_\eta = (0.7 \text{ to } 0.8) \left(\frac{Q_{opt}}{Q_{Ref}} \right)^{0.06}$ $Q_{Ref} = 1 \text{ m}^3/\text{s}$	
Pressure coefficient at shut-off	Vortex		$\psi_0 = 1.38 - 0.0112 n_q/n_{qRef}$	
	$z_{La} = 1$		$\psi_0 = 1.45 - 0.0046 n_q/n_{qRef}$	
	$z_{La} = 2$		$\psi_0 = 1.36 - 0.0035 n_q/n_{qRef}$	
	$z_{La} = 3$		$\psi_0 = 1.39 - 0.002 n_q/n_{qRef}$	
Pressure coefficient at BEP	Vortex		$\psi_{opt} = 1.7 \exp(-0.02 n_q/n_{qRef})$	
	$z_{La} = 1$		$\psi_{opt} = 1.03 \exp(-0.0095 n_q/n_{qRef})$	
	$z_{La} = 2$		$\psi_{opt} = 1.1 \exp(-0.0087 n_q/n_{qRef})$	
	$z_{La} = 3$		$\psi_{opt} = 1.17 \exp(-0.008 n_q/n_{qRef})$	
Typical slip factors for $n_q < 60$ from Eq. (T3.2.9)	$z_{La} = 1$		$\gamma_{opt} = 0.48 \text{ to } 0.6$	
	$z_{La} = 2$		$\gamma_{opt} = 0.53 \text{ to } 0.65$	
	$z_{La} = 3$		$\gamma_{opt} = 0.67 \text{ to } 0.75$	
Impeller tip speed	Vortex		$u_{2,max} [\text{m/s}] = 45 - 0.42 n_q/n_{qRef}$	
	$z_{La} = 1$		$u_{2,max} [\text{m/s}] = 42 - 0.32 n_q/n_{qRef}$ up to $n_q = 60$	
Minimum to reduce risk of clogging: $u_{2,min} = 12 \text{ m/s}$	$z_{La} = 2$		$u_{2,max} [\text{m/s}] = 38 - 0.15 n_q/n_{qRef}$	
	$z_{La} = 3$		$u_{2,max} [\text{m/s}] = 42 - 0.15 n_q/n_{qRef}$	

much lower efficiencies than the closed designs at off-design conditions. This finding applies to single-channel and two-channel impellers. This difference in efficiency reached roughly 10% at $q^* = 0.5$ and at $q^* = 1.5$.

The ratio of the axial thrust of the closed impeller to that of the open impeller was about 0.5 at shutoff and 0.3 near BEP. Overall, closed impellers give better performance and lower axial thrust than open impellers.

Impeller tip speed: The impeller tip speed of single-channel impellers must be limited in order to keep the excitation forces due to the hydraulic unbalance

(Chap. 9.3.9) within manageable limits. Values for the maximum tip speeds are given in Table 7.2. A minimum tip speed of about 12 m/s should be considered to avoid deposits in the volute.

Since the blades are not well adapted to the flow, vortex pumps are prone to cavitation noise which increases with the flow rate. Therefore, the tip speed of vortex pumps must also be limited as stipulated in Table 7.2.

Blade design: The blade thickness for impellers with 1, 2 or 3 channels can be estimated from Eq. (T14.3.1) in Chap. 14.1. At the inlet, the blades of impellers with 2 or 3 blades must be thickened to approximately $e_{\max}/d_2 = 0.05$ and well profiled to prevent textiles or plastic foils from getting caught on the leading edges. For single-channel impellers the leading edge thickness is sometimes higher: in [60] $e_{\max}/d_2 = 0.1$ was used. An even thicker leading edge profiled on the pressure side was found to reduce the risk of clogging by long fibers (up to 300 mm), [64]. The blade angles at the impeller inlet are chosen relatively small to improve the partload and noise characteristics. The range of wrap angles of the blades is given in Table 7.2.

Impact of hydraulic design on clogging: Open impellers are proposed by some manufactures as being less prone to clogging than closed impellers. This may be true if the blade is designed that way that stringy matter tends to move towards the outer streamline where it is caught in the clearance between wear plate and blades. There, the textile matter can be chopped into smaller pieces if a cutting groove is provided in the wear plate.

How the blades should be designed in order to reduce the risk of clogging is somewhat speculative. Injecting a sufficient load of rags, every pump will clog. On the other hand, unfavorable (highly loaded) blade designs will attract any kind of stringy matter, while lightly loaded blades are likely to remain clean if the concentration of rags is moderate.

An attempt to calculate the forces acting on a rag appears to be questionable since these forces cannot be quantified with sufficient confidence. Thinking of a piece of a plastic bag sticking on the leading edge of the blade, one would have to integrate the pressure distribution on the blade pressing the rag onto the blade surface. Together with the surface properties of blade and rag, the shear stresses between rag and blade and between the fluid and the rag must be determined. A piece of plastic bag on a smooth surface tends to stick there also because of capillary forces.

Qualitatively, the following tendencies could be postulated:

1. Blades which are highly loaded near the leading edge (i.e. large inlet angles) are apt to attract rags.
2. Large inlet angles with the stagnation point on the pressure surface lead to flow separation on the suction surface (as well as cavitation). Trash or rags are likely to be caught in a zone of stalled fluid (think of a dead-water zone in a river).

3. In general terms, the flow in an impeller is decelerated on the pressure surface of the blade. The resulting reduced velocities imply lower shear stresses acting on any rag wrapped around the leading edge. As a consequence, the flow exerts lower forces on the rag.
4. In contrast, the flow is accelerated on the suction surface. Higher velocities, creating higher drag, are likely to better be able to carry away a rag. However, one can always imagine a situation where the forces acting on the rag are the same on the pressure and the suction surface that way that equilibrium is established. In this sense, a probabilistic approach would be needed. The rag may eventually be swept away by fluctuations in the approach flow.
5. Low inlet angles, i.e. blades lightly loaded near the leading edge with the stagnation point on the suction surface, tend to eliminate the flow separation and cavitation on the suction surface.
6. A strong sweep of the blade introduces radial flow components which might be able to shift rags along the leading edge. Once the rag is set in motion, it is more likely to be swept away from the leading edge. A high ratio d_{1a}/d_{1i} at the radial leading edge is apt to induce stronger secondary flows which transport rags radially outwards.
7. A thick leading edge with a short elliptical profile should be used. The profile length may be $L_p = (1 \text{ to } 1.5) e_{\max}$ if $e_{\max}$ is the maximum blade thickness.
8. An oval-shaped leading edge would have the effect of a sweep back on the outer streamline as well as on the hub. Both outer and inner streamline are unloaded that way. Fluids (hence rags) are displaced towards the mid streamline. The mid streamline is then to be designed with a low inlet angle - hence a stagnation point on the suction surface - in order to sweep the rags away.

In summary, it is suggested tentatively that the blade should be designed according to the features listed below in order to reduce the risk of clogging by fibrous matter or rags - even though hard experimental proof is difficult to obtain.

- Low inlet angles, stagnation point on the suction surface. The blade development becomes S-shaped in order to achieve the required ball passage.
- Introduce radial movement of the flow by sweeping the leading edge
- Secondary flow is apt to set rags radially in motion. A lean of the blade may have similar effects as the sweep.
- Small blade diameters d_{1i} at the inner streamline or a high ratio of d_{1a}/d_{1i} are apt to create radial flow components which can set a rag in motion.
- An oval-shaped leading edge appears worth a try. Some manufacturers seem to have this feature. Near the mid-streamline very low inlet angles may be developed to prevent rags from getting caught there.
- The leading edge profile should be thick and short.

Figure 7.21 shows a sewage pump designed with a 3-bladed impeller. The rear shroud includes expeller blades in order to reduce the axial thrust and to keep the impeller sidewall gap free of coarse solids. Wear plates are installed adjacent to the front and rear shrouds to prevent casing wear.

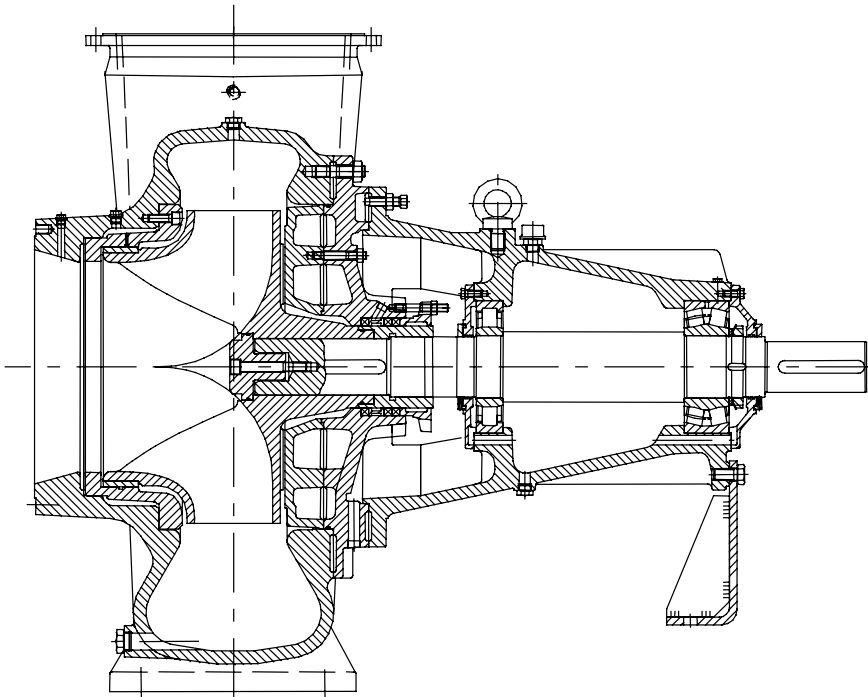


Fig. 7.21 Sewage pump $n_q = 71$; $Q_{\text{opt}} = 1.1 \text{ m}^3/\text{s}$, $H_{\text{opt}} = 18 \text{ m}$ at $n = 590 \text{ rpm}$, Sulzer Pumps

7.5 Semi-axial Impellers

As discussed in Chap. 2, semi-axial pumps are designed with specific speeds of about $n_q = 30$ to 200. This wide range is given rather by design than by hydraulic requirements. Hydraulically, the semi-axial design is at an optimum between about $n_q = 40$ and $n_q = 160$. With regard to the shapes of the impeller, the transition from radial to semi-axial and axial pumps is continuous (Fig. 3.8). Semi-axial impellers with specific speeds below $n_q = 50$ differ little from radial pumps in their hydraulic characteristics; both are therefore designed according to the same principles using slip factors.

Assuming an optimum design, semi-axial pumps are superior to pumps with radial impellers above $n_q \approx 60$ since the discharge flow from radial impellers becomes more non-uniform with increasing specific speed due to the flow deflection in the meridional section and growing blade span. At specific speeds above $n_q \approx 150$, semi-axial pumps have characteristics similar to propeller pumps and the calculation according to the slip factor concept becomes uncertain due to the widely spaced blades.

Large cooling water pumps (CWP) in power stations or chemical process plants are one important application of semi-axial hydraulics. The rotor speed of CWPs

must be selected on the available NPSH and on mechanical considerations. The $NPSH_A$ of CWP is given by the atmospheric pressure and the submergence, resulting typically in $NPSH_A = 12$ to 15 m (unless the pumps are installed at higher elevations). The required $NPSH_3$ at maximum flow must be sufficiently below the $NPSH_A$. Considering this requirement and suction specific speeds in the range of $n_{ss} = 220$ to 260 at BEP, the maximum rotational speed of the pump can be estimated from Eq. (7.11a). This gives a first estimate; the actual speed selected then depends on the standard motor speed, head, specific speed, available models and the specific requirements of the project. The maximum head can be checked against Eq. (7.11b). Equations (7.11a) and (7.11b) represent an upper band of CWP in service, but no absolute limit. These limits can be exceeded by providing a very robust design with suitable blade thicknesses, adequate separation margins to eigen frequencies, suitable line-shaft bearings and mechanical components.

$$n = n_{Ref} \left(\frac{Q_{Ref}}{Q_{opt}} \right)^{0.45} \quad \text{with } Q_{Ref} = 10,000 \text{ m}^3/\text{h} \text{ and } n_{Ref} = 650 \text{ rpm} \quad (7.11a)$$

The maximum head for this type of application can be assessed from:

$$H_{opt} = H_{Ref} \left(\frac{n_{q,Ref}}{n_q} \right)^{1.1} \quad \text{with } H_{Ref} = 27 \text{ m and } n_{q,Ref} = 100 \quad (7.11b)$$

Semi-axial impellers are basically designed according to Chap. 7.2; however, a number of special features must be considered which result from the radius ratio d_{2a}/d_{2i} at the impeller outlet. The geometrical features of a stage are shown in Figs. 7.22 and 7.4.

1. The head coefficient $\psi = \psi_{2a}$ is selected from Fig. 3.24 and the blade diameter d_{2a} on the outer streamline calculated from Eq. (T7.1.3).
2. The shape of the impeller in the meridional section is strongly influenced by the angle ε_s formed by the outer streamline with a normal to the rotor axis. This angle increases with the specific speed approximately according to Eq. (7.12); for axial impellers it becomes 90° by definition.

$$\varepsilon_s = 90^\circ \left(\frac{n_q}{n_{q,Ref}} \right)^{0.74} \quad n_{q,Ref} = 200 \quad (7.12)$$

Executed pumps are in the range of $\pm 5^\circ$ of the angles calculated from Eq. (7.12), [B.2, B.18, 25].

3. The mean impeller outlet diameter can be written as:

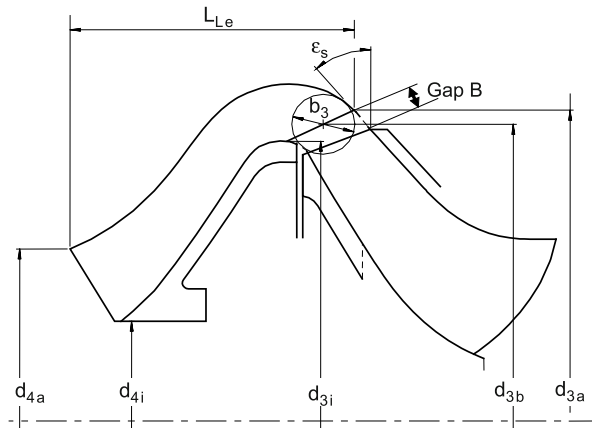


Fig. 7.22 Some design parameters of a semi-axial stage

$$\frac{d_{2m}}{d_{2a}} = e^{-0.04 \left(\frac{n_q}{n_{q,Ref}} - 1 \right)} n_{q,Ref} = 30 \quad (7.13)$$

The diameter of the inner streamline is obtained from:

$$\frac{d_{2i}}{d_{2a}} = \sqrt{2 \left(\frac{d_{2m}}{d_{2a}} \right)^2 - 1} \quad (7.14)$$

Equations (7.13 and 7.14) produce a ratio $d_{2i}/d_{2a} = 0.52$ for $n_q = 200$, which, according to Fig. 7.24, quite accurately corresponds to the hub ratio of an axial pump of this specific speed. The empirical correlation provided by Eq. (7.13) consequently produces a smooth transition from semi-axial to axial impellers.

4. As in Sect. 7.2.1 and Table 7.1 the impeller inlet is calculated according to the required suction specific speed. The blade inlet angles are generally in a range of 12 to 18°. With specific speeds above $n_q = 150$ the explanations in Chap. 7.6 concerning axial pumps must also be observed. According to [B.18] the flow coefficient in the design point can be selected as follows:

$$\varphi_1 = (0.18 \text{ to } 0.27) \left(\frac{n_{q,Ref}}{n_q} \right)^{0.3} n_{q,Ref} = 200 \quad (7.15)$$

The upper limit must be selected for optimum efficiency and the lower limit for high suction specific speeds.

5. Number of blades: For specific speeds below $n_q \approx 140$ the criteria given in Chap. 7.2 apply. In the lower n_q -range, 7 blades are a good choice; nonetheless, 5 to 7 blades can be selected throughout the range $20 < n_q < 140$. Table D7.3

(Chap. 7.6.2) gives an indication for the blade number for $n_q > 140$. However, the solidity L/t is the more important criterion. The solidity required at the outer streamline can be estimated from Eq. (7.16) according to [B.18].

$$\left(\frac{L}{t}\right)_a = 0.64 \left(\frac{n_{q,Ref}}{n_q}\right)^{0.74} \quad n_{q,Ref} = 200 \quad (7.16)$$

As further indication of the required blade length on the outer streamline the following relationship can be derived from data given in [B.18]:

$$\left(\frac{L}{d_2}\right)_a \geq 1.1 \left(\frac{\beta_{2a}^\circ}{25^\circ}\right) \left(\frac{n_q}{n_{q,Ref}}\right)^{0.4} \quad n_{q,Ref} = 200 \quad (7.17)$$

The blade length resulting from Eq. (7.17) is considered the minimum necessary to forestall an instability of the Q-H-curve.

6. The impeller outlet width can be determined according to Fig. 7.2.
7. The impeller outlet angle on the outer streamline $\beta_{2B,a}$ is determined as in Chap. 7.2. The following information for the selection of the blade angles is given in [B.18]:
 - The highest efficiencies are obtained when the blade outlet angle on the mean streamline is selected as $\beta_{2B,m} = 20$ to 26° .
 - On the streamline with the diameters d_{1b} and d_{2b} , inlet and outlet angles should not deviate by more than $\pm 2^\circ$. In the blade development the mean streamline will then almost result in a straight line.
 - In addition, $\beta_{1B,a} < \beta_{2B,a}$ should be executed at the outer streamline and $\beta_{1B,i} > \beta_{2B,i}$ at the inner streamline.

To determine the distribution of the impeller blade outlet angle over the blade height an assumption has to be made with regard to the energy transfer on the various streamlines. In principle, this energy distribution can be selected arbitrarily. However, to obtain reproducible results the designer usually selects a specific distribution of the angular momentum on the impeller outlet according to: $u \times c_u = f(r)$ or $c_u = c_{u,a}(r_a/r)^m$.

As per Table D7.2 three possibilities are compared in the following:

- A. In the free vortex design with $m = 1$ (that is $u \times c_u = u_a \times c_{u,a} = \text{constant}$) the same blade work is transmitted theoretically on each streamline. The circumferential component of the absolute velocity grows from the blade tip to the hub according to: $c_u(r) = c_{u,a} \times r_a/r$.
- B. When selecting the exponent $m = 0$, the absolute velocity over the blade height becomes constant: $c_u(r) = c_{u,a}$.
- C. The fluid rotates with constant angular velocity like a solid body if $m = -1$ is chosen: $c_u(r) = c_{u,a} \times r/r_a$.

Table D7.2 Swirl distribution at impeller outlet			The figures at the sketches are valid for $v = 0.5$			
			Angular momentum constant: $m = 1.0$		Constant swirl velocity: $m = 0$	Solid body rotation $m = -1.0$
Circumferential component of absolute velocity	$\frac{c_u}{c_{u,a}} = \left(\frac{r_a}{r} \right)^m$				$\frac{r}{r_a}$	
Theoretical head	$\frac{H_{th}}{H_{th,a}} = \frac{r c_u}{r_a c_{u,a}}$				$\frac{r^2}{r_a^2}$	
Static pressure	$\frac{p_a - p}{\frac{\rho}{2} c_{u,a}^2}$				$1 - \frac{r^2}{r_a^2}$	
Total pressure	$\frac{p_{tot,a} - p_{tot}}{\frac{\rho}{2} c_{u,a}^2} = \frac{p_a - p}{\frac{\rho}{2} c_{u,a}^2} + 1 - \frac{c_u^2}{c_{u,a}^2}$				$2 \left(1 - \frac{r^2}{r_a^2} \right)$	
Absolute flow angle at impeller outlet	$\frac{\tan \alpha}{\tan \alpha_a} = \frac{c_{u,a}}{c_u}$				$\frac{r_a}{r}$	
Relative flow angle at impeller outlet	$\frac{\tan \beta}{\tan \beta_a} = \frac{1 - \frac{\psi_{th}}{2}}{\frac{r}{r_a} - \frac{c_u}{c_{u,a}} \frac{\psi_{th}}{2}}$					

These three options for allocating the angular momentum over the blade height at the impeller outlet yield different distributions of the specific work, the static pressure, the total pressure and the flow angles in the relative and absolute reference frames. The corresponding equations, which were derived assuming constant meridional velocity and swirl-free approach flow, can be found in Table D7.2.

If the blading is designed according to the law of the free vortex, the radial equilibrium is inherently satisfied. This implies that the axial velocity (apart from boundary layer effects) remains roughly uniform since no forces normal to the main flow direction cause a re-distribution of the fluid. However, the strong flow deflection on the hub requires a correspondingly high blade twisting. The hub diameter must be selected sufficiently large in order to limit the twist and avoid flow separation, see Chap. 7.6. The static pressure drops significantly from the tip to the hub while the total pressure over the blade height remains constant. At the hub, kinetic energy is at its maximum and exceeds the value on the outer streamline by the factor $(r_a/r_i)^2$. The diffuser approach flow angle is smaller on the hub than on the outer streamline. Consequently, the flow deflection to be achieved in the diffuser is greater on the hub than on the tip, which is likely to cause flow separation near the hub. Furthermore, a large amount of kinetic energy has to be converted into static pressure on the hub to avoid losses due to secondary flows and turbulence. Due to these unfavorable flow distributions, the diffuser is an altogether critical element in free vortex designs since the diffuser vanes are subjected to extreme loading at the hub. The recoverable kinetic energy at the diffuser inlet diminishes with rising specific speed because the ratio of the circumferential velocity c_{2u} to the axial velocity c_{2m} decreases.

If the impeller is designed for constant absolute outlet velocity, the blade work diminishes from the outer to the inner streamline in proportion to the radius. Radial equilibrium is no longer ensured so that radial flows are induced and a constant meridional velocity cannot be expected. Static pressure and total pressure decrease slightly towards the hub, but to a significantly lesser degree than with free vortex blading.

When the blading is designed according to the forced vortex, the radial equilibrium is disturbed most severely. The twist of the impeller blades is minimized in this design. Energy transfer and total pressure rise on the hub are lowest. According to [B.18], the blading designed for the forced vortex has advantages when trimming the impeller, since the loss in efficiency is reduced.

As extensively discussed in Chap. 5, stability of the Q-H-curve, power consumption and head at shut-off operation greatly depend on the meridional section and the position of the leading and trailing edges. The explanations in Chap. 5.6 should therefore be observed when selecting these design parameters. It is important to manufacture the impellers of high specific speeds with good surface finish and to profile the trailing edges on the suction surface, [B.18].

7.6 Axial Impellers and Diffusers

7.6.1 Features

Axial (or “propeller”) pumps are used for large flow rates with low heads. Typical applications are irrigation, sewage treatment plants, cooling water pumps and drainage systems. As a rule, these pumps draw the water from an open sump. Vertical pumps are used in wet pit and dry pit installations. Smaller units are also equipped with submersible motors.

Since suction is taken from surface waters, the atmospheric pressure and the necessary submergence (Chap. 11.7.3) determine the available $NPSH_A$ which amounts to 10 to 14 m in the majority of applications. This circumstance has a significant effect on pump selection and impeller design.

The blades of high-quality propeller pumps are designed as slim airfoils. They are individually attached to the hub unless a cast impeller is used which is the case for pumps of small or moderate size. For mechanical reasons and because of the given $NPSH_A$, the impeller tip speeds are limited to about $u_2 = 25$ to 28 m/s (apart from special designs).

Axial pumps are mainly employed in the range above $n_q \approx 150$. Although axial pumps can be designed for lower specific speeds, semi-axial pumps are superior to axial pumps in the range up to approximately $n_q = 170$ unless the plant layout or other design reasons call for a purely axial design. The majority of axial pump applications is in the range $180 < n_q < 300$. These are single-stage machines with axial approach flow to the suction bell.

As a result of flow separation due to excessive deceleration and incidence, the Q-H-curves of propeller pumps, as a rule, have a pronounced saddle-shaped instability which restricts the allowable operating range (see Figs. 4.10 and 5.25, also Chap. 11).

Since the impeller inlet and outlet diameters of an axial pump are identical by definition, the energy is only transmitted through deflection of the absolute flow and deceleration of the relative velocity, see definition of Y_{sch} in Eq. (T3.3.2). During operation near the best efficiency point, there is no centrifugal effect. As discussed in Chap. 5, centrifugal forces become effective only with pronounced partload recirculation. The achievable head coefficients are therefore low, Fig. 3.24. Accordingly, the deflection of the flow is minor. Consider a propeller pump of the specific speed $n_q = 230$ with axial approach flow, a head coefficient of $\psi_{opt} = 0.22$ and a hydraulic efficiency of 88% as an example. The required flow deflection is calculated from Eqs. (T3.3.9), (T3.2.11 and T3.2.15) as follows:

$$\Delta\beta = \beta_2 - \beta_1 = \arctan \frac{\tan \beta_1}{1 - \frac{\psi}{2\eta_h}} - \beta_1 \quad (7.18)$$

Assuming $\beta_1 = 12^\circ$, flow turning is only $\Delta\beta = 1.7^\circ$ on the outer streamline in the present example. As a consequence, axial pumps react very sensitively to minor errors in the blade angles which would hardly be noticed with a radial impeller. It is therefore useful to design the blade fixation on the hub so that the blade position can be modified if the required performance data are not achieved. This design also allows performance adjustments when operating requirements change.

Because of the sensitivity to minor geometrical deviations and since the blades do not form actual channels due to a lack of overlap, calculation methods according to the slip factor concept are unsuitable for axial impellers. For this reason, propellers are designed according to the airfoil theory unless numerical methods are employed. CFD is particularly suitable for this application insofar as strong centrifugal and Coriolis forces are acting normal to the main flow direction. The blade height is large compared with the flow path length through the impeller; hence streamline theory has its limitations. Nevertheless, the streamline theory is useful to make the first design before a numerical optimization can be started.

As mentioned above, cavitation limits the impeller tip speed, hence the rotor speed for a given set of performance data. In order to reliably predict the cavitation characteristics in the design stage, the pressure distributions on the blades should be calculated. This calls for numerical methods if high-quality pumps are to be designed. Not only Navier-Stokes programs but also simpler - and considerably faster - methods (e.g. singularity methods) can be employed for this type of calculation. With the frequently used free vortex blading, simple methods may be used with success since the radial equilibrium is satisfied. If the blades are designed for different vortex distributions, 3D-programs should be considered, Chap. 7.5.

7.6.2 Calculation and Selection of Main Dimensions

In contrast to radial or semi-axial impellers, the conditions at the impeller inlet and outlet of propeller pumps cannot be treated independently. Even the speed is not necessarily needed when specifying the pumping task. In the following we consider therefore the general case where only flow rate Q_{opt} , head H_{opt} and the suction conditions in the system (i.e. NPSH_A) are specified. Since the head is low, losses in inlet, column pipe, outlet elbow and valves can take a considerable fraction of the head. Such additional losses should therefore be considered when establishing the design head to be developed by the impeller since they have an effect on the optimization of the hydraulics, [25].

The first task of the design is to determine the speed n , the impeller outer diameter d_2 , the hub ratio $v = d_n/d_{1a}$ (or hub diameter d_n), the specific speed n_q , the head coefficient ψ , the flow coefficient φ and the cavitation coefficient σ . These seven unknowns must be selected such that the specified head is achieved and the

pump can be safely operated with the specified $NPSH_A$ value. Seven equations according to Table 7.3 are available to determine these quantities.³

Explanations to Table 7.3

- Axial approach flow with $\alpha_1 = 90^\circ$ and blading according to $c_u \times r = \text{constant}$ are assumed (Chap. 7.5).
- The hydraulic efficiency (e.g. from Table 3.9) must be assumed for the initial calculation and verified after designing the blading.
- Corresponding to the specified $NPSH_A$ a cavitation criterion or the required $NPSH_R$ of the pump must be chosen. To this end, a safety margin F_{NPSH} according to Table 6.2 can be determined to calculate $NPSH_R = NPSH_A / F_{NPSH}$. An alternative criterion would be to assume $NPSH_A = NPSH_{(\Delta\eta=0)}$. The pump would then operate with that extent of cavitation which is just below the level where efficiency would begin to drop. This assumption allows to calculate $NPSH_{(\Delta\eta=0)}$ from Eq. (T7.3.6) which originates from [28]. However, no information is available about the scatter of the test data around this correlation.
- Equation (T7.3.2) gives an indication of the maximum head coefficients to be selected in practice. With $f_\psi = 1$ it approximately constitutes the upper limit of the curve given for ψ_{opt} in Fig. 3.24 which reflects the data of various publications (among these [B.22, 25]). If cavitation must be optimized, larger impeller diameters are required; the head coefficients and the constant f_ψ must be reduced accordingly (e.g. $f_\psi = 0.9$). The head coefficients given by Eq. (T7.3.2) can be exceeded in cases where a targeted high head takes precedence over cavitation and partload characteristics.
- Hub ratio: Like the relative outlet width b_2^* of radial impellers, the ratio $v = d_n / d_2$ is a most important design parameter which has an effect on the efficiency and the blade twist at the impeller inlet and outlet. Various criteria can be utilized here: (1) The blade outlet angle on the hub $\beta_{2B,i}$ should be less than 90° . (2) The deceleration ratio of the relative velocity on the hub should not be below an allowable value, i.e. $w_{2i} / w_{1i} \geq 0.6$. (3) In a swirling flow, a core of stalled fluid is formed when the diameter ratio drops below a specific limit. The criterion for core stall is $c_m / c_{2u,i} \geq 1$. For blading according to $c_u \times r = \text{constant}$ this leads to the condition stated by Eq. (T7.3.5) which is used subsequently. (4) The hub diameter can also be calculated according to the values for ψ_i given in Fig. 3.24. For the blading according to $c_u \times r = \text{constant}$ this produces a hub ratio of $v = (\psi_a / \psi_i)^{0.5}$. The attainable head coefficient is limited by the hub ratio. Higher head coefficients can also be achieved with axial impellers when the hub ratio v is increased.

³One could also formulate an equation $d_2 = f(n)$ by inserting Eqs. (T7.3.1 and T7.3.2) into Eq. (T7.3.3). Further equations for $v = f(n)$ and $NPSH = f(n, d_2)$ could be derived to obtain three more complicated equations to determine the three quantities n , d_2 and v .

- The flow coefficient is calculated from Eq. (T7.3.4), if Eq. (T7.3.5) is substituted for v in Eq. (T7.3.1) and the resulting quadratic equation is solved for ϕ .
- Instead of calculating $NPSH_{(\Delta\eta=0)}$ from Eq. (T7.3.6a), it is possible to use the coefficients λ_c and λ_w according to Eq. (6.10) and Eq. (T7.3.6.b). Following this approach, the λ -values have to be selected in accordance with the selected cavitation criterion. The dependency of the λ -values on the flow coefficient and the geometry should also be considered. It may be derived from test data gathered on similar pumps and impeller designs.

The equations of Table 7.3 supply a single solution for given values of Q_{opt} , H_{opt} and $NPSH_A$, which can be found iteratively. However, a more direct and obvious method than the iteration is to treat the speed as an independent variable and to plot the $NPSH_R$, the circumferential speed and the approach flow angle against the rotor speed. If the speed range is selected sufficiently large, a minimum value in the curve $NPSH_R = f(n)$ will be found as shown in Fig. 7.23. Entering the diagram at the given $NPSH_R$ value (e.g. at $NPSH_A/F_{NPSH}$), one obtains exactly the speed which is required to accomplish the specified pumping task. At the same time, the seven equations are satisfied and the selected NPSH-margin is maintained. When the speed has been determined in this manner, all other quantities (d_2 , v , ϕ , ψ) can be calculated from Table 7.3.

If an electric motor is employed as a direct drive, the standard speeds must be used according to the frequency f of the electrical power supply. The synchronous speed is calculated with the number of pole pairs $p = 1, 2, 3$, from Eq. (7.19).

$$n_{syn} = \frac{60f}{p} \quad (7.19)$$

The slip of asynchronous motors is 1 to 1.5% of the synchronous speed. The speed of asynchronous motors is thus obtained by multiplying n_{syn} from Eq. (7.19) with a factor between 0.985 and 0.99.

However, additional criteria must be observed when selecting the speed:

- The approach flow angle β_1 must not be selected too large. According to [25] the best cavitation characteristics are obtained at $\beta_1 = 10$ to 12° . The larger the approach flow angle, the higher will be the flow rate at which the Q-H-curve becomes unstable. For this reason, a small approach flow angle is also propitious for achieving a wide operating range. In general, the approach flow angle should hardly exceed $\beta_1 = 15^\circ$, but $\beta_1 = 10$ to 12° is recommended when good cavitation characteristics are important. At $n_q < 250$, this requires smaller head coefficients than given by Eq. (T7.3.2) and correspondingly larger pumps. A factor $f_\psi < 1$ must then be employed in optimization calculations according to Table 7.3. It is also possible to select the approach flow angle (or the flow coefficient) and determine the head coefficient iteratively from Eq. (T7.3.8), while maintaining the hub ratio as given by Eq. (T7.3.5).

Table 7.3 Main dimensions of axial impellers. Note: n_q with rpm, m^3/s , m		
Assumptions	axial approach flow: $\alpha_1 = 90^\circ$ Constant angular momentum: $c_u \times r = c_{2u,a} \times r_2$	Eq.
Given/selected quantities	$Q_{opt}, H_{opt}, NPSH_A, F_{NPSH}, \eta_h$	
Required $NPSH_R$	$NPSH_R = NPSH_A / F_{NPSH}$ F_{NPSH} acc. Table 6.2	
Desired quantities	$n, d_2, v, n_q, \psi, \varphi, \sigma$	
Specific speed	$n_q = n \frac{\sqrt{Q_{opt}}}{H_{opt}^{0.75}}$ $n_q = 158 \frac{\sqrt{\varphi(1-v^2)}}{\psi^{0.75}}$	7.3.1
Pressure coefficient ($f_\psi = 1$ gives about maximum achievable values)	$\psi \equiv \frac{2gH}{u_{2a}^2} = 0.29 f_\psi \left(\frac{n_{q,Ref}}{n_q} \right)^{1.44}$ $n_{q,Ref} = 180$	7.3.2
Impeller outer diameter	$d_2 = \frac{60}{\pi n} \sqrt{\frac{2gH}{\psi}} = \frac{84.6}{n} \sqrt{\frac{H}{\psi}}$	7.3.3
Flow coefficient	$\varphi \equiv \frac{c_m}{u_{2a}} = \frac{a}{2} + \sqrt{\frac{a^2}{4} + \left(\frac{\psi}{2\eta_h} \right)^2}$ $a = \left(\frac{n_q}{158} \right)^2 \psi^{1.5}$	7.3.4
Hub ratio	$v = \frac{\psi}{2\eta_h\varphi}$	7.3.5
Cavitation coefficient Note: $\lambda_w = f(\varphi)$	$\sigma(\Delta\eta=0) = 0.14 + 1.14\varphi^2 + 2\varphi^3$	7.3.6a
	$\sigma = (\lambda_c + \lambda_w)\varphi^2 + \lambda_w$	7.3.6b
Required $NPSH_R$	$NPSH_R = \sigma \frac{u_2^2}{2g} \quad (\text{note: } u_2 = u_1)$	7.3.7
Calculation of pressure coefficient for selected flow coefficient	$\psi = \varphi^{\frac{4}{3}} \left\{ \varphi \left(\frac{n_q}{158} \right)^2 + \frac{\sqrt{\psi}}{4\eta_h^2} \right\}^{-\frac{2}{3}}$	7.3.8
Plot in a graph: $NPSH_R, u_2$ and $\beta_1 = \arctan \varphi$ as a function of rotor speed. If pump is driven directly by electro motor: use standardized speeds		

Fig. 7.23 Selection of the speed of an axial pump for a given NPSH

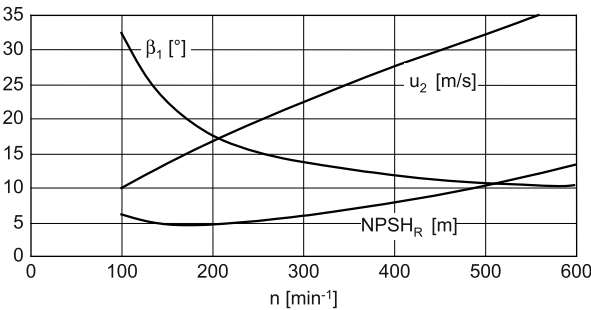


Fig. 7.24 Dimensionless coefficients of axial pumps. (The curves for φ and $\sigma_{(\Delta\eta = 0)}$ virtually coincide)

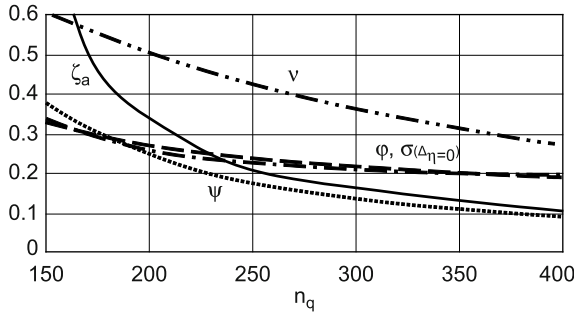


Table D7.3 Impeller blade numbers for $n_q > 140$

n_q	140 to 170			160 to 230	220 to 290	> 290
z_{La}	5	6	7	4	3	2
z_{Lc}	7	8	9	7	5	5
	8	11	10	9	7	7
	9	13	11,12	11	10	9

- The hydraulic efficiency drops with increasing specific speed. If a high efficiency is desired, the speed must be selected lower than would be permissible with respect to cavitation in order to avoid too high a specific speed.
- As mentioned above, the impeller tip speed should also be limited for mechanical reasons.

By means of the equations in Table 7.3, the dimensionless parameters ψ , φ , v and σ can be calculated and plotted as functions of the specific speed. Figure 7.24 shows the characteristics obtained, which are based on the head coefficients given by Eq. (T7.3.2). Figure 7.25 supplies the parameters when a flow coefficient of $\varphi = 0.22$ (or $\beta_1 = 12.4^\circ$) is specified. Optimization calculations in [62] produced similar (at high n_q up to 10% higher) hub ratios as Figs. 7.24 and 7.25.

If a maximum impeller tip speed is assumed, the attainable head and the necessary $NPSH_R$ can also be plotted against n_q as done in Fig. 7.26. These diagrams provide a quick overview of the design options for a specific pumping task (subject to the assumptions made for the calculations). The curves depend especially on the head coefficients selected according to Eq. (T7.3.2).

Impeller blade number z_{La} : The optimum number of blades z_{La} decreases with increasing specific speed as shown by Table D7.3. The optimum blade number is determined by the blade loading so that the final selection has to be verified during the blade design process.

7.6.3 Basic Properties of Airfoils

Airfoil geometry: As mentioned above, the blades of axial pumps (with the possible exception of small inexpensive machines for uncritical service) are

Fig. 7.25 Hub ratio, lift and pressure coefficients of axial pumps calculated for $\varphi = 0.22$

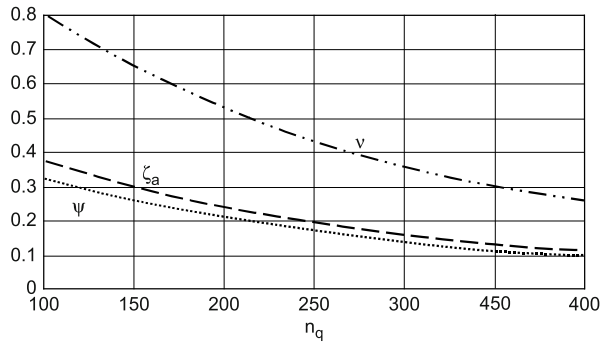
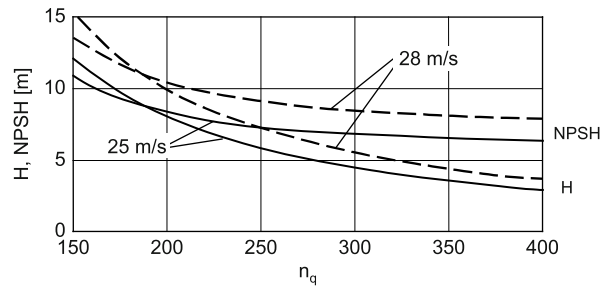


Fig. 7.26 Approximate head and NPSH $_{(\Delta\eta=0)}$ of axial pumps at $u_{2,\max} = 28$ and 25 m/s



designed as airfoils whose geometrical characteristics are explained with reference to Table 7.4. An airfoil can be imagined as a camber line (with maximum camber f) on which a symmetrical profile is superimposed according to a specific thickness distribution. The resulting coordinates of the profile, upper (or suction) surface $y_o(x)$ and the lower (or pressure) surface $y_u(x)$, completely describe the airfoil. The chord on the camber line (NACA airfoils) or the profile tangent (Göttinger airfoils) is used as reference axis or x -axis. All dimensions are based on the profile length L . The following geometric parameters have an effect on the flow characteristic (see figure in Table 7.4):

- maximum thickness ratio d/L
- relative thickness position x_d/L
- camber f/L and its position x_f/L on the profile chord
- nose radius r_k/L
- shape of camber line (e.g. S-shaped)

Airfoil flow: If an airfoil is exposed to the undisturbed approach flow velocity w under the angle of attack δ_A , a stagnation point is formed near the leading edge, Fig. 7.27. At $\delta_A = 0$ the stagnation point is right on the leading edge, but it moves away from the edge with increasing incidence (hence there is a flow around the leading edge at $\delta_A \neq 0$).

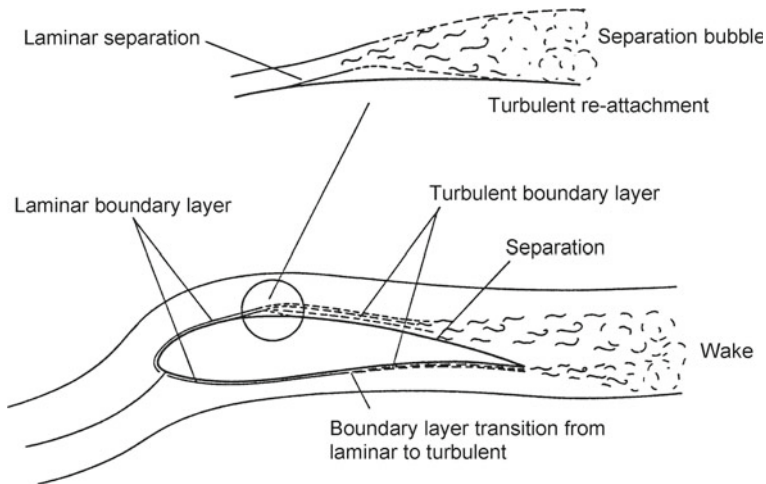


Fig. 7.27 Flow around an airfoil according to [15]

On the suction surface of the airfoil the flow is accelerated and the static pressure drops accordingly. Downstream of the maximum airfoil thickness, where the profile tapers off, the flow is decelerated and the static pressure rises up to the surrounding pressure p_∞ .

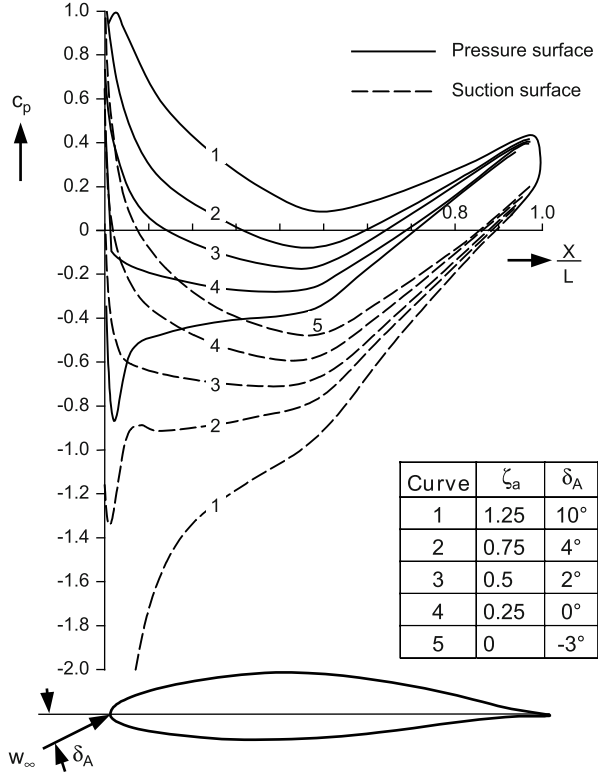
Near the nose of the airfoil the boundary layer is laminar, but it becomes turbulent after a specific flow path length. This transition depends on the Reynolds number, the turbulence of the approach flow and the roughness. Flow separation occurs where the boundary layer becomes too thick (or the deceleration too high). The wake is formed by the boundary layers and stalled fluid.

The local velocities determine the pressure distribution which, when integrated, yields the forces acting on the airfoil. The pressure distribution also determines the onset and extent of cavitation, Chap. 6. According to Bernoulli's equation the local pressure coefficient is obtained from $c_p = 1 - (w/w_\infty)^2$. As per Eq. (T7.4.4), c_p represents the distribution of the local static pressure referred to the stagnation pressure. With increasing incidence the low-pressure peak grows, the flow around the profile nose increases and the location of the pressure minimum moves closer to the leading edge, Fig. 7.28. The magnitude of $c_{p,min}$ grows with increasing thickness and curvature ratio of the profile. The pressure remains constant in the separated flow.

Lift and drag: The fluid movement exerts a force F on the airfoil. This force is obtained by integrating the pressure and shear stress distribution on the airfoil.

The resultant force is made up of a component F_a normal to the approach flow direction, the "lift" and a force F_w in the direction of the approach flow, the "drag" or "resistance". The resistance consists of the "skin friction drag" and the "pressure drag" (or "form drag"). The friction drag practically does not change with the

Fig. 7.28 Pressure distribution on airfoil NACA 65₂-415 according to [50]



incidence while the form drag increases with rising angle of attack and grows severely when the flow separates from the airfoil.

The flow characteristics of an airfoil are described by the lift and drag coefficients, the pressure distribution and the minimum pressure coefficient $c_{p,min}$. With a given profile these values primarily depend on the angle of attack δ_A , which constitutes the angle between the direction of the undisturbed approach velocity w and the profile chord (or the profile tangent). The *local* incidence $i_l = \beta_{lB} - \beta_l$ between the velocity vector at the leading edge and the camber angle must be distinguished from the angle of attack δ_A between w_∞ and the profile chord. The incidence i_l determines the pressure distribution near the leading edge; thus it is relevant for cavitation. The incidence of shockless entry ($i_l = 0$) is described as δ_{SF} ; it corresponds to the “design angle” of the profile. The angle between the lift and the resulting force F is called the “gliding angle” λ . The ratio $\varepsilon = F_w/F_a = \tan \lambda$ is termed “drag-to-lift ratio”. The drag-to-lift ratio of commonly used airfoils is in the range of 0.01 to 0.04 depending on the angle of attack. It can be seen as a measure for the hydraulic quality of the profile. At a specific approach flow angle - the “zero lift direction” - the airfoil is not subjected to any lift.

Fig. 7.29 Polar diagram for NACA65₂-415 and NACA65-209, $Re = 6 \times 10^6$

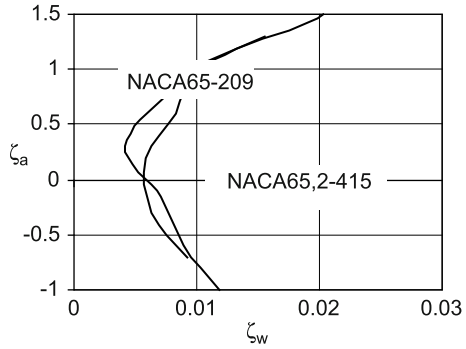
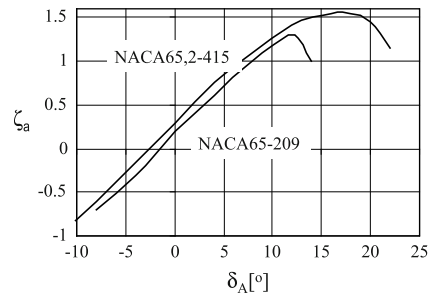


Fig. 7.30 Lift coefficients as function of the angle of attack δ_A , $Re = 6 \times 10^6$



Profile coordinates and flow characteristics of airfoils are published in airfoil catalogues and manuals based on measurements and calculations, e.g. [1, 15, 50]. The lift coefficients are often presented in “profile polars” according to Fig. 7.29 as a function of the drag coefficient. A line drawn from the coordinate origin to any point on the curve represents the dimensionless resultant force acting on the profile with the components of lift and drag. The tangent from the origin to the polars defines the operation point with the optimum gliding angle or minimum loss. The design point of airfoils is preferably selected close to this optimum.

For application in pump design (where the resistance is less significant than with aircrafts) the presentation of the lift coefficients as a function of the angle of attack according to Fig. 7.30 is more convenient. From this plot, it is possible to read the value of δ_A required for a specific lift coefficient. In the domain of small angles of attack, the lift coefficient grows virtually in proportion to δ_A . The maximum lift is obtained just prior to the onset of stall. The gradient is mostly in the range of $d\zeta_a/d\delta_A = 0.09$ to 0.11 . If δ_{NA} stands for the zero-lift angle, the lift coefficient can be calculated from Eq. (7.22):

$$\zeta_a = \frac{d\zeta_a}{d\delta_A}(\delta_A - \delta_{NA}) \quad (7.20)$$

Influence of the airfoil parameters on the aerodynamic coefficients

1. The maximum local velocity increases with the thickness ratio d/L roughly according to:

$$w_{\max}/w_{\infty} = 1 + f_p d/L \quad \text{with } f_p = 1.3 \text{ to } 1.8 \quad (7.21)$$

2. With increasing thickness ratio d/L the magnitude of $c_{p,\min}$ consequently increases according to:

$$c_{p,\min} = 1 - \left(\frac{w_{\max}}{w_{\infty}} \right)^2 = -f_p \frac{d}{L} \left(2 + f_p \frac{d}{L} \right) \quad (7.21a)$$

The pressure minimum remains at the same location. Tables for the velocity ratios $w(x)/w_{\infty}$ were established for many airfoils so that the pressure distribution can be calculated for a selected profile, [1]. The lift coefficient of a profile in incident flow increases slightly with the growing thickness ratio since the streamline curvature on the suction surface is amplified. The thickness ratio has only little effect on the gradient $d\zeta_a/d\delta_A$.

3. When the relative thickness position x_d/L is increased, the location of the pressure minimum and the transition laminar/turbulent are shifted further downstream.
4. If the curvature gets stronger, the lift and the magnitude of $c_{p,\min}$ increase.
5. The minimum drag coefficient is reached when the flow approaches in the zero-lift direction. The drag increases with the thickness ratio.
6. The point of attack of the resulting force (the “neutral point”) is located in the range of $x_N/L = 0.25$ to 0.28 .

The linear relationship between lift coefficient and angle of attack allows an approximate formula as follows:

$$\zeta_a = a \frac{d}{L} + b \delta_A \quad (7.22)$$

The gradients b of many profiles are within the range of $b = 0.09$ to 0.11 . The property of the airfoils described by Eq. (7.22) allows to multiply the y-coordinates of the profile geometry with a constant factor and to calculate the lift coefficient of the thickened (or slimmed down) profile from Eq. (7.22). In this way it is possible to thicken blades from the outer streamline to the hub without the need to select different profiles. After a profile has been selected, the applicable constant “ a ” can be easily calculated with Eq. (7.22) from the curve $\zeta_a = f(\delta_A)$, since the gradient b is known as well.

The zero-lift direction δ_{NA} depends on the camber and the thickness of the profile. For any profile δ_{NA} can be estimated either from Eq. (T7.4.5) or in graphic

form by drawing a line from the trailing edge through the point of maximum camber (see upper figure in Table 7.4: line NA through point W).

If the angle of attack $\delta_{A,o}$ is referred to the zero lift direction, the theoretical lift coefficient of a flat plate in incident flow is $\zeta_a = 2\pi \times \sin \delta_{A,o}$. According to the figure in Table 7.4, $\delta_{A,o} = \delta_A - \delta_{NA}$ applies (take note of the sign of the angles). Experience shows that the lift coefficient of any profile can be estimated according to the above theoretical relationship if a profile efficiency η_P is introduced which results in the approximation formula Eq. (T7.4.6). The drag coefficients, too, can be determined from an empirical relationship, Eq. (T7.4.7).⁴

The lift and drag coefficients depend on the Reynolds number. This is true especially in the range below the critical Reynolds number $Re = w_\infty \times L/\nu$, which is around 10^5 . The lift coefficient increases with the Reynolds number. The transition laminar/turbulent is shifted upstream with increasing Re while the resistance drops. Except for oils and similar viscous media, pumps operate above the critical range. Rough surfaces reduce the lift coefficient, especially with high Reynolds numbers.

7.6.4 Blade Design

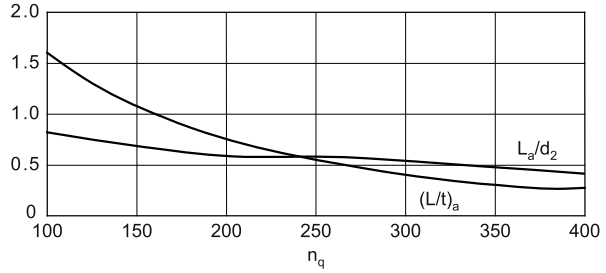
It is possible to design an axial pump for any swirl distribution at the impeller outlet (three possible distributions have been discussed in Chap. 7.5). The following discussion focuses on the design of the blading according to the free vortex concept with $c_u \times r = c_{2u,a} \times r_2$. Here, the same blade work $Y_{sch} = g \times H_{th}$ is transmitted on all streamlines and the axial velocity over the radius is constant. If other swirl distributions are used, the c_m -distribution should be calculated iteratively from the radial equilibrium. Apart from the variation of the axial velocity, the equations in Table 7.5 are written so that they can be used for any swirl distribution.

The blade design is described in Table 7.5 which includes the velocity triangles for axial approach flow. The step-by-step procedure is described below:

1. Given are the performance data. The hydraulic efficiency η_h can be estimated from Table 3.9.
2. Several streamlines (cylinder sections) are defined on which the calculation according to Eqs. (T7.5.4 to 7.5.8) is performed. The meridional velocity is assumed to be identical on all sections.
3. The approach flow angle in the relative system increases from the tip to the hub according to the ratio of the radii, Eq. (T7.5.2).
4. Define shape of the hub (axial or slightly semi-axial). The hub diameter at the outlet can be calculated from Eq. (T7.3.5).

⁴These approximation formulae originate from [B.1].

Fig. 7.31 Selection of the blade length at the outer streamlines of axial pumps



5. The circumferential velocity component at the outer streamline which is required to achieve the specified head is calculated from Eq. (T7.5.3).
6. At the outlet, the circumferential velocity component c_{2u} and flow outlet angle α_2 increase from the tip to the hub, Eqs. (T7.5.4 and 7.5.5). Equation (T7.5.5) supplies the flow outlet angle for which the blading has to be designed.
7. When designing the blades as airfoils, the approach flow velocity w_∞ and angle β_∞ are calculated from Eqs. (T7.5.6 and 7.5.7).
8. According to Eqs. (T7.5.8a or 7.5.8b) it is now possible to determine the flow deflection coefficient $\zeta_a L/t$ for each of the streamlines. In this formula ζ_a stands for the lift coefficient, L for the length of the chord and $t = 2\pi \times r/z_{L,a}$ is the blade pitch on the cylinder section considered (definitions for airfoils see Table 7.4). Equation (T7.5.8) can be derived from the conservation of angular momentum, Chap. 3.2, if the external torque on the rotor is calculated from the forces acting on the airfoil according to Table 7.4. The term $\cos \lambda$ in the numerator of Eq. (T7.5.8a) is frequently not written explicitly but set at $\cos \lambda \approx 1.0$ because the gliding angle λ of relevant airfoils is below 5° . This was done in the expression on the right-hand side of Eq. (T7.5.8a). Sometimes, $\sin(\beta_\infty + \lambda) \approx \sin \beta_\infty$ is assumed as an approximation resulting in Eq. (T7.5.8b). In this case the design implies some head margin since a higher deflection coefficient is specified.
9. The velocity triangles at inlet and outlet can now be calculated from Tables 3.1 and 3.2. As a result we get the velocities and the flow angles which are required to achieve the specified head.
10. Blade length and pitch (solidity) must now be selected so that the deflection coefficient calculated in the respective cylinder section is obtained. Initially, only the product of ζ_a and L/t is known, which constitutes the required deflection coefficient. One of these two quantities must be selected so that the second one can be calculated. One option is to estimate $(L/t)_a$ on the outer streamline from Fig. 7.31 as a function of the specific speed.⁵ Alternatively, the

⁵Figure 7.31 was derived from information provided in [B.22].

Table 7.5 (1) Blade design of axial impellers			
A. Calculation of velocities required to achieve the specified head			
Assumptions and definitions	* $c_m = \text{constant}$ * $x \equiv (r_2/r)^m$	Constant angular momentum: $m = 1$ Solid body rotation: $m = -1$ Constant whirl velocity: $m = 0$	$c_u = c_{2u,a} \times r_2/r$ $c_u = c_{2u,a} \times r/r_2$ $c_u = \text{constant}$
Given or selected quantities	$Q_{\text{opt}}, H_{\text{opt}}, \eta_h, n, d_2, v, n_q, \psi, \phi, r_1 = v \times r_2$		Eq.
Meridional velocity	$c_m = \frac{4Q}{\pi d_2^2 (1 - v^2)}$		7.5.1
Flow angle at impeller inlet as function of radius	$\tan \beta_1 = \frac{r_2}{r} \tan \beta_{1a}$		7.5.2
Circumferential component of absolute velocity at outer streamline	$c_{2u,a} = \frac{g H}{\eta_h u_{2,a}} + c_{1u}$		7.5.3
Circumferential component of absolute velocity	$c_{2u} = c_{2u,a} \times$		7.5.4
Impeller flow outlet angle of relative velocity	$\tan \beta_2 = \frac{c_m}{u - x c_{2u,a}}$		7.5.5
Approach flow angle	$\beta_\infty = \arctan \frac{c_m}{u - \frac{1}{2}(c_{2u} + c_{1u})}$		7.5.6
Approach flow velocity	$w_\infty^2 = c_m^2 + \left[u - \frac{1}{2}(c_{2u} + c_{1u}) \right]^2$		7.5.7
Flow deflection coefficient	$\zeta_a \frac{L}{t} = \frac{2g H_{th} c_m \cos \lambda}{u w_\infty^2 \sin(\beta_\infty + \lambda)} = \frac{2 \Delta c_u \sin \beta_\infty}{w_\infty \sin(\beta_\infty + \lambda)}$		7.5.8a
	$\zeta_a \frac{L}{t} \approx \frac{\psi_{th}}{\sqrt{\phi^2 + (1 - 0.25 \psi_{th})^2}} \approx \frac{2 \Delta c_u}{w_\infty}$		7.5.8b
Hydraulic efficiency	$\eta_{sch} \equiv 1 - \frac{Z_{sch}}{H_{th}} = 1 - \frac{w_\infty \varepsilon}{u \sin(\beta_\infty + \lambda)}$		7.5.9
Sliding coefficient relevant for pump	$\varepsilon_{res} = 0.02 + \frac{0.008}{1 - v}$		7.5.10
	$\varepsilon_{res} = \varepsilon + \frac{0.02 t}{\zeta_a r_a (1 - v)} + 0.018 \zeta_a$		7.5.11
Diffusion factor Outer streamline max. $D_{fz,a} = 0.45$ Hub maximum: $D_{fz,i} = 0.6$	$D_{fz} = 1 - \frac{w_2}{w_1} + \frac{w_{1u} - w_{2u}}{2w_1} \left(\frac{t}{L} \right)$		7.5.12
Length to pitch ratio (solidity) [41][B.22]	$\left(\frac{L}{t} \right)_a = 3.2 \psi_a$		7.5.13

Table 7.5 (2) Blade design of axial impellers according to [32]

B. Determine incidence, deviation angle and blade angles

The complete velocity triangles and w_∞ , β_∞ are known from Table 7.5 (1)		
Select	z_{La} , z_{Le} from Table D7.3	Eq.
Blade tip clearance (tentative)	$\Delta D/d_2 = 0.004$	7.5.14
Select solidity at outer streamline from Fig. 7.31. Eq. (7.5.15) gives higher length in order to reduce blade loading	$\sigma_a = \left(\frac{L}{t}\right)_a = 0.7 \left(\frac{n_{q,Ref}}{n_q}\right)^{0.46}$ $n_{q,Ref} = 200$	7.5.15
Select solidity or blade length at inner streamline $L_i/L_a = 0.7$ to 0.9	$\sigma_i = \left(\frac{L}{t}\right)_i$	7.5.16
Relative blade height, measured from hub	$h = \frac{r - r_i}{r_a - r_i}$	7.5.17
Design blade length in a way that smooth leading and trailing edges are achieved. The factors f_2 and f_3 can be suitably selected; f_1 follows from Eq. (7.5.19)	$\frac{L}{L_a} = \frac{L_i}{L_a} + f_1 h + f_2 h^2 + f_3 h^3$	7.5.18
	$f_1 = 1 - \frac{L_i}{L_a} - f_2 - f_3$	7.5.19
Select airfoil type and thickness ratio d/L at hub and tip. Thickness distribution over relative blade height from Eq. (7.5.20) if linear. Tangent on blade thickness at tip is zero. Alternatively a_1 and b_1 can be selected	$\frac{d}{L} = \left(\frac{d}{L}\right)_i + a_1 h + b_1 h^2$ $b_1 = \left(\frac{d}{L}\right)_i - \left(\frac{d}{L}\right)_a$ $a_1 = -2 \left[\left(\frac{d}{L}\right)_i - \left(\frac{d}{L}\right)_a \right]$	7.5.20
Incidence factor taking into account the thickness ratio d/L , [32], Fig. 186, p. 236 ($K_i = 1.0$ for $d/L = 0.1$)	$K_i = 18.88 \left(\frac{d}{L}\right) - 118 \left(\frac{d}{L}\right)^2 + 292 \left(\frac{d}{L}\right)^3$	7.5.21
Design incidence for zero-cambered profile with $d/L = 0.1$. Simplified correlation, [32], Fig. 187, p. 237	$i_o = 5 \sigma \left(1 - \frac{\beta_1 - 20^\circ}{70^\circ}\right)$	7.5.22
Effect of 3D-flow in impeller on incidence, compared to 2D-cascade data; [32], Fig. 201, p. 247	$(i_c - i_{2D}) = 2.16 - 7.24h + 2.277h^2$	7.5.23
Effect of 3D-flow in impeller on deviation angle compared to 2D-cascade data; [55], Fig. 202, p. 247 According to tests in [55] and [56] typically 2° should be added to the values given by Eq. (7.5.24)	For $h < 0.355$: $(\delta_c - \delta_{2D}) = 1.55 - 5.5h$ For $h > 0.355$: $(\delta_c - \delta_{2D}) = -0.4$	7.5.24
[32], Fig. 195, p. 242	$m = 3.488 \times 10^{-5} \beta_1^2 - 5.777 \times 10^{-3} \beta_1 + 0.4107$	7.5.25
[32], Fig. 196, p. 243	$b = -9.435 \times 10^{-5} \beta_1^2 + 0.01594 \beta_1 + 0.2812$	7.5.26
Correction factor for deviation angle derived from experiments with axial pumps	$K_{exp} = 1.25$ tentatively	

Table 7.5 (3) Blade design of axial impellers according to [32]

Factor for deviation angle taking into account the thickness ratio d/L , [32], Fig. 193, p. 241 ($K_\delta = 1.0$ for $d/L = 0.1$)	$K_\delta = 6.6 \left(\frac{d}{L} \right) + 34 \left(\frac{d}{L} \right)^2$	7.5.27
Deviation angle for zero-cambered profile with $d/L = 0.1$. Simplified correlation [32], Fig. 194, p. 242	$\delta_o = a_\delta \sigma^y$ $a_\delta = 4.4103 - 0.1075 \beta_1 + 0.000706 \beta_1^2$ $y = 0.0000006258 \beta_1^4 + 0.0001034 \beta_1^3$ $-0.006011 \beta_1^2 + 0.137 \beta_1 - 0.1438$	7.5.28
$n = -0.68 + 0.007\beta_1 - \sigma^2(-0.0000191 \beta_1^2 + 0.00152 \beta_1^2 + 0.0179)$ $-\sigma(-0.1305 - 0.0103 \beta_1 + 0.0002124 \beta_1^2 - 0.0000009543 \beta_1^3)$		7.5.29
$\frac{d_\delta}{di} = a_i e^{b_i \beta_1} \left\{ 1 - 1.4075\sigma + 0.6419\sigma^2 - 0.0885\sigma^3 \right\}$ $a_i = 1 + 0.58 \sigma + 0.1273 \sigma^2$ and $b_i = 0.0047 \sigma^2 - 0.0322 \sigma$		7.5.30
Required camber angle (m from Eq. 7.5.25):	$\gamma = \frac{\beta_2 - \beta_1 + K_{\exp} K_\delta \delta_o - K_i i_o + (\delta_c - \delta_{2D}) - (i_c - i_{2D}) \left\{ 1 - \left(\frac{d\delta}{di} \right)_{2D} \right\}}{1 - \frac{m}{\sigma^b} + n}$	7.5.31
Incidence	$i_c = K_i i_o + n \gamma + (i_c - i_{2D})$	7.5.32
Deviation angle $\delta_c = \beta_{2B} - \beta_2$	$\delta_c = K_{\exp} K_\delta \delta_o + \frac{m}{\sigma^b} \gamma + (\delta_c - \delta_{2D}) + (i_c - i_{2D}) \left(\frac{d\delta}{di} \right)_{2D}$	7.5.33
Blade inlet angle	$\beta_{1B} = \beta_1 + i_c$	7.5.34
Blade outlet angle	$\beta_{2B} = \beta_2 + \delta_c$	7.5.35
The stagger angle is determined after selection of the profiles:		
Direction of zero lift in degrees referred to chord	calculate δ_{NA} from Eq. (T7.4.5)	7.5.36
	for comparison calculate δ_{NA} from: $\delta_{NA} = -\text{atan} \left(\frac{f}{1 - x_f} \right)$	7.5.37
Angle of attack from required lift and selected gradient $d\delta_a/d\delta_A$	$\delta_A = \delta_{NA} + \left(\frac{\zeta_a}{d\zeta_a/d\delta_A} \right)$	7.5.38
Stagger angle, use average δ_A from Eqs. (T7.4.5) and (T7.5.38)	$\beta_s = \beta_\infty + \delta_A$	7.5.39
Camber line (mean line), [1], p. 74, Eq. (4.26) for uniform blade loading	$y_c = -\frac{\zeta_a}{4\pi} \{ (1-x) \ln(1-x) + x \ln x \}$	7.5.40
Calculate the camber required for a specific lift ζ_a from a known profile designed for $\zeta_{a,base}$; applicable to any blade loading	$y_c = y_{c,base} \frac{\zeta_a}{\zeta_{a,base}}$	7.5.41
Calculate slope dy_c/dx of actual camber line from tabled slope for base	$\left(\frac{dy_c}{dx} \right)_{\text{actual}} = \left(\frac{dy_{c,base}}{dx} \right)_{\text{base}} \frac{\zeta_a}{\zeta_{a,base}}$	7.5.42

Table 7.5 (4) Blade design of axial impellers

Blending function for final trailing edge thickness, $x_{BL} = 1 - x_{TE}$	$\Delta y = \frac{e_{TE}}{2L} \frac{x - x_{BL}}{1 - x_{BL}}$		7.5.43
Thickness distribution of <i>symmetrical</i> profile without TE correction	$y_{sp} = \frac{(d/L)_{actual}}{(d/L)_{base}} y_{sp,base}$		7.5.44
Thickness distribution of <i>symmetrical</i> profile with TE correction	$y_{sp} = \frac{(d/L)_{actual}}{(d/L)_{base}} y_{sp,base} + \frac{e_{TE}}{2L} \frac{x - x_{BL}}{1 - x_{BL}}$		7.5.45
Blade coordinates, <i>suction</i> surface	$x_{SS} = x - y_{sp} \sin \Theta$ $y_{SS} = y_c + y_{sp} \cos \Theta$	$\Theta = \text{atan}\left(\frac{dy_c}{dx}\right)$ from Eq. (7.5.42)	7.5.46
Blade coordinates, <i>pressure</i> surface	$x_{PS} = x + y_{sp} \sin \Theta$ $y_{PS} = y_c - y_{sp} \cos \Theta$		7.5.47
$\sigma = L/t$ solidity Dimensionless representation $x = X/L$, $y = Y/L$ X and Y are the absolute coordinates in mm or m			
Forces, stresses, natural frequency			
Forces acting on a section of the blade			
average height of blade section	$\Delta r = 0.5(\Delta r_a + \Delta r_i)$		7.5.48
average radius	$r = 0.5(r_a + r_i) - 0.5\Delta r$		7.5.49
lift acting on blade section	$\Delta F_a = 0.5\rho w_\infty^2 \zeta_a L \Delta r$		7.5.50
axial force	$\Delta F_{ax} = \Delta F_a \cos\beta_\infty$		7.5.51
tangential force	$\Delta F_u = \Delta F_a \sin\beta_\infty$		7.5.52
bending moment on root of blade	$\Delta M_B = \Delta F_a (r - r_{i,m})$		7.5.53
Total forces and bending moment are obtained as the sum of all blade sections: $F_a = \Sigma \Delta F_a$ etc.			
Mean bending stress at root of blade W = section modulus at the hub	$\sigma = \frac{M_B}{W}$		7.5.54
Mass per unit length A = area of blade section $f_{am} = 1.5$ accounts for the effect of the added mass of the surrounding water	$\mu = \rho_{mat} A + f_{am} \rho A$		7.5.55
First natural frequency of blade I = area moment of inertia at the hub	$f_n = \frac{3.52}{2\pi} \sqrt{\frac{EI}{\mu L^4}}$		7.5.56

lift coefficient ζ_a can be determined from Figs. 7.24 and 7.25. Executable values of ζ_a are slightly above the head coefficient so that $\zeta_a = (1.2 \text{ to } 1.35) \times \psi_{a,opt}$ can serve as a basis for the outer streamline. The blade length at the outer streamline is believed to be critical for achieving good performance in terms of the allowable operation range. Low blade loading is beneficial with respect to cavitation and the point of stall.

11. An airfoil profile is selected according to the criteria given in Chap. 7.6.5.
12. In order to determine the *blade* angles necessary to accomplish the flow deflection calculated from step 8, the camber angle $\gamma = \beta_{2B} - \beta_{1B}$ must be determined. This can be done according to the procedure given in [32], Table 7.5(2). Incidence, deviation angle and blade angles follow from Eqs.

- (T7.5.21 to T7.5.35). The procedure yields incidences which become *negative* at the outer streamline and positive at the inner streamline. However, it is necessary to design the blades for zero-incidence at the outer streamline in order to realize the specified BEP flow rate. With negative incidence the optimum flow rate would be below the design, as found for example in [35].
13. The deviation angle predicted from [32] may be too low. This can be derived from test data in [55, 56] where the pressure rise predicted was not achieved. To compensate this deficiency, a correction factor K_{exp} has been introduced in Eq. (T7.5.31). This factor is to be derived from test data with axial pumps; tentatively it can be set to $K_{\text{exp}} = 1.25$.
 14. As a result we have now the blade inlet and outlet angles, Eqs. (T7.5.34 and 7.5.35).
 15. The camber determines the lift of the blade section. Consequently a suitable camber line (called “mean line” in [1]) is selected according to the lift coefficient required for the considered blade section.
 16. The required angle of attack δ_A and the stagger angle follow from the lift coefficient ζ_a as given by Eqs. (T7.5.37 to T7.5.39).
 17. A symmetrical profile close to the required thickness (e.g. $d/L = 0.06$) is selected; e.g. NACA 65-006. The tabled coordinates of the symmetrical profile are thus known, e.g. [1].
 18. The actual thickness required for any given blade section is obtained by correcting the blade thickness coordinates by the factor $f_d = (d/L)_{\text{actual}}/(d/L)_{\text{base}}$. This slightly modifies the profile characteristics, but the influence is neglected.
 19. For manufacturing reasons a final trailing edge thickness e_{TE} is required; e.g. $e_{\text{TE}}/d_{2a} = 0.001$. The corresponding blade thickness correction is added to the blade coordinates. The trailing edge thickness is blended into the profile coordinates over a selected length $\Delta x_{\text{TE}}/L$; for example over 20% of the chord: $\Delta x_{\text{TE}}/L = 0.2$. Using a linear blending function gives Eq. (T7.5.43).
 20. The coordinates of the symmetrical profile are calculated from Eq. (T7.5.45).
 21. Finally, the coordinates of the suction and the pressure surface are obtained by superposing the thickness distribution of the symmetrical profile on the camber line, Eqs. (T7.5.46 and T7.5.47).
 22. All geometric parameters must vary smoothly over the blade height. The present procedure ensures that blade length, thickness and camber fulfill this requirement, Eqs. (T7.5.18 to T7.5.20).

Various recommendations for selecting these parameters can be found in the literature. Unfortunately, most data originate from tests on axial compressors (notably [32, 38, 39]) where the blading is significantly different from that of propeller pumps. The extent to which these recommendations can be applied to pumps is therefore uncertain. Nevertheless, the methods given in [32] are often applied for designing axial pumps.

It is possible to derive the relationships for ζ_a and L/t given as Eq. (T7.5.13) from the literature. Summarizing the various data the following is obtained:

- The lift coefficient in the design point ought to be selected in the range of the lowest gliding angle: $\zeta_a = 0.2$ to 0.8 but not greater than: $\zeta_a = (1.2 \text{ to } 1.35) \times \psi_a$
- Maximum lift coefficient (at the inner streamline): $\zeta_{a,\max} = 1.25$
- Solidity $L/t = 0.4$ to 1.2
- Maximum deflection coefficient (at the inner streamline): $(\zeta_a L/t)_{\max} \approx 1.5$
- On the hub, ψ_i must also be verified according to Fig. 3.24.

These recommendations must be observed primarily for the outer section of the blades. Compromises may be required near the hub. The greater the required flow deflection $\Delta\beta = \beta_2 - \beta_1$, the greater L/t must be selected.

The lift and drag coefficients given in the literature apply to individual airfoils of infinite span b (frequently measured for $L/b = 5$). The flow from the pressure to the suction surface around the tip reduces the lift of an airfoil of finite span. In an axial pump this flow is restricted by the narrow gap between the casing and the tip of the blade so that the profile in the pump tends to supply a slightly greater lift. In a cascade there is some interaction between the blades. In principle, some correction for the effects of span and cascade would therefore have to be made when applying data from individual profiles to the impeller of an axial pump. Since generally applicable rules for the cascade correction are not known, it seems to be common practice to ignore these corrections when $t/L \geq 1$, [B.1, 25]. Span correction is generally ignored as well since its effect is minor and to some extent compensates for the effect of the neighboring blades.

The hydraulic losses consist of the actual blade losses which can be determined from the airfoil data plus friction losses on casing, hub, inlet and diffuser. The actual blade losses are obtained from Eq. (T7.5.9) when the drag-to-lift ratio of the profile is inserted. The hydraulic efficiency of the entire pump can also be estimated by means of Eq. (T7.5.9) if the resulting drag-to-lift ratio ε_{res} is employed according to the correlations (T7.5.10 or 7.5.11), [B.1]. This calculation is performed for the mean diameter d_m of the blading:

$$d_{1m} = d_2 \sqrt{0.5(1 - v^2)} \quad (7.22a)$$

The allowable deceleration of the relative velocity on the various cylinder sections can be verified by means of the diffusion factor D_{fz} suggested by Lieblein [37], Eq. (T7.5.12). According to [7, 9] a value of $D_{fz,a} = 0.45$ on the outer streamline and $D_{fz,i} = 0.6$ on the hub should not be exceeded. In [B.3] it is recommended to limit the value on the outer streamline to $D_{fz,a} = 0.35$.

Forces and moments acting on the blade can be calculated from the lift and drag forces according to Table 7.5 for various cylinder sections. Integrating these over the blade height allows to determine the bending stresses at the root of the blade. In addition to the hydraulic forces, centrifugal forces are acting on the blade. The lowest natural frequency of the blade must be calculated to ensure that it is

sufficiently above the excitation frequency $z_{Le} \times n/60$, Chap. 10.7.1. The axial force acting on the impeller can be determined by integrating the axial force components over the blade height (unless the approximation formulae according to Chap. 9.2 are preferred). Comments to calculation of forces and stresses:

- The maximum head at partload should be used for checking the blade strength. To this end the ratio $H_{\max}/H_{\text{opt}}$ is applied to the stresses calculated for BEP conditions. This factor may be set to $H_{\max}/H_{\text{opt}} = 1.6$ to 2.0 .
- The mean (steady) bending stress and alternating stresses are considered. At partload close to stall and flow recirculation the alternating lift forces are expected to be high. The dynamic lift components can be assumed to amount to 35% of the steady forces.
- The section modulus W is calculated for the blade section at the hub. In doing so, the camber is conservatively neglected. The area moment of inertia is calculated for the blade section at the hub. Again the camber is neglected making the calculation conservative with respect to the natural frequency of the blade.
- The tapering of the blade thickness from the hub to the outer streamline is neglected. This makes the calculating conservative by typically 2 to 4%.
- For geometrically similar impellers the natural frequency is proportional to $f \sim 1/d_2$.

7.6.5 Profile Selection

The camber determines the lift of the blade section. According to the theory of thin blade sections, *the lift is directly proportional to the camber*. Consequently, the camber y_c required to achieve a specific lift ζ_a at a specific blade section can be calculated by means of Eq. (T7.5.41) from any known (NACA or other) baseline profile having the lift coefficient $\zeta_{a,\text{base}}$. The camber line of the selected baseline profile is given by $y_{c,\text{base}} = f(x)$ as tabled in [1]. $\zeta_{a,\text{base}}$ is the design lift coefficient of the tabled profile. Criteria for the selection of the profiles:

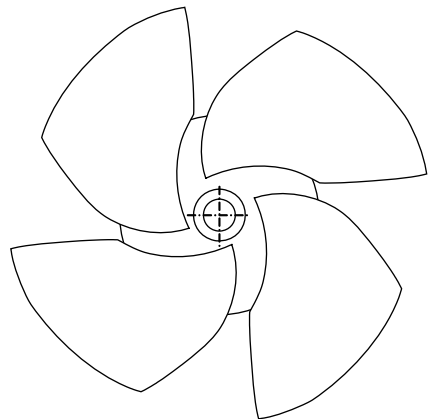
1. *The thickness distribution and the camber line both determine the pressure distribution.* In order to achieve good cavitation characteristics, a profile with a flat pressure distribution and a low pressure coefficient $c_{p,\text{min}}$ should be selected. With such profiles, the maximum thickness is sufficiently far away from the leading edge; $x_d = 0.5$ appears to be close to optimum (e.g. the NACA65 series). Higher values of x_d/L are not recommended since this would increase the risk of flow separations towards the rear of the profile. Lift coefficient (blade loading), low-pressure peak $c_{p,\text{min}}$ and sensitivity to cavitation increase with growing profile thickness and camber.
2. Shaping the camber line as a circular arc is unfavorable because such profiles are sensitive to excessive incidence. A parabolic camber line is superior in terms of

cavitation since the first section of the blades can be configured so that a flatter pressure distribution is obtained.

3. Although sharp profiles promise minimal losses in shockless approach flow (which would be favorable at BEP), they are sensitive to incidence and unsuitable in terms of partload and cavitation characteristics.
4. The low-pressure peak $c_{p,min}$, and consequently the cavitation sensitivity, increase with growing profile thickness d/L . For this reason, the profile thickness is usually not selected larger than $d/L = 0.15$ to 0.18 .
5. A pillar subjected to inclined approach flow produces smaller low-pressure peaks than a structure in perpendicular approach flow. Leading edges that appear radial in the plan view are thus not optimal. Consequently, sweep-back similar to Fig. 7.32 (see for instance [61]) is frequently used in order to reduce blade loading on the hub streamline and secondary flows. Sweep-back can significantly improve the stability of the Q-H-curve, the cavitation characteristics (reducing the $NPSH_R$ by typically 10 to 30%) and even the efficiency, see Fig. 5.43 and [21]. The sweep-back also prevents textiles, plastic foils and all kinds of fibers from being caught on the blade leading edges. Sweep-back is indispensable for sewage applications, but such contaminations also occur in irrigation and drainage systems. The effects of sweep-back have been investigated in [18, 20, 35, 48].
6. For mechanical and hydraulic reasons, the profile thickness has to increase from the tip to the hub. The thickness coordinates of a profile may be multiplied by a constant factor to achieve this.

Based on the above criteria, the profiles of the NACA 6 series can be selected for axial pumps since the camber lines are designed for uniform hydraulic loading from the leading edge up to a point a/L . The designation of the profiles indicates their flow characteristics. The profile “NACA 65₂-415, $a = 0.7$ ” may serve as an example: (a) The first Fig. (6) stands for the profile series and its design principles. (b) The second Fig. (5) indicates the position of the pressure minimum for the

Fig. 7.32 Axial impeller with sweep-back, similar to [61]



symmetrical basic profile with zero lift in terms of $10 \times x_{cp,min}/L$. In the present case this means that the point of the lowest pressure under the above conditions is situated at 50% of the profile length. (c) The third figure indicates the domain of the lift coefficients around the design value in which favorable pressure gradients prevail on the upper and lower surface of the profile. (d) The figure after the dash signifies the design lift coefficient in terms of $10 \times \zeta_a$; in the present example this means that the profile is designed for $\zeta_a = 0.4$. (e) The last two figures signify the maximum profile thickness in percent, i.e. $d/L = 0.15$. (f) The additional quantity “a” describes the hydraulic load of the camber line; in the present case the hydraulic load is uniform from the leading edge up to 70% of the profile length; downstream it decreases to zero in a straight line up to the trailing edge.

The characteristics of the profiles NACA 65-209 and NACA 65₂-415 are shown in Figs. 7.28, 7.29 and 7.30. The pressure distribution (according to [50]) for various lift coefficients is shown in Fig. 7.28, where the approximate angle of attack required to achieve the corresponding lift coefficient has been entered for illustration. The pressure minimum on the suction surface (according to the design) is located at 50% of the profile length. The pressure distribution remains flat up to an angle of attack of 2° but a clear low-pressure peak appears on the suction surface when the angle of attack exceeds 4° . Figure 7.29 shows the profile polars $\zeta_a = f(\zeta_w)$ for both profiles which differ in the thickness ratios and the design lift coefficients. However, both polars are very similar. The minimum of the flow resistance is situated roughly at the design lift. Figure 7.30 shows the lift coefficients as function of the angle of attack according to the data in [1]. Both curves confirm the mentioned linear increase of the lift with the angle of attack up to a maximum lift which, with the thicker, more cambered profile “425”, is higher than with profile “209”. In the linear range both profiles reach the same lift when the angle of attack of the thinner profile is increased by about one degree.

7.6.6 Design of Axial Diffusers

The absolute impeller outlet velocity is $(c_2/u_2)^2 = (c_{2u}/u_2)^2 + (c_m/u_2)^2 = 1/4 \psi_{th}^2 + \varphi^2$. If a diffuser is arranged downstream of the impeller, the flow is decelerated therein to the outlet velocity $c_4 = c_m/A_R$ (A_R being the ratio of the annular cross sections of the diffuser outlet to the diffuser inlet). The diffuser also serves to reduce the circumferential velocity component to a value as close as possible to zero because any swirl at the pump outlet would imply additional losses.

If ζ_{Le} is the pressure loss coefficient of the diffuser, the pressure recovery in the diffuser is obtained from the Bernoulli equation:

$$\Delta\psi_{Le} = \left(\frac{c_2}{u_2}\right)^2 (1 - \zeta_{Le}) - \left(\frac{c_4}{u_2}\right)^2 = \frac{\psi_{th}^2}{4} (1 - \zeta_{Le}) + \varphi^2 \left(1 - \zeta_{Le} - \frac{1}{A_R^2}\right) \quad (7.23)$$

For cost reasons an expansion of the channel is frequently avoided so that $A_R = 1.0$. The lower the head coefficient, the lower will be the possible pressure recovery because of unavoidable hydraulic losses. If, for instance, $\beta_1 = 12.4^\circ$ (i.e. $\varphi = 0.22$), the pipe friction coefficient $\lambda = 0.04$ and the channel length $L/D_h = 3$ ($\zeta_{Le} = \lambda \times L/D_h = 0.12$), no pressure recovery can be achieved in the diffuser with $A_R = 1$ if the head coefficient is $\psi_{th} = 0.15$, Eq. (7.23). At specific speeds above approximately 270 it must therefore be carefully examined if a diffuser is able to produce any noticeable pressure recovery at all - or if it would even cause a loss in efficiency. If a vaned diffuser brings no advantages, a conical diffuser can be employed which opens up in the direction of the hub and/or to the outside (design according to Chap. 1.6).

If a diffuser is required to achieve a good efficiency, the vanes can be designed as airfoils. The equations in Table 7.5 can be converted accordingly. Whether the cost for the manufacture of profiled vanes is justified or not, has to be examined in each individual case. Velocity and flow angle distribution at the diffuser inlet are also afflicted with some uncertainties. This is true especially at the hub where complicated flow conditions prevail already in the impeller. Flow separation is frequently observed near the diffuser hub and careful analyses and optimization steps are required to avoid or limit stall in this region. On the outer streamline the flow distribution also deviates from the inviscid calculation due to boundary layer effects and the tip clearance flow (see Chap. 5.7 for details). If the diffuser is designed with unprofiled vanes, the calculation is performed according to the channel model:

1. The absolute velocities and flow angles at the impeller outlet are known: $c_{3m} = c_m$, $c_{3u} = c_{2u}$, $\alpha_3 = \alpha_2$. These correspond to the values at the diffuser inlet, since radii as well as cross sections are constant. All of these quantities are functions of the radius.
2. The number of vanes is often selected between 5 and 8 while the criteria according to Chaps. 7.2.1 and 10.7.1 have to be taken into account. Axial length and vane number of the diffuser must be matched so that the ratio of vane length L to pitch ("solidity") is sufficiently large in order to achieve the intended flow deflection. This means approximately $L/t = 1$ to 1.5.
3. To limit dynamic vane loading and pressure pulsations a distance of $a/L = 0.05$ to 0.15 is required between impeller blades and diffuser vanes (L is the chord length of the impeller blades). With regard to the efficiency and head coefficient there is presumably an optimum distance even though generally applicable design rules are lacking. Excessive distances produce unnecessary friction losses and impair efficiency and head coefficient. At distances $a/L < 0.05$ the efficiency drops again according to the tests in [2], presumably because the non-uniformities of the impeller outlet flow are not sufficiently smoothed out at

the diffuser inlet. Greater losses generated by fluctuations of the approach flow velocity at the diffuser inlet are the consequence.

4. The vane angles at the diffuser inlet can be calculated according to Eqs. (3.18 and 3.19). The vane blockage is generally low but can be considered in c_{3m}' .
5. To limit the pressure losses in the column pipe and the outlet bend, the flow velocity in the column pipe should not exceed 5 m/s. The annular cross section at the diffuser outlet must be dimensioned accordingly.
6. Since the design aims at a swirl-free discharge flow with $\alpha_4 = 90^\circ$ and the flow in the diffuser cannot be vane-congruent, the vane angle should be selected at $\alpha_{4B} = 94$ to 96° .
7. The vanes can be designed as arcs of circles. The leading edges can be rounded or shaped as elliptical profiles. The trailing edge is designed blunt (rounding on the outlet would be unfavorable because of the formation of vortex streets, Table 10.13).

7.7 Inducers

To reduce the required $NPSH_R$ of a pump it is possible to install an inducer upstream of the actual impeller. As an example, Fig. 2.1 shows the section of a single-stage process pump with inducer. An inducer typically allows a reduction in the value of the required $NPSH_R$ of a pump to half of the value without inducer.

The inducer increases the static pressure upstream of the impeller. Cavitation bubble development on the impeller blades is reduced or suppressed accordingly. As a consequence, the pump can be operated at a higher speed or a lower $NPSH_A$ than without an inducer, provided that the inducer does require significantly less $NPSH_R$ than the impeller.

Inducers for industrial application typically reach suction specific speeds of $n_{ss} = 400$ to 700 , Chap. 6.2.4. These high suction specific speeds are achieved by small approach flow angles β_{1a} (refer to Fig. 6.21), thin leading edge profiles with low blade blockage and long channels in which the cavitation bubbles generated in the inducer implode at least partially before the fluid enters the actual impeller. The low approach flow angle and the small flow coefficient ϕ_1 at the inducer inlet require a considerably enlarged inlet area compared with the impeller. This is achieved by reducing the hub ratio and by enlarging the inlet diameter.

The recommendations and data for the inducer design discussed in the following largely originate from work done by the NASA published in [19, 29, 30, 45], and in [B.16, B.24]. The information is tailored for industrial applications with suction specific speeds of up to $n_{ss} \approx 600$. Inducers with far higher suction specific speeds are employed in the aerospace technology where even two-stage inducers are used at times. The design methods discussed in the following do not fundamentally differ from those of the high-performance inducers.

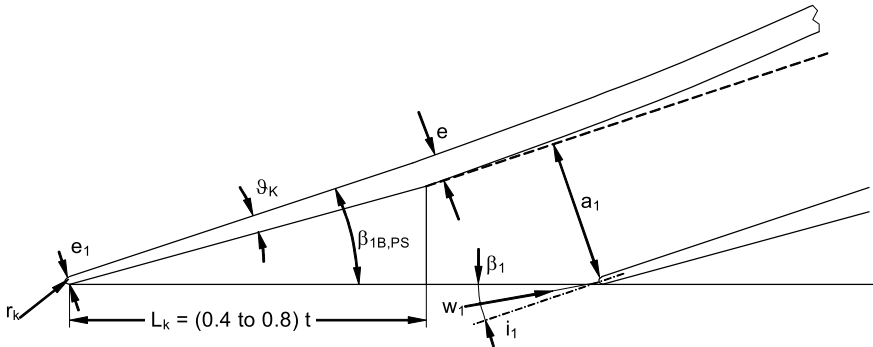


Fig. 7.33 Selection of the blade geometry at the inducer inlet

As is evident from the example in Fig. 2.1, the flow in the inducer is essentially axial. Often the inlet diameter is slightly larger than that of the downstream impeller. The hub diameter at the inducer inlet is designed as small as possible so that the hub contour is generally conical with the hub radius growing towards the impeller inlet. As with axial impellers, the static pressure in the inducer is increased in that the relative velocity is decelerated, Eq. (T3.3.8). This must be considered when deciding on one of two possible blading concepts:

- Blades with constant pitch designed as screw surfaces with blade angles $\beta_{1B} = \beta(L) = \beta_{2B} = \text{constant}$ on cylindrical sections. These blades only generate pressure when the incidence $i_1 = \beta_{1B} - \beta_1$ is greater than zero or for as long as the flow in the blade channel is decelerated according to $w_{1q} < w_1$. The advantage of this blading is its simple manufacture.
- Blades with variable pitch (Fig. 7.34, right) where the angle on cylinder sections grows from the inlet to the outlet as with a propeller pump: $\beta_{2B} > \beta(L) > \beta_{1B}$.

7.7.1 Calculation of Inducer Parameters

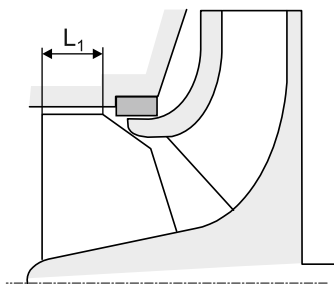
Determination of main dimensions and inlet conditions, Table 7.6

1. **Basis:** Known parameters are the pump selected for the relevant application, its speed n , flow rate Q_{opt} , NPSH_R curve and the dimensions of the impeller. In addition, the suction specific speed to be achieved by the inducer or its NPSH_R is specified. Furthermore, the boundary conditions for the mechanical design must be ascertained. Usually the approach flow angle is $\alpha_1 = 90^\circ$ and the axial velocity is assumed as constant over the inlet cross section.
2. **Design flow rate:** Like axial impellers, inducers have a steep Q-H-curve. The pressure rise therefore drops rapidly to zero above the shockless entry (depending on the type). For this reason, the maximum required flow rate of the

pump must be taken into account when determining the design flow rate of the inducer by selecting $Q_{\text{ind}} = (1.1 \text{ to } 1.15) \times Q_{\text{opt}}$ or by making use of Eq. (7.6.7).

3. **Blade number $z_{\text{VL}a}$:** Inducers usually have 2 to 4 blades.
4. **Inducer inlet diameter d_1 :** The inlet is designed for the selected suction specific speed. The optimum flow coefficient for the required suction specific speed can be selected from Eq. (T7.6.3). The appropriate inlet diameter can then be determined. Alternatively, the optimum inlet diameter can be calculated for selected coefficients λ_w and λ_c from Eq. (T7.6.1). The expected suction specific speed is obtained from Eq. (T7.6.4). Finally, it is also possible to calculate $\phi_{1,\text{opt}}$ from Eq. (T7.6.2). All of these options yield the same result. For inducers with axial inlet $\lambda_c = 1$ to 1.1 and $\lambda_w = 0.02$ to 0.08 can be assumed. The inlet diameter must also be matched to the inlet nozzle and the downstream impeller, which can imply considerable restrictions on the design options in case of standard pumps.

Iteration may be required with respect to the hub ratio. Two uncertainties must be considered in this calculation: (a) As discussed in Chap. 6.3.2, the coefficients λ_c and λ_w depend on the geometry; the values assumed in the calculation will be attained only if the blade design is as good as in the measurements from which the coefficients had been derived. (b) The NPSH_R of the pump is not only governed by the inducer but also by the downstream impeller. If the impeller has not been specifically designed for operation with the inducer, the design suction specific speed of the inducer may not be reached under certain conditions, Chap. 7.7.3. This risk is particularly high in the range of large flow rates ($q^* > 1$); it grows with increasing suction specific speed of the inducer.



Arranging an inducer in the meridional section



Fig. 7.34 Impeller with variable-pitch inducer

Table 7.6 Inducer design, $\alpha_1 = 90^\circ$ (Fig. 7.33)			Eq.
Optimum inlet diameter for given λ_w	$\lambda_c = 1.1$ $\lambda_w = 0.02$ to 0.08	$d_{1,opt} = 3.25 \left(\frac{Q_{La}}{n k_n} \right)^{\frac{1}{3}} \left(\frac{\lambda_c + \lambda_w}{\lambda_w} \right)^{\frac{1}{6}}$	7.6.1
Optimum flow coefficient for given λ_w	$\phi_{1,opt} = 0.1$ to 0.18	$\phi_{1,opt} = \sqrt{\frac{\lambda_w}{2(\lambda_c + \lambda_w)}}$	7.6.2
Optimum flow coefficient for given n_{ss} $n_{ss,Ref} = 400$	$\phi_{1,opt} = 0.15 \left(\frac{n_{ss,Ref}}{n_{ss}} \right)^{0.93}$	$d_{1,opt} = 2.9 \left(\frac{Q_{opt}}{k_n n \tan \beta_1} \right)^{\frac{1}{3}}$	7.6.3
Suction specific speed $n_{ss} = 350$ to 700	$\phi_1 = \begin{cases} n_{ss,Ref} \\ n_{ss} \end{cases}^{0.93}$ $n_{ss,Ref} = 52$	$n_{ss} = \frac{98}{(\lambda_c + \lambda_w)^{0.25}} \left(\frac{k_n}{\lambda_w} \right)^{0.5}$	7.6.4
Flow inlet angle at outer streamline	$\beta_{1,a,opt} = \arctan \phi_{1,opt}$	$\beta_{1,a,opt} = \arctan \frac{240 Q_{opt}}{\pi^2 d_{1,opt}^3 n k_n}$	7.6.5
Pressure surface blade inlet angle at outer streamline $i_1 = 2$ to 4°	Criterion: design incidence: $i_{1,opt} = (0.35 \text{ to } 0.45) \times \beta_{1B,PS}$	$\beta_{1B,a,DS} = (1.5 \text{ to } 1.8) \times \beta_{1,a,opt}$	7.6.6
	Criterion: incidence at maximum flow rate: $i_{1,Qmax} = 1^\circ$	$\beta_{1B,a,DS} - 1^\circ \geq \beta_{1,max} = \arctan \frac{240 Q_{max}}{\pi^2 d_1^3 n k_n}$	7.6.7
Blade inlet angles over blade height; same incidence for all streamlines		$\tan \beta_{1B,DS}(r) = \frac{r_a}{r} \tan \beta_{1B,a,DS}$	7.6.8
Blade leading edge profile (s. Fig. 7.33)		$\vartheta_k = (0.35 \text{ to } 0.5) \times \beta_{1B,a,DS}$	7.6.9
Check also: $i_1 \geq \vartheta_k + 0.5^\circ$		$L_k = (0.4 \text{ to } 0.8) \times t$	
		$r_k = (0.03 \text{ to } 0.05) \times e$	
Cavitation coefficient only for $i_1 > 0$ $k_3 = 0.22$ to 0.65	$\sigma_3 = k_3 k_n \left\{ \frac{\phi_1}{\sqrt{k_n}} - \frac{\phi_1^2}{\tan \beta_{1B}} \left(1 - 0.36 \frac{k_n}{\tau_1} \right) + \frac{3.18 \times 10^{-3}}{\tau_1 \left(\tan \beta_{1B} - \phi_1 \tau_1 \sqrt{k_n} \right)} \right\}$		7.6.10
Influence of tip clearance on n_{ss}	$\frac{n_{ss}}{n_{ss,s=0}} = 1 - k_4 \sqrt{\frac{s}{b_1}}$	$k_4 = 0.5$ to 0.65	7.6.11
Blade length		$(L/t)_a = 1$ to 2.5	7.6.12
Hydraulic efficiency		$\eta_h = 1 - 0.11 \times L/t$	7.6.13
Deviation angle δ_2 Flow angle $\beta_2 = \beta_{2B} - \delta_2$		$\delta_2 = \left(2 + \frac{\beta_{2B} - \beta_{1B}}{3} \right) \left(\frac{t}{L} \right)^{1/3}$	7.6.14
Static pressure rise in inducer. The calculation is done for the mean streamline		$\psi_p = \eta_h \left[1 - \phi_1^2 \left\{ \frac{A_1^2}{A_2^2 \sin^2(\beta_{2B} - \delta_2)} - 1 \right\} \right]$	7.6.15
Note: Use <i>pressure surface</i> blade angle $\beta_{1B,PS}$ rather than camber angle.			
Number of inducer blades: $z_{Ind} = 2$ to 4 . Sweep-back $\epsilon_{sb} = 65$ to 90° according to Fig. 7.37			
Hub diameter (given by mechanical design)	Inlet: Select hub diameter close to minimum: $v_{1,min} \approx 0.15$ Outlet: Match to impeller hub; a large hub diameter helps to increase the head developed by the inducer (centrifugal forces)		

5. **Inlet diameter at the inner streamline d_{1i} :** The lower limit for d_{1i} is obtained from a minimum hub ratio of $v_{1,\min} \approx 0.15$. Smaller values of d_{1i} are unfavorable since excessive blade inlet angles would result at the inner streamline.
6. **Blade inlet angle β_{1B} :** When the inducer inlet diameter has been determined according to step (4), all quantities of the inlet velocity triangle according to Table 3.1 can be calculated. The first section of the inducer blades is designed according to Fig. 7.33. The calculation is based on the angle on the pressure surface of the blade $\beta_{1B} = \beta_{1B,PS}$ which is obtained from the flow angle plus an incidence of $i_1 = 2$ to 4° from Eqs. (T7.6.5 to 7.6.7). In this way, the flow and blade angles are calculated for outer, mean and inner streamline. When determining the inlet angles, it must be observed that the steep rise of the $NPSH_R$ -curve (depending on the blade configuration) occurs near the flow rate of shockless entry. In addition to the design point it is therefore necessary to consider the maximum required flow rate at which the incidence should not drop below one degree to avoid cavitation on the pressure surface of the blades, Eq. (T7.6.7).
7. **Inlet throat area A_{1q} :** As with impellers (see Chap. 7.2.1, item 15), the deceleration of the relative velocity vector w_1 to the mean flow velocity w_{1q} in the throat area must be limited in order to avoid premature inlet recirculation. Conversely, the fluid must not be accelerated at the maximum flow rate with respect to overload cavitation. The throat area at the inducer inlet must therefore be selected so that the ratio w_{1q}/w_1 in the design point is in the range of 0.6 to 0.75 but does not exceed the value $w_{1q}/w_1 = 1$ at the intended maximum flow rate.
8. **Inducer outlet diameter $d_{2,a}$ and $d_{2,i}$:** The diameters on the outer streamline and on the hub are obtained from the dimensions of the downstream impeller. A conical hub of the inducer with $d_{2i} > d_{1i}$ is favorable since the head is increased through centrifugal forces and since the flow on the hub requires less severe deflection with increasing d_{2i} (lower tendency to stall). If the inducer diameter at the outlet is smaller than at the inlet, $d_{2,a} < d_1$, the first cylindrical part (i.e. dimension L_1 in Fig. 7.34) must be designed sufficiently long as to cover at least the throat area of the inducer.

Relationships at the outlet of the inducer: The blade outlet angle must be chosen so that the static pressure in front of the impeller increases sufficiently in the specified operating range to cover the NPSH requirement of the impeller. The effect of the static pressure rise in the inducer on the NPSH requirement of the system inducer/impeller is explained in Fig. 7.35 which was calculated on the basis of measurements on a process pump. The measured $NPSH_{3,imp}$ of the impeller alone was plotted against the flow rate, curve 1.

The static pressure rise H_p in the inducer is plotted as curve 3. Curve 3 passes through zero at Q_B . The inducer operates as a pump at $Q < Q_B$, but as a brake at $Q > Q_B$, (Chap. 12). If the static pressure rise in the inducer (curve 3) is subtracted from the $NPSH_{3,imp}$ of the impeller (curve 1: without inducer), curve 5 is obtained which passes through zero at Q_u . In the domain with $Q < Q_u$, (that is at

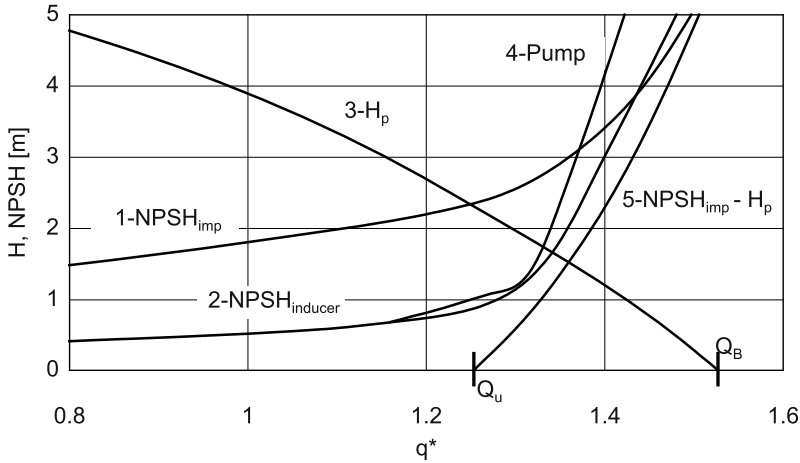


Fig. 7.35 Inducer characteristics

$H_p > \text{NPSH}_{3,\text{imp}}$) curve 5 is situated below the flow rate axis. In this domain the NPSH_R of the system inducer/impeller is largely determined by the $\text{NPSH}_{3,\text{inducer}}$ of the inducer. In the region $Q_u < Q < Q_B$ the control of the NPSH_R of the pump passes from the inducer to the impeller which now obtains insufficient pressure. The operating limit is located to the left of curve 5 and left of the point in which curves 1 and 2 intersect. However, it is not easy to accurately predict the steep rise of curves 2 and 5. In brake operation $Q > Q_B$, curve 5 is located above curve 1. Here the pump cannot be operated under any circumstances because the impeller receives less inlet pressure than without an inducer. The calculated NPSH requirement $\text{NPSH}_{3,\text{inducer}}$ of the inducer is plotted as curve 2 while curve 4 constitutes the measured NPSH_3 of the system inducer plus impeller.

These relationships determine the procedure for calculating the inducer outlet as follows:

9. The measured or calculated $\text{NPSH}_{3,\text{imp}}$ of the impeller (without inducer) is plotted as a function of the flow rate, curve 1.
10. The curve of the required $\text{NPSH}_{3,\text{inducer}} = f(Q)$ of the inducer is estimated according to Eq. (T7.6.10) [30]. Compared with other correlations this relationship has the advantage that it produces a steep rise of the curve with increasing flow rate when the incidence tends to zero (the nominator of the term on the right side of this equation then tends to zero). Since the constant k_3 varies over a wide range, it is recommended to estimate the suction specific speed at the design point from Eq. (7.23a) and use Eq. (7.6.10) only for the shape of the curve $\text{NPSH}_3 = f(Q)$.
11. A blade outlet angle β_{2B} on the outer streamline is selected and the static pressure rise in the inducer calculated. This is done using Eq. (T7.6.15) which can be derived from Table 3.3. The head coefficient is calculated on the mean

streamline. The flow deflection is taken into account by way of the deviation angle using Eq. (T7.6.14). The hydraulic efficiency of the inducer can be estimated from Eq. (T7.6.13). Depending on the flow rate ratio q^* and the geometry, it is in the range of $\eta_h = 0.7$ to 0.9 . For this calculation it would also be possible to use correlations for the head coefficient given in the literature. When applying correlations, it must be considered that individual geometrical parameters such as the hub diameters on the inlet and outlet of the inducer cannot be assessed in their effect on the pressure increase. Furthermore, the geometries of the inducers examined in the literature are not adequately known. In addition, the *static* pressure rise (not the head rise or the total pressure rise) must be used as a criterion since it is solely responsible for the NPSH improvement.

12. The rise in the static pressure caused by the inducer must be calculated for the flow rate range between shockless entry or $H_p = 0$ and the flow rate Q_u where H_p is greater than $NPSH_{3,imp}$ of the impeller alone. At partload the calculation becomes uncertain due to growing inlet recirculation. H_p increases due to the centrifugal effect (streamline displacement) so that roughly linear characteristics $H_p = f(Q)$ are obtained which tend towards the point $\psi_{p(Q=0)} = 1.0$.
13. The difference between $NPSH_{3,La}$ (curve 1) and the static pressure rise H_p (curve 3) gives an indication of the maximum flow rate which is definitely to the left of the resulting curve 5.
14. To facilitate assessing this limit, the curve $NPSH_{3,inducer} = f(Q)$ of the inducer (curve 2) is also taken into account.
15. If the maximum flow rate does not meet the requirements, the calculation according to steps (10) to (13) must be repeated with another outlet angle. With respect to partload operation, over-sizing should be avoided.
16. When the outlet angle has been established, all parameters of the outlet velocity triangle according to Table 3.2 can be determined. To verify the configuration, w_2/w_{1a} should be calculated too. To avoid premature flow separation and efficiency loss, the deceleration ratio w_2/w_{1a} should not fall below the value of 0.7 ("de Haller criterion"). The lift coefficient according to Eq. (T7.5.8) should not exceed the value $\zeta_{a,a} = 0.5$ on the outer streamline.
17. The blade angles on the mean and inner streamlines are selected according to $u \times c_u = \text{constant}$ while a compromise is made on the hub (if required) to avoid excessive blade angles, Chaps. 7.5 and 7.6. The larger the hub diameter, the easier it is to configure this free-vortex blading.

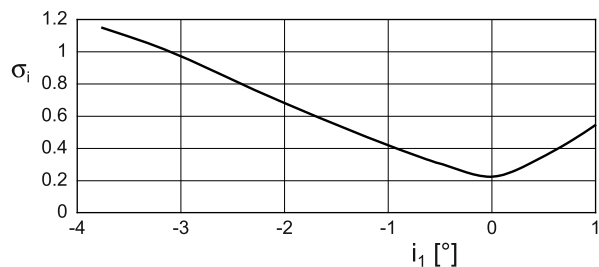
The calculation of the pressure rise in the inducer is quite uncertain since flow deflection and hydraulic losses are affected by intense secondary flows which occur as the result of thick boundary layers and centrifugal forces, Fig. 5.7.

7.7.2 Design and Shaping of an Inducer

After determining the inlet and outlet diameters as well as the blade angles, the inducer can be designed. The design follows Chap. 7.2.2. The following information can help to select appropriate blade shapes and hub contours:

- Blade profile at inlet: The first section of the blade is usually designed wedge-shaped with a leading edge as thin as possible. The minimum required blade thickness is given by blade stresses and natural frequencies. The length L_k of the wedge-shaped portion is between 50 and 80% of the pitch, Fig. 7.33. The wedge angle ϑ_k , which makes up the profiling on the suction surface, is selected as $\vartheta_k = (0.3 \text{ to } 0.5) \times \beta_{1B,a,PS}$. The thin inlet profile is important to obtain good suction capability. It also means low cavitation inception coefficients ($\sigma_i \approx 0.2$ to 0.3) in shockless approach flow. However, σ_i increases rapidly even with a minor incidence, as shown in Fig. 7.36.
- The blade angle β_{1B} is best kept constant over the first part of the blade length, approximately up to $x/L = 0.25$, in order to produce less deflection of the flow and consequently reduce the low-pressure peak and the generation of cavitation bubbles.
- If the pressure distribution is calculated, it should be checked that the blade design ensures that the static pressure in the throat area on the blade suction surface exceeds the vapor pressure, that is $p_{ss(x=t)} > p_v$.
- A conical hub fosters the pressure build-up in the inducer due to centrifugal forces. At the same time the tendency towards flow separation decreases with the amount of hub taper that can be designed.
- The blade lengths are selected in a relatively large range $L/t = 1.5$ to 2.5 . If the overhung of the inducer permits, values in the range of 1.5 to 1.8 should be the target. If the blades are made too short, the pressure rise in the inducer is impaired since the deviation angle increases. Moreover, the suction capability suffers and there is a growing tendency for pressure pulsations at partload. Blades with $L/t > 2.5$ do not seem to bring about any advantages.
- The blade outlet angles on the outer streamline generally are below 20° , but this depends on the inducer and impeller inlet diameters. The blades are slightly tapered towards the outlet.

Fig. 7.36 Visual cavitation inception of an inducer as a function of the incidence



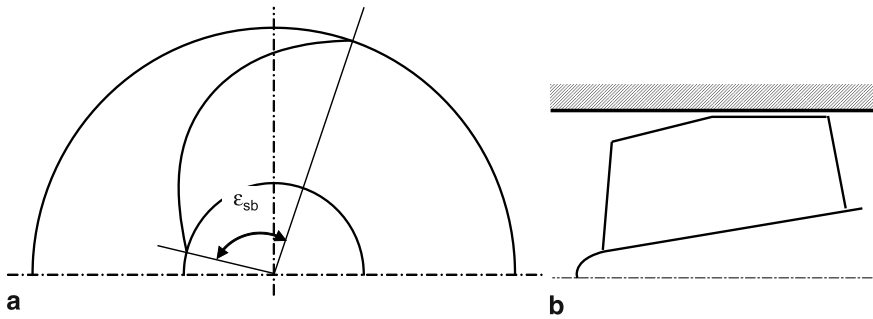


Fig. 7.37 Sweep-back of the inducer blade leading edge

- Efficient inducers are nearly always designed with a “sweep-back” according to Fig. 7.37a. A sweep-back does not only increase the suction specific speed but also moderates the tendency to pressure pulsations. During the tests in [3] the optimum wrap angle ϵ_{sb} (defined according to Fig. 7.37) lay between 65 to 90° on the outer streamline. At half the blade height ϵ_{sb} should be selected roughly half as large as on the outer diameter. Compared with a basic test with $\epsilon_{sb} = 29^\circ$, a suction specific speed increase from $n_{ss} = 280$ to $n_{ss} = 510$ was obtained with optimum sweep-back. The cavity extensions observed were correspondingly smaller and the pressure pulsations were also lower. It should be noted that the blade leading edge in Fig. 7.37 is already strongly inclined towards a radial line on the hub, i.e. it does not join the hub radially.
- The outer contour of the inducer is often tapered at the inlet, Fig. 7.37b. At the inducer inlet there is therefore a large gap between the blades and the casing. This design also reduces the pulsations at low partload and low $NPSH_A$.

7.7.3 Inducer Performance Prediction

If no test data for the blade parameters (λ_w and λ_c) are available for the type of design intended, a statistical correlation can be used to determine the expected n_{ss} and $NPSH_R$ from selected geometric parameters. In order to establish such a correlation, inducer tests from various sources (including [82–85]) were analyzed. The results are shown in Fig. 7.37-1 where the cavitation coefficient is plotted as function of the quantity x_σ which contains the flow coefficient and geometric data. The correlation, Eq. (7.23a) is based on the data near design flow (symbol x), while the graph also shows data at off-design conditions (symbol Δ). Most of the data points are within $\pm 15\%$ of the correlation, while the scatter is larger at off-design operation (not surprisingly). In view of the low absolute $NPSH_R$ the uncertainty of the test data is presumably even greater than $\pm 15\%$. Furthermore, the data from literature are often incomplete and not always easy to interpret. The plot demonstrates

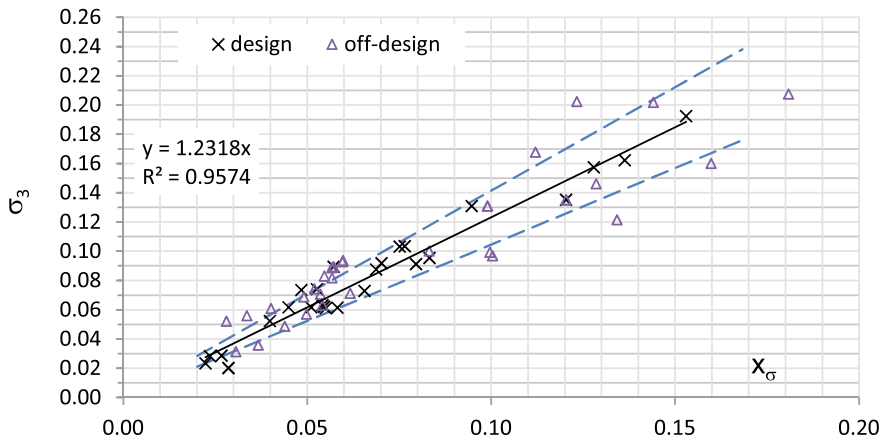


Fig. 7.37-1 Cavitation coefficient for inducers correlated by Eq. (7.23a). The dashed lines mark $\pm 15\%$ of the correlation

that the correlation can also be used to predict the $NPSH_R$ as a function of the flow rate, which is important because the BEP of an inducer is not well defined (the BEP of the pump is given by the volute and the impeller - and with low n_q mostly by the volute). An evaluation of the factor k_3 in Eq. (7.6.10) from the same test data gave no useful correlation.

$$x_{\sigma} = \varphi_1^{0.43} \varphi_{1,SF}^{0.75} \left(\frac{L}{t}\right)_a^{-0.2} \left(\frac{d_{1i}}{d_{1a}}\right)^{0.3} \left(\frac{d_{2a}}{d_{2i}}\right)^{-0.1} \left(1 + \frac{\varepsilon_{sb}}{100}\right)^{-0.47} \left(\frac{z_v - L_a}{3}\right)^{0.24}$$
$$\sigma_3 = 1.26x_{\sigma} \tag{7.23a}$$

For $L/t > 2$, set $(L/t)_{\max} = 2$ in the correlation since blades longer than that are not expected to bring much further improvement. The correlation covers the following range of design data ($n_{ss} = 280$ to 800).

Table D7.3-1 Inducer data to Fig. 7.37-1												
	z_{La}	d_{1a}	d_{1i}/d_{1a}	d_{2i}/d_{1i}	ϕ_1	i_{1a}	$\beta_{1B,a}$	$\beta_{2B,a}$	$(L/t)_a$	ε_{SB}	$\phi_{1,SF}$	$\phi_1/\phi_{1,SF}$
		mm	—	—	—	°	°	°		°	—	
Average			0.27	1.4	0.15	2.8	11	14	2.0	32	0.20	0.78
Maximum	4	300	0.53	2.9	0.30	7.5	19	22	3.3	93	0.34	1.20
Minimum	2	50	0.13	1.0	0.057	−1.4	6.5	8	1.0	0	0.11	0.42

- The main parameters according to the correlation are:
- (1) The inlet flow coefficient φ_1 : Low $NPSH_R$ require low φ_1 .

- (2) The coefficient of shockless entry flow $\varphi_{1,SF} = \tan\beta_{1B,a}/\tau_{1a}$ accounts for the effect of the blade inlet angle at the outer streamline: low $NPSH_R$ requires low $\beta_{1B,a}$ in order to reduce blade loading and cavitation.
- (3) Length to pitch ratio $(L/t)_a$: Longer blades reduce blade loading and tend to reduce cavitation. However, $(L/t)_a > 2.5$ to 3 must not be surpassed. Above $(L/t)_a > 1.5$ to 2 little or no improvement is expected. Even though extra-long blades have little effect on the head produced by the inducer, cavitation bubbles may diminish in long channels.
- (4) Number of blades: reducing the blade number increases the triangular section up to the throat area and reduces blade blockage; both effects reduce the $NPSH_R$. However, the axial length of the inducer increases for a selected ratio $(L/t)_a$. For this reason, 3 blades are often selected even though two blades would promise lower $NPSH_R$.
- (5) Reducing the hub ratio at the inlet, d_{1i}/d_{1a} increases the flow area and reduces $NPSH_R$.
- (6) Reducing the hub diameter d_{2i} at the inducer outlet (increasing the ratio d_{2a}/d_{2i}) seems to be beneficial - maybe due to larger channels.
- (7) Sweep back ε_{SB} : improves $NPSH_R$ and reduces pressure pulsations and instabilities of the cavity flow.

Overall, these parameters and their trends are plausible (in fact most act similar to impellers). Attempts to make a correlation with the ratio $\varphi_1/\varphi_{1,SF}$ failed because the absolute blade angle or stagger angle has a dominating influence on the blade loading, hence the pressure minimum at the leading edge and cavitation.

7.7.4 Inducer Design Systematic

Similar to the concept introduced in Chap. 7.14 for impellers, a design systematic can be developed for a range of inducers. The idea is to use a set of rules and equations by which an inducer can be designed for any given flow rate, speed and inlet diameter. The latter is given by the pump into which the inducer shall be installed. The objectives and concepts are:

- “Design systematic” means that the inducer *geometries are fully described by analytical equations*. This approach will reduce the uncertainty of $NPSH_R$ -prediction although the endeavor is limited by the effect of the impeller downstream of the inducer
- The performance curves are given by: $\sigma_R = f(\varphi_1)$ and $\psi_p = f(\varphi_1)$
- The inducer geometry is fully described in dimensionless terms. Therefore, the inducers can be applied to any pump (independent of n_q) by:
 - Scaling of any inducer selected from the range to any size and speed
 - Develop inducer geometry for any new φ_1 not yet designed

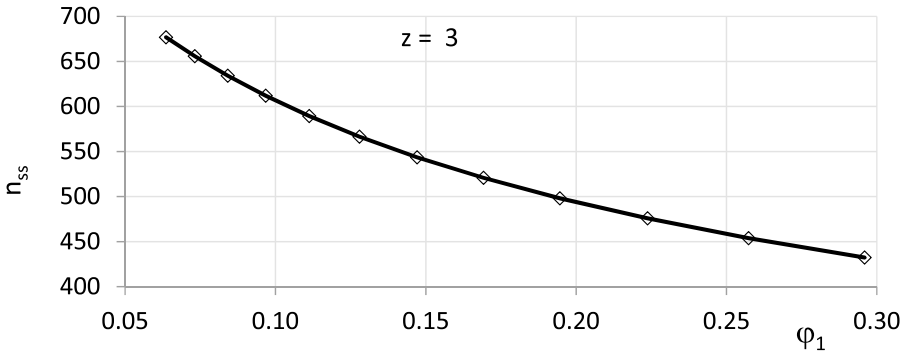


Fig. 7.37-2 Suction specific speed for a series of inducers with 3 blades

- Since the geometry follows from analytical equations, the new geometry is consistent with the design of the adjacent geometries (designed for lower resp. higher flow coefficients)
- The range covers the flow coefficients used for industrial inducers. As an example, Fig. 7.37-2 shows the suction specific speeds as function of the inlet flow coefficient.

For any given application n , Q_{opt} and impeller inlet diameter d_{1a} are specified. Selecting the inducer inlet diameter (equal to d_{1a} or d_{sp} for example) and the hub diameter at the inlet, ϕ_1 and n_{ss} can be calculated. If the n_{ss} is not sufficient, the inducer inlet diameter, must be increased.

7.7.5 Matching the Inducer to the Impeller

Frequently the inducer is mounted upstream of the impeller of a standard pump whose $NPSH_R$ is unable to satisfy the specific requirements of the application. If an impeller of moderate suction capability is combined with an inducer of high suction specific speed, there is, however, no guarantee that the expected reduction of the $NPSH_R$ will actually be achieved. Especially the maximum flow rate can be limited through excessive incidence of the downstream impeller if cavitation occurs on the blade pressure surfaces. To check this, the flow angle β_2 at the inducer outlet is calculated from $\beta_2 = \beta_{2B} - \delta_2$ with δ_2 from Eq. (T7.6.4) (or according to Table 3.2). If β_2 is larger than the blade inlet angle β_{1B} of the downstream impeller, the blade leading edges of the latter can be adjusted to a certain extent by cutting back the blades and profiling them on the pressure surface. To keep the discharge angle as low as possible, the pressure increase in the inducer should not be chosen unnecessarily large.

Figure 7.38 shows measured flow velocities and angles at the outlet of an inducer. Characteristically the meridional velocity near the outer streamline is far

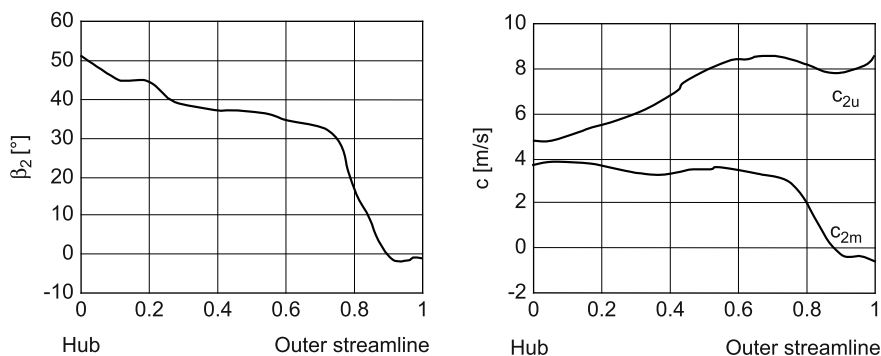


Fig. 7.38 Outlet flow angle and absolute velocity at the outlet of an inducer, Sulzer Pumps

below the mean value while the circumferential velocity is above average. Consequently, the discharge angles are considerably smaller on the outer streamline and far greater near the hub than would be expected from the calculation. These three-dimensional effects are already pronounced in the design point and are reinforced at partload (see also Chap. 5.2 and Fig. 5.7). The deficit in the meridional velocity near the outer streamline is also caused by the tip clearance flow and the boundary layer thickness. With growing tip clearance the deviation angle or the difference between blade and flow angles increases. Owing to the distorted flow conditions the calculation of the flow angles at the inducer outlet becomes quite uncertain.

An acceleration of the fluid from the inducer outlet to the impeller inlet causes a corresponding drop in static pressure and, consequently, an increased tendency to cavitation. The ratio of the velocity w_{1q} in the throat area of the impeller to the approach flow velocity w_1 (which is identical to the relative velocity w_2 at the inducer outlet) should therefore also be verified, Chap. 7.2.1 item (15): at the maximum flow rate $w_{1q}/w_2 = 1$ should not be significantly exceeded.

7.7.6 Recommendations for Inducer Application

The large diameter ratio d_1/d_{1i} causes partload recirculation to start near the design point. At low flow rates recirculation then becomes very intense inducing a strong swirl in the inlet pipes of end-suction pumps. As a consequence of the rotation a parabolic distribution of the static pressure $p(r)$ builds up at the inducer inlet. At low suction pressures a vapor-filled vortex can be generated in the center of the suction pipe. As discussed in Chap. 6.5.1, such vortices can cause severe pulsations. By a suitable design of the inducer it is possible to largely avoid such kinds of pulsations. Favorable in this respect seem to be: long blades (high solidity L/t), variable pitch design, sweep-back and a tapered outer contour of the inducer. However, it is not yet

possible to provide generally valid design criteria for mastering this highly three-dimensional two-phase flow. For this reason it is necessary to prove by way of experiment that no impermissible pulsations and excitation forces occur in the demanded operating range. To this end, the water in the test circuit has to be largely de-aerated since the pulsations are greatly attenuated in air-saturated water. If at all possible, the test should be done with the same circumferential velocity as in the planned operation.

As a rule, pumps for industrial application are required to operate also at low partload. To be able to satisfy this requirement without excessive recirculation, the suction specific speeds of industrial inducers are limited to approximately $n_{ss} = 500$ to 700. If this range is markedly exceeded, the $NPSH_3$ at partload increases so that the inducer may no longer satisfy its actual purpose of lowering $NPSH_3$ throughout the operating range.

Obviously the target should be to restrict the partload operation as far as possible and to entirely avoid operation at $q^* < 0.3$ (except during start-up).

Major cavitation zones are created in the inducer in most applications. Despite the low $NPSH_A$ - i.e. low impllosion pressure - the impllosion energy can therefore be considerable. While hardly any cavitation damage is to be expected when pumping hydrocarbons, the circumferential velocities of the inducers when pumping water have to be limited in order to reduce the risk of cavitation damage (especially with de-aerated cold water, refer also to Table 6.2). For pumping water it is therefore advisable to use high-alloy steels with suitable cavitation resistance. The limit for the circumferential velocity cannot be determined generally; it may be within a range of 25 to 30 m/s when pumping water and 35 to 40 m/s with hydro carbons. These limits can be inferred from published applications, e.g. [33]. Initial cavitation bubbles (corresponding to $NPSH_i$) are created in the gap between casing and inducer.

Inducers are able to some extent to transport gas-laden liquids. Inducers combined with closed impellers are claimed to handle gas fractions of up to 25% of the flow rate at the inducer inlet; combined with semi-open impellers 35% and with open impellers even 40%. The gas-handling capabilities depend on the geometry and should be confirmed by testing, see also Chap. 13.2.

The Q-H-curve and the efficiency virtually do not change near the best efficiency point when installing an inducer. Due to the energy transfer in the inducer the circumferential component of the absolute velocity at the inducer outlet is increased, but the downstream impeller has to perform correspondingly less work in accordance with the Euler equation, Eq. (T3.3.1). The efficiency of the pump tends to fall since the hydraulic losses in the inducer are proportionally slightly higher than in the impeller, but this effect is sometimes hardly measurable.

However, an inducer increases the shut-off head by several percent due to the intense inlet recirculation and the associated rise of the centrifugal head component, Chap. 5.4.1. In the overload range the head drops slightly compared with the pump without inducer but this is likely to become noticeable only when the inducer operates in the range $H_p < 0$ ($Q > Q_B$), Fig. 7.35.

Inducers have relatively thin blades with a large blade height. When the inducers are cast possible blade deformations must be monitored. In particular it should be

Table 7.7 Diffuser and volute design calculations			Eq.
Given	n, Q _{opt} , H _{opt}		
Diffuser or volute design flow rate	Q _{Le} = Q _{opt} + Q _{s3} + Q _E		7.7.1
Circumferential component of absolute velocity at impeller outlet	c _{2u} = $\frac{g H}{\eta_h u_2} + \frac{u_{1m} c_{1u}}{u_2}$		7.7.2
Circumferential component of absolute velocity at diffuser inlet	c _{3u} = $\frac{d_2 c_{2u}}{d_3}$		7.7.3
Meridional component of absolute velocity at diffuser inlet. For volutes set: τ ₃ = 1.0	c _{3m} ' = $\frac{Q_{Le} \tau_3}{\pi d_3 b_3}$		7.7.4
Flow angle at diffuser inlet with blockage	tan α ₃ ' = $\frac{c_{3m}'}{c_{3u}}$		7.7.5
Diffuser vane inlet angle	α _{3B} = α ₃ ' + i ₃	incidence i ₃ = ±3°	7.7.6
Throat area of a volute of circular section or equivalent circle of a volute with an arbitrary section	$X_{Sp} = \frac{Q_{Le}}{\pi c_{2u} r_2} \frac{\epsilon_{sp}}{2\pi}$ $d_{3q} = X_{Sp} + \sqrt{2 d_z' X_{Sp}}$ $A_{3q} = \frac{\pi d_{3q}^2}{4}$		7.7.7
Factor for diffusers or volutes with rectangular cross section	X _{Le} = exp { $\frac{Q_{Le}}{b_3 c_{2u} r_2 z_{Le}}$ }		7.7.8a
Outer radius r _a of the end section (throat area) of a rectangular volute	r _a = r _z ' X _{Le}		7.7.8
Height of the end section (throat area) of a rectangular volute	a ₃ = r _a - r _z ' = r _z ' exp { $\frac{Q_{Le}}{b_3 c_{2u} r_2} \frac{\epsilon_{sp}}{2\pi} - 1$ }		7.7.9
Diffuser inlet width f _{a3} = 1.1 to 1.3	a ₃ = f _{a3} $\frac{d_3}{2}$ { X _{Le} - 1 }		7.7.10
Diffuser inlet width calculated for a selected deceleration ratio c _{3q} /c ₂	a ₃ = $\frac{Q_{Le}}{z_{Le} b_3 \sqrt{c_{2m}^2 + c_{2u}^2}} \left(\frac{c_2}{c_{3q}} \right)$		7.7.11
Return channel outlet width	b ₆ = $\frac{Q_{Le}}{\pi d_6 c_{6m}}$	c _{6m} = (0.85 to 0.9) c _{1m}	7.7.12
Meridional component of absolute velocity at return channel inlet	c _{5m} = $\frac{Q_{Le}}{\pi d_5 b_5}$		7.7.13
Circumferential component of absolute velocity at return channel inlet	c _{5u} = $\frac{c_{4u} r_{4m}}{r_{5m}}$		7.7.14
Flow angle at return channel inlet	α ₅ = arctan $\frac{c_{5m}}{c_{5u}}$		7.7.15
Gap A and overlap x _{ov} (Fig. 9.1)	$\frac{\text{gapA}}{r_2} = 0.007 \text{ to } 0.01$	$\frac{x_{ov}}{\text{gapA}} = 2 \text{ to } 4$	7.7.16
Gap B	s. Table 10.2		
ε _{Sp} is given in radians; for volutes with equal spacing: ε _{Sp} = 2 π/z _{Le}			

checked that the desired blade outlet angles are achieved: if β_{2B} is too small, the necessary pressure increase is not attained and the steep rise of the NPSH occurs at too low a flow rate. The thin blades also require a verification of the natural frequencies and stresses in order to prevent blade fractures.

When installing an inducer in an end-suction pump, the impeller overhung increases. This has to be considered when evaluating the shaft deflection and the critical speed. Also some additional excitation forces in radial direction must be expected. Radial excitation forces can (for instance) be created by the tip clearance flow or through periodically variable flow conditions that might develop in the channels of the inducer as the result of varying cavity volumes, [63].

7.8 Volute Casings

7.8.1 Calculation and Selection of Main Dimensions

In the volute casing the kinetic energy present at the impeller outlet is converted into static pressure with as few losses as possible. The volute casing then directs the fluid into the discharge nozzle or, in multistage pumps, into the following stage. Prior to calculating a volute, the boundary conditions for the casing design must be established. In particular it must be decided whether a single or double volute is required. The selection criteria are:

- **Single volutes** are the least expensive solution in terms of manufacturing costs; they are best accessible for the dressing of the cast channels. Their disadvantage consists in considerable radial forces which develop during operation at off-design conditions according to Chap. 9.3 through disturbances of the circumferential symmetry of the flow in the casing. The radial forces generate bearing loads, bending stresses in the shaft and shaft deflection which can threaten the reliability of the machine. The heads up to which single volutes can be sensibly used depends on the specific speed, the design of the pump (especially of the bearing housing), the shaft thickness and the bearing span or the overhung of the impeller. When pumping water, the limit is about $H_{opt} = 80$ to 120 m at $n_q < 40$. At high specific speeds a double volute may be indicated already for heads above $H_{opt} = 60$ to 80 m. When pumping liquids with densities considerably lower than water, the limit is correspondingly higher since the radial forces are proportional to $\rho \times g \times H$.
- **Double volutes** are employed when bearing loads, shaft stresses and shaft deflection become impermissibly high without measures for radial thrust reduction so that their control would require an excessively high design effort and cost. The rib between the inner volute and the outer channel also reduces the casing deformation under internal pressure, which facilitates the casing design at high specific speeds. The pressure to be applied during the hydrostatic test of the casing must also be considered.

- **Twin volutes** differ from double volutes mainly in that both partial volutes end in separate channels - and not in a common discharge nozzle as is the case with double volutes. Apart from special designs they are found in multistage volute pumps or vertical pumps where the partial volutes end in a central column pipe.
- **Three or four volutes** are sometimes used in borehole pumps (for example) instead of semi-axial diffusers. Their application allows smaller casing diameters to be used.

When volute type and design boundary conditions have been established, the main dimensions must be selected or calculated according to Table 7.7 in Chap. 7.9.2:

1. Wrap angle of partial volutes: In the case of double or twin volutes the wrap angle of the partial volutes generally amounts to $\epsilon_{sp} = 180^\circ$. In this case an even impeller blade number must be avoided in order to reduce pressure pulsations, Chap. 10.7.1. If nevertheless a 6-bladed impeller (for example) is to be employed it is advisable to reduce the wrap angle of the inner volute ϵ_{sp} to 165° or 170° so that two blades never pass the cutwaters simultaneously. A similar procedure should be employed in the case of multiple volutes. The selection of a wrap angle of 165° may reduce the risk of flow instabilities and alternating stall as discussed in Chaps. 5.8 and 10.7.3. This design option could be recommended for general use since the increase in static radial thrust as compared to 180° is moderate and a defined radial force might be advantageous with respect to bearing design and vibrations.

In the case of axial-split pumps wrap angles of less than 180° are sometimes selected to avoid that the center rib passes through the split flange of the casing (matching difficult because of casting inaccuracies). The radial thrust increases according to Chap. 9.3 and Table 9.4.

Double volutes with wrap angles under 90° must not be used since they are unable to reduce the radial forces. Wrap angles over 180° should be avoided as a matter of principle since the long outer channel causes additional losses, and high radial thrusts must be expected at $q^* > 1$ (see Chap. 9.3.4).

2. Casing design flow rate Q_{Le} : To ensure that the actual best efficiency flow coincides with the design flow rate, the volute must be designed for Q_{opt} (if dimensioning were done for another flow rate, the best efficiency point would be

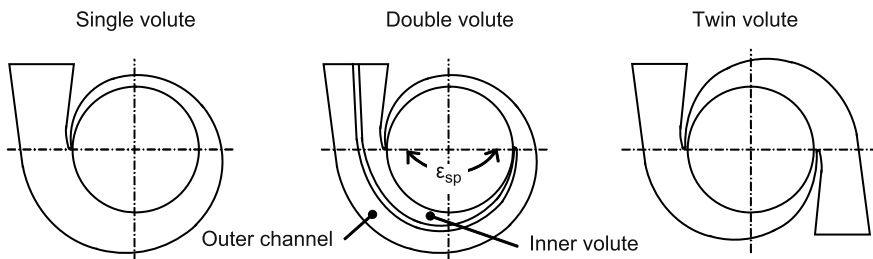


Fig. 7.39 Volute casings

shifted according to Chaps. 3.7 and 4.2). The design flow rate must be augmented by possible leakages (Q_E and Q_{s3}) which may flow *through the volute*, Eq. (T7.7.1). Note that the leakage losses at the impeller inlet do not flow through the volute.

3. Inlet velocity: The circumferential component of the absolute velocity at the impeller outlet is calculated from Eq. (T7.7.2) or Table 3.2. Downstream of the impeller it develops in accordance with the conservation of angular momentum as per $c_{3u} = c_{2u} \times r_2/r_3$, Chap. 3.7. With some types of pumps a diffuser or stay vanes according to Chap. 7.11 are arranged between the impeller and the volute. In this case the circumferential velocity c_{4u} at the outlet of these components must be used as inlet velocity to the volute.

4. Cutwater diameter d_z^* : Between impeller and cutwater a minimum clearance (gap B) must be maintained to limit pressure pulsations and hydraulic excitation forces to allowable levels, Chap. 10. The ratio of the cutwater diameter $d_z^* = d_z/d_2$ is calculated from the formula given in Table 10.2 as a function of n_q and H_{st} .

5. Shape of the volute cross sections: The shapes must be selected so as to suit the pump type, taking into account casing stresses and deformations as applicable. Figure 7.40 shows some design options for the volute cross section. When designing the casing, also the requirements of economic manufacture of patterns and castings must be taken into account. With the rectangular and trapezoidal basic shapes, for instance, all corners must be well rounded for casting reasons and flat surfaces designed with a slope (“draft angle”) in accordance with the pattern split plane. Concrete volutes have specific shapes to facilitate shuttering, Fig. 7.40f to h.

Rectangular and trapezoidal shapes offer the advantage to be developed on surfaces of revolution, something that facilitates design and manufacture. Volute casings with circular cross sections that cannot be arranged on surfaces of revolution are employed, for instance, when the volute is welded together from segments. Semi-axial impellers may have asymmetric cross sections as per Fig. 7.40d.

In double volutes the cross sections of the inner volute and the outer channel must be matched so that the casing outer wall is given an easily producible shape, and a simple transition to the circular discharge nozzle is achieved, Fig. 7.40e.

Basically, the designer has considerable freedom to configure the cross section without risking major losses in efficiency. It is expected that flat cross sections (similar to flat elbows) result in less intense secondary flow than circular cross sections, thereby generating fewer losses. For this reason, flat cross sections with a ratio of width B to height H in the range of $B/H = 2$ to 3 may be considered the optimum.

6. Inlet width b_3 : The inlet width is obtained from the impeller outlet width b_2 , the necessary width of the impeller sidewall gaps and requirements of the casing design. In particular the transition from the volute to the discharge nozzle should be smooth. This requirement means a relatively large b_3 in single volutes so that the ratio h/b at the volute throat area is close to 1.0. In a double volute the ratio h/b is rather near 0.5. The ratio b_3/b_2 depends on the specific speed; it can be selected in wide limits without major effect on the efficiency. Open impeller sidewall gaps

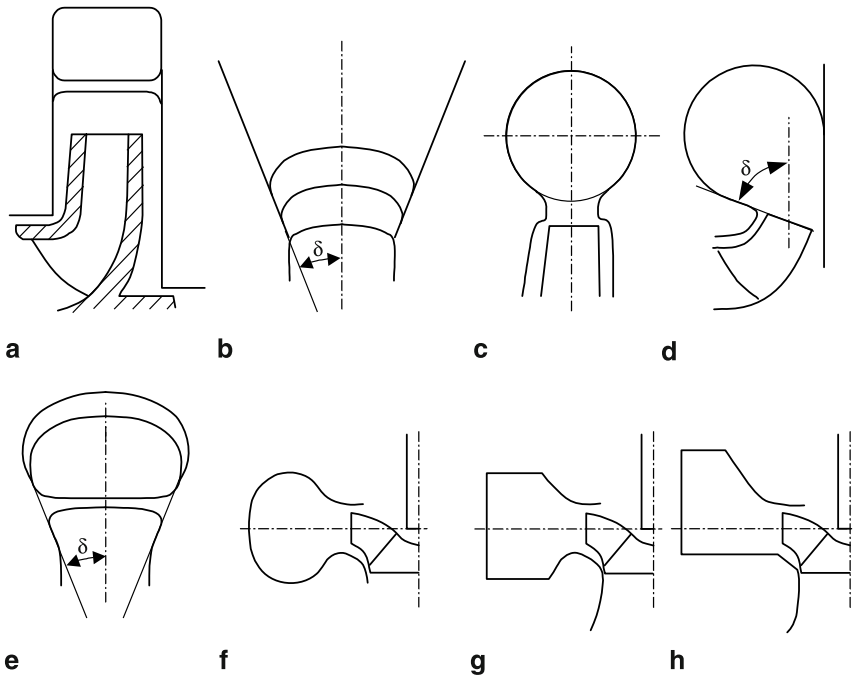


Fig. 7.40 Cross section shapes of volute casings. **a** Rectangular volute, **b** trapezoidal, **c** circular, **d** asymmetric (semi-axial impeller), **e** double volute, **f** concrete volute 1, **g** concrete volute 2, **h** concrete volute 3

according to Fig. 7.40a are favorable with respect to efficiency and radial thrust. At small specific speeds they result in values of $b_3/b_2 = 2.0$ to 4.0 . In some applications vibration problems were reported on pumps with impeller sidewall gaps wide open to the volute, see Chaps. 10.7.3 and 9.1 for design guidelines. However, large values of b_3/b_2 are prohibited for design reasons at high specific speeds and relatively wide impellers; $b_3/b_2 = 1.05$ to 1.2 may then be selected. At high n_q large ratios of b_3/b_2 are unfavorable because strong secondary flows and turbulent dissipation losses would result.

7. Cutwater profile: The cutwater should be profiled elliptically in order to render it as insensitive as possible to changes of the approach flow direction with changing operating conditions. In practice such profiling is feasible for large pumps. The cutwater is highly stressed and any profile must be short and thick (except the nose radius) to reduce the risk of cracks forming in the cutwater. The leading edge thickness e_3 of the cutwater can be defined by a circle; its diameter should be approximately $0.02 \times d_2$. In the case of sewage pumps or abrasive wear at least twice that cutwater thickness should be selected. For double volutes the required thickness of the center rib is obtained from stress calculations (and the minimum casting wall thickness required).

The cutwater has a camber angle α_{3B} with the circumferential direction which has to be designed to match the approach flow angle according to Eqs. (T7.7.3 to 7.7.6). As discussed in Chap. 3.7, it is recommended to design the cutwaters with low camber angles and incidences.

It is even advantageous to shape the cutwater with an asymmetric profile on the low-pressure side (i.e. the side facing the impeller). Then the incidence becomes negative, the pressure distribution is flatter and the tendency to flow separation at partload is reduced. This design feature can help with the Q-H-curve stability of double volutes. The resulting thick cutwater profiles are also beneficial with respect to mechanical stresses.

In order to reduce noise and vibrations an oval-shaped leading edge of the cutwater may be selected (Table 10.2). This design could also help to reduce erosion near the cutwater caused by horse shoe vortices (slurry, dredge or waste water pumps).

8. Volute throat area A_{3q} : As explained in Chaps. 3.7 and 4.2, volute casings are mainly designed in accordance with the theorem of angular momentum conservation. Different approaches to the volute design did not produce any measurable improvements in efficiency, [17]. A partial volute with the wrap angle ε_{sp} must be designed for the flow $Q_{Le} \times \varepsilon_{sp}/2\pi$ where $\varepsilon_{sp} = 2\pi$ applies to single volutes ($z_{Le} = 1$). $\varepsilon_{sp} = \pi$ is used for twin volutes or double volutes ($z_{Le} = 2$) with $2 \times 180^\circ$ and $\varepsilon_{sp} = 2/3\pi$ for a triple volute ($z_{Le} = 3$). If all partial volutes have the same wrap angle, $\varepsilon_{sp}/2\pi = 1/z_{Le}$ applies. The throat area of each partial volute must satisfy Eq. (3.14); this is obtained from:

$$\int_{r'_z}^{r_a} \frac{b}{r} dr = \frac{\varepsilon_{sp} Q_{Le}}{2 \pi c_{2u} r_2} \quad (7.24)$$

Although the thickness of the volute cutwater causes local accelerations and excess velocities, it has only a minor impact on the best efficiency point flow rate (this may be visualized by comparison to a Venturi nozzle which has scarcely any effect on the flow rate through a long pipe at a given driving pressure differential). Integration therefore proceeds from the stagnation point radius $r'_z \approx r_z + e_3/2$ to the outer limitation of the volute cross section at the radius r_a , without taking into account the cutwater thickness, [68]. The integral of Eq. (7.24) can be solved analytically for rectangular cross sections of constant width $b = b_3$. The outer volute radius r_a is obtained from Eq. (T7.7.8) and the height of the throat area $a_3 = r_a - r'_z$ from Eq. (T7.7.9).

An analytical solution can also be given for volutes with circular cross sections. The diameter of a circular throat area d_{3q} is obtained from Eq. (T7.7.7). Also, d_{3q} can serve as an approximation for the rapid estimation and assessment of cross sections of any shape if considered as an equivalent cross section giving the actual throat area of the casing. As discussed in Chap. 4.2, large hydraulic losses which occur downstream of the volute in the diffuser shift the best efficiency point to

smaller flow rates than would be obtained according to the conservation of angular momentum. In the case of double and twin volutes with low specific speeds the volute must therefore be designed for a slightly higher flow rate by substituting, for instance, $Q_{Le} = (1.05 \text{ to } 1.25) \times Q_{opt}$ in Eq. (T7.7.1).

7.8.2 Design and Shaping of Volute Casings

After the main dimensions have been established and the decisions in terms of type and basic shape of the volute have been made as per Chap. 7.8.1, the volute and subsequent diffuser can be designed:

1. Volute cross sections designed according to the conservation of angular momentum: The area distribution of the partial volutes is best developed on surfaces of revolution which are inclined at an angle δ relative to a normal to the rotor axis, Figs. 7.40b and 7.41. As mentioned in Chap. 7.8.1 item (5), the shape of the cross section is defined by straight lines and arcs of circles for casting reasons. The shape cannot be described analytically by a single expression. It is therefore generally expedient to design the cross sections so that they can be optimally integrated into the desired casing shape and to finally calculate the circumferential angle at which the cross section thus designed satisfies the angular momentum conservation, Fig. 7.41. If this design is done manually, the integral according to Eq. (7.24) is replaced by a summation of finite surface elements $\Delta A = b \times \Delta r$, where b and Δr have to be taken from the drawing. The circumferential angle on which the cross section under consideration has to be arranged is obtained according to Eq. (3.14) from:

$$\varepsilon = 360^\circ \frac{c_{2u} r_2}{Q_{Le}} \int_{r_z'}^{r_A} \frac{b}{r} dr = 360^\circ \frac{c_{2u} r_2}{Q_{Le}} \sum_{r_z'}^{r_A} \frac{b}{r} \Delta r \quad (7.25)$$

Step-by-step procedure (Fig. 7.41):

1. Use b_3 , r_z' and δ to design the two surfaces of revolution on which the volute cross sections are to be developed.
2. Draw the outer limitation for various cross sections: AF1 to AF3 in Fig. 7.41. The height to which these cross sections are required can be estimated from the throat area of an equivalent volute with circular cross section from Eq. (T7.7.7).
3. Divide the cross sections into elements of height Δr located on the radii r . The width b of these surface elements can then be measured.
4. Enter the values b , r and Δr of all surface elements located in the cross section considered into a table.
5. Calculate the sum $\Sigma(r \times \Delta r/b)$ of all elements.

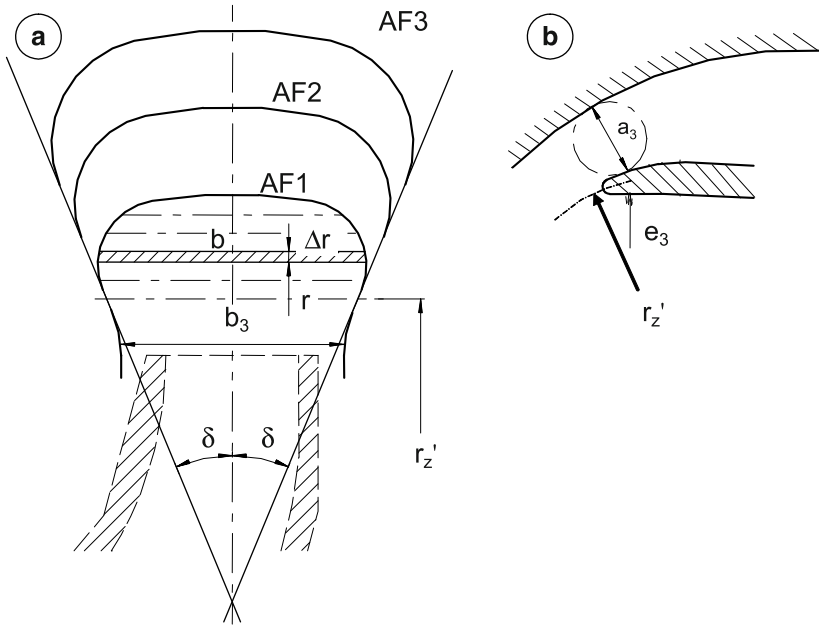


Fig. 7.41 Design of the volute sections. **a** Meridional section; **b** cutwater details

6. Enter this sum into Eq. (7.25) and calculate circumferential angle ε on which the designed cross section has to be arranged.
7. The outermost point of the cross section on radius r_a is obtained at the same time so that the layout of the outer volute shape $r_a(\varepsilon)$ can be drawn.

As a rule, this task will be completely or at least partly performed on the computer describing the cross sectional shapes by a set of equations.

2. Volute cross sections designed for constant velocity: Alternatively, volute casings are frequently designed by assuming a constant velocity in all cross sections over the circumference, [B.2]. The volute throat area A_{3q} is again calculated in accordance with the conservation of angular momentum from Eq. (7.24) or Eqs. (T7.7.7 to 7.7.9) - otherwise the specified best efficiency point would be missed. The velocity in the throat area then amounts to:

$$c_{3q} = \frac{Q_{Le}}{A_{3q}} \quad \text{with} \quad A_{3q} = \int_{r_z'}^{r_A} b \, dr \quad (7.26)$$

All the other cross sections of the volute $A(\varepsilon)$ for any wrap angles ε are now calculated from this velocity:

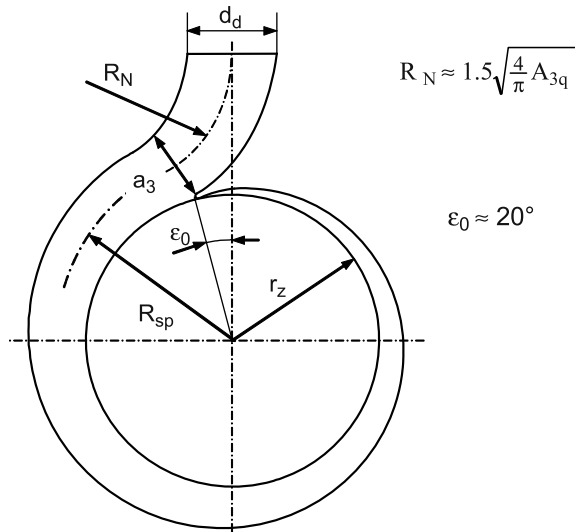
$$A(\varepsilon) = \frac{Q(\varepsilon)}{c_{3q}} = \frac{Q_{Le}}{c_{3q}} \frac{\varepsilon}{\varepsilon_{sp}} = A_{3q} \frac{\varepsilon}{\varepsilon_{sp}} \quad (7.27)$$

Thus the cross sections are directly proportional to the circumferential angle. The efficiencies of the volutes designed for constant velocity are at least on a par with the ones designed according to the conservation of angular momentum. At low specific speeds even a slight advantage might be expected since the flow velocities in the first part of the volute are reduced and lower friction losses can thus be expected. In contrast, volutes designed according to the conservation of angular momentum might have an advantage at high specific speeds because of reduced secondary flow in cross sections which match perfectly. Furthermore, the casing dimensions become noticeably smaller at high n_q if the volute is designed to the theorem of angular momentum conservation. Following these arguments it appears that an optimum philosophy may be:

- Design volutes for $n_q < 25$ to 35 for constant velocity as per Eqs. (7.26 to 7.27)
- Design volutes for $n_q > 25$ to 35 according to the conservation of angular momentum as per Table 7.7.

3. Cutwater correction: Because of its finite thickness, the cutwater interferes with the flow, generating local excess velocities and deviations of the streamlines from the pattern given by the conservation of angular momentum. The interference increases with the thickness of the cutwater. When thick cutwaters are required, this effect cannot be neglected (e.g. sewage, dredge or slurry pumps). The relative blockage increases as the volute cross sections get narrower with falling specific speed. Consequently, an allowance for the cutwater thickness must be applied to the

Fig. 7.42 Volute casing with radial discharge nozzle; R_{sp} represents a mean streamline



throat area of all volute casings; it can be estimated from Eq. (7.28), suggested in [67], where e_3 and a_3 are measured at the throat, see Fig. 7.41b:

$$\frac{\Delta a_3}{a_3} = 0.2 \left(\frac{e_3}{a_3} \right)^2 \quad (7.28)$$

The cross sections upstream and downstream of the throat area must be adjusted in accordance with the cutwater correction so that continuous shapes are obtained. Inevitably this implies local deviations from the cross sections calculated from the conservation of angular momentum.

4. Discharge nozzle/diffuser: As a rule, a diffuser follows the volute throat area. In the case of single-stage pumps the diffuser simultaneously forms the discharge nozzle. At high specific speeds (above $n_q \approx 80$) the fluid should not be further decelerated downstream of the volute throat area - otherwise the discharge nozzles would become uneconomically large. In contrast, at low specific speeds the deceleration in the diffuser is significant and every care should be applied to design smooth hydraulic contours which minimize the losses. Maximum allowable diffuser opening angle, pressure recovery and losses are calculated according to Chap. 1.6. The diffuser length and the area ratio have to be matched so that the pressure recovery (or the c_p -value) is maximized. In the case of non-circular cross sections the optimization is done using the concept of an equivalent diffuser as per Eq. (1.45). The diffuser is designed for standardized discharge nozzle sizes. The length of the diffuser/discharge nozzle is limited by cost considerations and by the moments and stresses induced by pipeline forces (since these moments are proportional to the nozzle length). Experience shows that the data of straight diffusers according to Chap. 1.6 can be applied to curved discharge nozzles without correction, presumably since the secondary flow in the volute has a favorable effect on the boundary layers. In this way, the negative effect of a slight to moderate curvature is largely compensated for. A more prudent approach would be to apply the curvature correction according Eq. (1.54).



Fig. 7.42-1 Short and long cross-over for multistage pumps

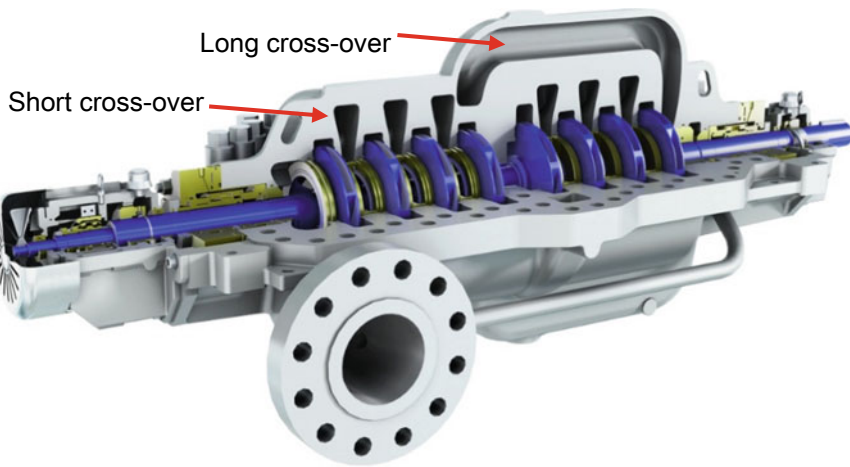


Fig. 7.42-2 8-stage pump with double volutes, Sulzer

In addition to the tangential discharge nozzle according to Fig. 7.39, radial discharge nozzles as per Fig. 7.42 and intermediate shapes with a slight offset are frequently found. According to [25], the volute should start roughly at an angle of $\varepsilon_o = 60^\circ$ for tangential nozzles and at about $\varepsilon_o = 20^\circ$ for radial nozzles. The mean radius of radial discharge nozzles can be estimated from Eq. (7.29), [25]:

$$R_N \approx 1.5 \sqrt{\frac{4}{\pi} A_{3q}} \quad (7.29)$$

These recommendations can only give some hints; actually the angle ε_o and the radius R_N result from the design process.

Excessive local curvatures often result at the transition from the volute to the discharge nozzle if the calculated cross sections are strictly adhered to. In order to prevent distorted velocity profiles and local flow separation, the volute contour should be adjusted so that a smooth transition between volute and diffuser is obtained. The resultant local increase in flow area near the throat usually has only little effect on the best efficiency flow rate - provided the remainder of the volute is dimensioned correctly. Neither would it be possible to shift the best efficiency point towards lower flow rates by way of a local area reduction, without suitably adjusting the remainder of the volute. A local area reduction would be comparable to the effect of the cutwater thickness as discussed in Chap. 7.8.1 item 8.

5. Double volutes: At small and medium specific speeds the fluid is decelerated in the outer channel downstream of the throat area. Again this diffuser is dimensioned according to Chap. 1.6. Since the flow resistances of the two partial volutes with their unlike outlet channels generally differ, different flow rates are obtained in both channels. The individual flow rates in both volutes/channels can be estimated

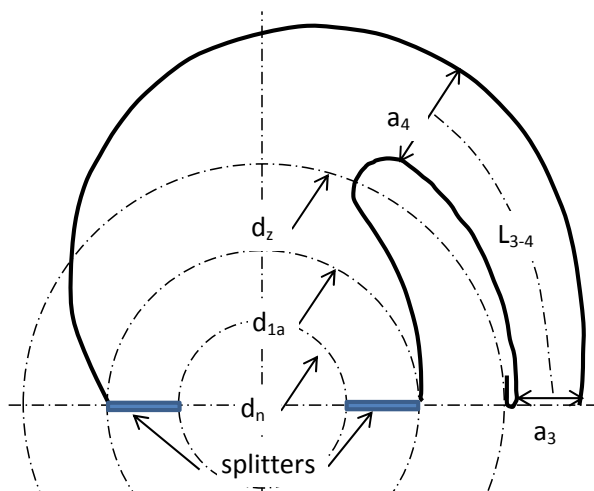


Fig. 7.42-3 Short cross-over design

according to Table 1.5. The pressure loss coefficients of the partial volutes and the various diffusers are calculated from Eqs. (T3.8.21) and (T3.8.22) to this end. The friction and deflection losses in the outer channel are calculated likewise from Table 1.4 and Chap. 1.5.1. Outer and inner volutes constitute two parallel flow resistances. After all resistances of the two channels have been combined into a total resistance according to Table 1.5, the two individual flows can be calculated. In this calculation it is assumed that both channels are subject to the same pressure difference, as given by the pressure at the discharge flange minus the pressure at the impeller outlet. In this way it is possible to evaluate the flow unbalance between both partial volutes and to optimize the geometry of the outer channel so that both partial volutes receive the same flow rates as far as possible. Since the impeller, in both partial volutes, works at different operating points (Chap. 9.3.3), the flow rates through the volutes adjust rather more favorably in reality than is borne out by the above estimate.

6. Multistage pumps with double volutes: In multistage pumps with double (or “twin”) volutes (as per Figs. 2.9 and 7.42-2), “short crossovers” are used to guide the fluid from one stage to the next, Fig. 7.42-1 (left). Downstream of the volute throat area, the fluid is decelerated in a diffuser from the throat velocity c_{3q} to the velocity c_4 at the inlet to the cross-over which has the same function as the return channels in a vaned diffuser pump. With the back-to-back impeller arrangement, the “long cross-over” is employed to guide the fluid from the last stage of the 1st group of stages to the 2nd stage group, Fig. 7.42-1 (right).

Recommendations for the design of twin volutes for multistage pumps, Fig. 7.42-3:

- a. The volutes and throat areas (A_{3q} , a_3) are designed according the equation for conservation of angular momentum as for any other type of volute, Table 7.7.
- b. However, it may be useful to increase the design flow rate Q_{Le} of the volute by an amount similar to the factor f_{a3} in Table 7.7. In this way, high losses occurring downstream of the throat area could be compensated for to some extent. Such losses tend to move the BEP to lower flow and increasing the factor f_{a3} may be necessary to offset this effect. In this way, the throat area can be optimized to minimize the overall losses in the casing and to achieve the specified BEP flow rate.
- c. Downstream of the throat area at a_3 the flow is decelerated in a diffuser to a_4 , see Fig. 7.42-3.
- d. It is most important to optimize this diffuser according to the rules set forth in Chap. 1.6: The allowable opening angle as per Eq. (1.44) must not be exceeded and the pressure recovery c_p should be maximized with the aid of Fig. 1.19 (or equivalent data) by properly selecting the area ratio A_R and the length ratio $L_{3-4}/R_{1,eq}$.
- e. Since the diffuser is curved, Eq. (1.54) can be applied to estimate the pressure recovery and losses.
- f. The area should increase in a linear fashion from a_3 to a_4 in Fig. 7.42-3. Specifically, flow deceleration should start right from the diffuser inlet in order to reduce friction losses in the region where velocities are highest.
- g. After the diffuser outlet at a_4 , the channel turns sharply and the area expands. Flow distortion and separation downstream of a_4 is the almost unavoidable consequence. In order to keep the associated losses as small as feasible, the velocity head in the area a_4 should be a low fraction of the head per stage: $c_4^2/(2gH_{opt,st}) = 0.02$ to 0.04 . This, because the losses are directly proportional to this fraction. Optimization of the return channel has been attempted. An additional guide vane near the inner streamline has been studied in [73]. The effect would be to reduce the non-uniformity of the flow, losses and pressure pulsations. The splitter must not be too long within the diffuser in order to reduce friction and shock losses in the region with high flow velocities. Other studies attempted to reduce the curvature of the inner streamline by an increased wrap angle, [74, 75].
- h. Effective flow splitters in the axial split-plane of the pump casing are essential to reduce the pre-rotation to the subsequent impeller, to stabilize the flow and help to prevent the Q-H-curve from drooping towards shut-off.
- i. In multistage pumps, the cutwater distance ratio d_z/d_2 is usually made smaller than for single-stage pumps (see Table 10.2), but it should not be made smaller than for diffuser pumps.
- j. If the impellers are arranged back-to-back as in Fig. 2.9, the head loss in the long cross-over has to be taken into account. In order to limit these losses, the velocity head should be made as low as compatible with the compactness of the design; again a range of $c_4^2/(2gH_{opt,st}) = 0.02$ to 0.04 could be envisaged. Typically, the long-cross-over (Fig. 7.42-1, right) consists of a flow turn at the inlet with a loss coefficient of $\zeta_{in} = 0.5$, a straight channel with $\zeta_{ch} = \lambda_R L/d_h = 0.2$, another flow

turn radially inwards $\zeta_b = 0.3$ and the inlet losses to the next stage $\zeta_E = 0.2$. The total loss coefficient is the sum of these individual loss components: $\zeta_{L-XO} = 1.2$. The head loss in the long cross-over $\Delta H_{L-XO}/H_{opt,st} = 0.024$ to 0.048 referred to one stage which would result in a head and efficiency loss of a 10-stage pump 0.24 to 0.48% assuming $c_4^2/(2gH_{opt,st}) = 0.02$ to 0.04 . These figures can just give a rough indication; for any given design, the losses must be estimated from the actual geometry

7. Concrete volutes: are sometimes used for pumps with flow rates in excess of about $10 \text{ m}^3/\text{s}$ and for heads up to 30 m . If the volutes are constructed without steel lining, the maximum local flow velocity must be limited to 10 m/s to avoid erosion of the concrete. For radial impellers, cross sectional shapes according to Fig. 7.40f, g are preferred, while shapes with a flat base according to Fig. 7.40h are used for semi-axial pumps. Shapes g and h are less costly to build than shape f. However, they involve a loss of efficiency of approximately 1% at $n_q = 70$ to 100 (at still higher specific speeds the differences in efficiency are insignificant), [36, 59].

7.8.3 Influence of the Volute Shape on Hydraulic Performance

When shaping volute and discharge nozzle, design requirements derived from the overall machine concept must be taken into account. Even the manufacturing process of the volute casing has an impact on the design: Consider for example the difficulties of core support for long and narrow channels of double volutes for low specific speeds, or the special requirements of welded designs or concrete volutes.

The effects of various geometrical parameters of the volute casing on the hydraulic behavior of pumps with specific speeds of up to approximately $n_q = 70$ are discussed in the following. The tendencies revealed originate from a number of experimental investigations such as [6, 17, 31, 44, 72]. The numerical values indicated illustrate the approximate magnitude of the effects as found in the mentioned tests; these figures may not be considered as generic.

- Wide impeller sidewall gaps according to Fig. 7.40a (or large b_3/b_2) foster pressure equalization over the impeller circumference, in this way reducing the radial thrust by up to 50% (Chap. 9.3). Wide impeller sidewall gaps yield clearly better efficiencies at partload than narrow gaps between impeller and casing (typically up to 3%). At Q_{opt} the efficiency is also slightly better (approximately 1%) even though shock losses due to sudden deceleration of the meridional velocity c_{2m} develop according to Eq. (T3.8.24). Wide impeller sidewall gaps lead to lower hydraulic losses since the low-energy boundary layer is able to flow from the volute into the impeller sidewall gap on the outer wall, as shown in Fig. 9.3. In as much as this fluid does not flow back to the

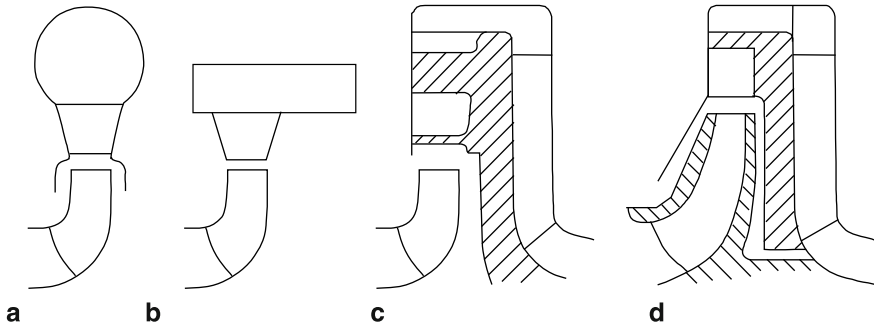


Fig. 7.43 Diffuser arrangements. **a** Diffuser with volute, **b** diffuser and annular casing as last stage, **c** overflow from diffuser to return vanes through vaneless annulus, **d** lateral overflow from diffuser to return vanes

suction casing in the form of annular seal leakage, it is re-accelerated on the rotating shroud. In this way the fluid enters the volute from the impeller shroud boundary layers with a high circumferential velocity contributing to the energy transfer.⁶

- The opening angle δ of the surfaces of revolution (Fig. 7.40b) has an influence on the height to width ratio H/B of the volute; it should be selected so as to minimize secondary flow, i.e. to get an optimum H/B ratio under the given design constraints.
- Cutwater distances of up to 10% of r_2 hardly have an effect on the efficiency in the case of thin cutwaters. If the distance is increased from 10 to 20%, the efficiency drops by approximately 1%. With thick cutwaters the efficiency falls rather continuously when increasing cutwater distance. If the impeller sidewall gaps are large, the effect of the cutwater distance on the radial force is small.
- A cutwater thickness of up to $e_3/d_3 = 0.04$ hardly has any effect on the efficiency at the best efficiency point. With thick cutwaters the partload efficiency increases since the cutwater becomes less sensitive to excessive incidence (provided the cutwater is properly profiled).
- A reduction of the cross sections in the outer channel of a double volute (or an increase of the resistance) yields greater radial forces - especially at $q^* > 1$.
- If the cutwater distances of partial volutes are not the same, the resulting asymmetries of the pressure distribution may lead to higher radial forces (the effect of casting tolerances must be considered).
- In radial discharge nozzles generally higher losses would be expected than in tangential nozzles because of the additional flow deflection. Tests reported in [31], for instance, yielded a loss in efficiency of 4% with $n_q = 23$ and measurements in [18] with $n_q = 45$ a loss of 1%. According to other authors, measurable differences should hardly be expected with optimally designed

⁶See also Chap. 3.10.3 for the effect of roughness on the pumping action of shrouds.

diffusers. The hydraulic conditions can be visualized as follows: The radial discharge nozzle produces a deflection in the opposite direction relative to the flow in the volute. This results in a certain equalization of the velocity profile at the volute outlet. The equalization is only efficient if both curvatures are correctly matched. At low specific speeds R_{sp}/a_3 is large while the curvature in the discharge nozzle is much more severe; the additional losses therefore predominate (R_{sp} is the radius of the mean streamline in the volute). At high specific speeds the conditions are much better: R_{sp}/a_3 falls with increasing n_q and moves closer to the curvature ratio R_N/a_3 in the nozzle; the counter curvature has a balancing effect in this case. In the light of this consideration, the above quoted test results on $n_q = 23$ and 45 are quite plausible. It can therefore be assumed that radial and tangential discharge nozzles roughly produce identical efficiencies at high specific speeds. Conversely, the losses in radial nozzles are expected to increase with falling specific speed implying a drop in efficiency and head.

7.9 Radial Diffusers with or Without Return Channels

7.9.1 Calculation and Selection of Main Dimensions

The diffuser serves to convert the kinetic energy present at the impeller outlet into static pressure with as little losses as possible. Various diffuser configurations are in use as shown in Fig. 7.43. In the case of single-stage pumps the diffuser discharges into a downstream volute casing or into an annular collector; the same applies to the last stage of multistage pumps. The diffusers of multistage pumps form a unit with the return channels which direct the fluid to the next stage (with the exception of the last stage).

The diffusers and return channels of multistage pumps are built in various types. While the diffuser inlet and the subsequent diffusing channel of all types are basically very similar in design, the overflow geometry from the diffuser outlet to the return vanes can be configured in quite different ways: (1) Diffuser and return vanes can be designed to form a single continuous channel according to Fig. 7.44 which is similar to a three-dimensional bend. This type of design tends to give the lowest hydraulic losses, but is more expensive in terms of design and manufacture. However, modern methods of 3D-CAD and numerically controlled manufacturing mitigate this disadvantage. (2) Diffuser and return channels can be separated as per Fig. 7.43c by a vaneless annulus. The fluid exiting radially from the diffuser is deflected in this annulus by 180° to enter the return channels radially. (3) The liquid flows laterally from the diffuser channels so that it enters the return channels with a 90° deflection (Fig. 7.43d). These basic shapes can be configured in quite different ways as shown by the examples in Fig. 7.45.

The diffuser designs discussed above feature an outer diffuser diameter d_{Le} which is typically 30 to 50% larger than the impeller diameter d_2 . More compact

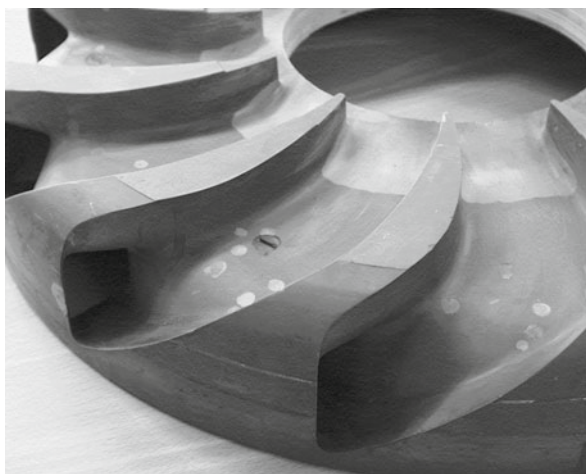


Fig. 7.44 Continuous diffuser and return vanes, Sulzer Pumps

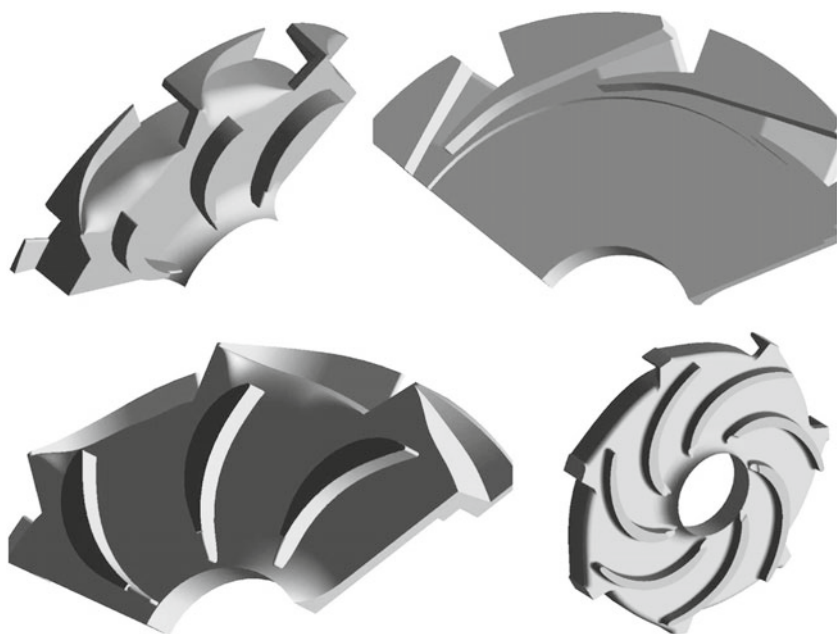


Fig. 7.45 Diffuser and return vanes as 3D-models, Sulzer Pumps

diffusers would be desirable in various applications, for example in submersible pumps. Figure 7.46 shows a most compact design where the radial impeller discharges axially into a diffuser whose outer diameter equals the impeller diameter, i.e. $d_{Le} = d_2$. The development of such diffusers is described in [51–54, 66]. After

impellers is 2 mm giving $d_3 = d_2 + 2$ mm, but only if this does not violate the criteria given in Table 10.2.

3. Inlet velocity: The circumferential component of the absolute velocity at the impeller outlet c_2 is calculated from Eq. (T7.7.2) or Table 3.2. The diffuser inlet flow parameters follow from Eqs. (T7.7.3 to 7.7.5).

4. Inlet angle α_{3B} and leading edge profile: The diffuser vanes are to be set at a camber angle α_{3B} with respect to the circumferential direction. α_{3B} is calculated from the approach flow angle as per Eq. (7.7.6). The incidence is selected in the range of $i_3 = \pm 3^\circ$.

As discussed in Chap. 3.7, it is recommended to design the diffuser vanes with low camber angles and incidences. The leading edge can be designed with an asymmetric profile on the low-pressure side (i.e. the side facing the impeller). Then the incidence becomes negative, the pressure distribution is flatter and the tendency to flow separation at partload is reduced. The asymmetric profile avoids accelerating the flow in the throat as might be the case with a symmetric profile.

5. Diffuser vane number z_{Le} : To reduce hydraulic excitation forces and pressure pulsations, it is necessary to carefully match diffuser vane and impeller blade numbers. When selecting the diffuser vane number, it is therefore imperative to consider the criteria given in Chap. 10.7.1. Together with the outer diameter d_{Le} and the throat width a_3 , the number of diffuser vanes has an influence on the shape and the length of the actual diffusing channel whose deceleration is critical for the hydraulic losses in the diffuser. Taking into account the castability (or the machining costs, if the diffuser is milled), excessively narrow channels are also unfavorable. Therefore, the parameters z_{Le} , a_3 and d_{Le} cannot be selected independently of each other. Commonly used blade/vane number combinations are: $z_{Le} = 12$ with $z_{La} = 7$ for specific speeds from approximately 20 to 35 and $z_{Le} = 8$ with $z_{La} = 5$ for $n_q < 20$. Circumferential velocity u_2 and impeller size d_2 should also be considered in this context. With high specific speeds and/or very large pumps 15 diffuser vanes are also sometimes employed. Further combinations are obtained from Table D7.4. When there is a risk of resonance between an impeller eigen frequency of the two-diameter mode and the excitation frequency $v_3 \times z_{Le} \times f_n$, all combinations resulting in $|v_3 \times z_{Le} - v_2 \times z_{La}| = 2$ should be avoided. Including this criterion leads to the preferred vane combinations (printed in italic face) in Table D7.4.

6. Inlet width b_3 : The inlet width b_3 is selected larger than the impeller outlet width; the ratio is in the range of $b_3/b_2 = 1.05$ to 1.3. Shock losses due to a mismatch of the impeller and diffuser channels should be avoided. Mismatch can be caused by manufacturing tolerances. Thermal expansion of multistage pumps must also be considered in this context. If the ratio selected for b_3/b_2 is small, a chamfer should be introduced to avoid the shock losses and associated hydraulic excitation forces, see figure in Table 0.2 (2). According to Chap. 5, the ratio b_3/b_2 has an effect on the partload characteristics: if b_3/b_2 is large, recirculation and shut-off head increase; if b_3/b_2 is reduced to values near 1.0, the risk of a saddle-type Q-H-curve

is reduced. The choice of the inlet width also has an effect on the overlap of the shrouds of impeller and diffuser. As discussed in Chaps. 5.4.2 and 9.1, it is recommended to decouple the impeller sidewall gaps from the main flow to avoid axial thrust excursions when recirculating fluid enters the impeller sidewall gap. According to [B.20], gap A between the shrouds should be $(0.007 \text{ to } 0.01) \times r_2$ and the overlap $x_{ov} = (2 \text{ to } 4) \times A$ (where A is the width of gap A), Fig. 9.1 and Table 0.2.

7. Throat width a_3 : With a given diffuser inlet width b_3 and vane number, the throat width a_3 largely determines the best efficiency point, Chap. 4.2. It also has an effect on the shut-off head and the shape of the Q-H-curve (saddle formation), Chap. 5. Equation (3.20) supplies a throat width $a_{3,th}$ which corresponds to the dimensioning in accordance with the conservation of angular momentum. The final throat width is designed according to Eq. (T7.7.10) as $a_3 = f_{a3} \times a_{3,th}$. Generally, $f_{a3} = 1.1$ to 1.3 should be assumed; in other words, the executed flow deceleration is greater than that corresponding to the conservation of angular momentum. Values $f_{a3} > 1$ also foster good overload characteristics at $q^* > 1$. With very wide impellers and low specific speeds it is sometimes necessary to design $f_{a3} < 1$ to achieve the specified best efficiency point.

Instead of using the conservation of angular momentum design as a base, it is also possible to calculate the throat width for a chosen deceleration ratio c_{3q}/c_2 from Eq. (T7.7.11). The deceleration ratio is in the range of $c_{3q}/c_2 = 0.7$ to 0.85 .

Because of the effect of the throat width on the partload and overload characteristics, the choice of the ratio c_{3q}/c_2 or of the factor f_{a3} is an optimization task for which no generally applicable rules are available. If the flow at the diffuser inlet is decelerated excessively, the stall occurs at too high flow and the Q-H-curve can become unstable. Insufficient deceleration may mean a loss of efficiency. It is also possible that the flow separation at partload occurs too *late* which can produce a saddle in the Q-H-curve, Chap. 5.5. The effect of the deceleration ratio c_{3q}/c_2 on the shape of the Q-H-curve is evident from the tests in Figs. 4.5, 4.6 and 5.19.

8. Diffuser outer diameter d_{Le} : With respect to size and costs one would try and make the outer diameter as small as possible. But low hydraulic losses require optimum diffuser length and sufficient space for flow turning; when choosing d_{Le} a compromise must be sought between efficiency and costs. The required outer diameter increases with growing design flow rate, i.e. with the specific speed. For multistage pumps it is in the range of $d_{Le}/d_2 = (1.05 \text{ to } 1.15) + 0.01 n_q$. The upper limit applies when the efficiency has priority over the pump size. If the diffuser discharges into a volute its outer diameter can be kept smaller. The diffuser channels require a sufficient overlap if it is intended to reduce the radial thrust.

9. Diffusing channel: Downstream of the throat area A_{3q} the channel opens similar to a plane or pyramidal diffuser in a pipe or channel (this section may be called “diffusing channel”, see Table 0.2). Maximum allowable diffuser opening angle, pressure recovery and losses are calculated according to Chap. 1.6. Diffuser length and area ratio are matched in such a way that the pressure recovery (or the c_p value)

is maximized. The diffuser losses can be estimated from Eqs. (T3.8.16 to 3.8.19). Since the cross sections are not circular, the calculation is done using the concept of an equivalent diffuser as per Eq. (1.45) unless the channel is designed as a plane diffuser. If the diffuser opens in the axial direction, $\vartheta_b = \vartheta_a$ may be selected with $\tan \vartheta_a = 0.5(a_4 - a_3)/L_{3-4}$. The diffusing channels are often designed with a slight curvature since this helps to achieve a greater diffuser length and, consequently, a higher pressure recovery or lower losses with a given outer diameter d_{Le} . Experience shows that the data of straight diffusers according to Chap. 1.6 can be applied to slightly curved diffusers (with up to 25° deflection) without correction - presumably since the unsteady approach flow of the diffuser fosters thinner boundary layers, in this way compensating for the potentially negative effect of the channel curvature. The final diffuser length and outlet width a_4 are obtained only when drawing the diffuser; iteration may be necessary.

10. Diffuser criteria: Once all parameters discussed above have been determined and a diffuser drawing has been made, the quality of the diffuser design should be verified according to the following criteria:

A. The optimum pressure recovery in a diffuser of given length L_{3-4} according to Fig. 1.19 is achieved when the area ratio $A_{R,opt}$ is approximately:

$$A_{R,opt} = 1.05 + 0.184 \frac{L_{3-4}}{R_{eq}} \quad \text{with } R_{eq} = \sqrt{\frac{a_3 b_3}{\pi}} \quad (7.30)$$

The pressure recovery coefficient will then be about:

$$c_{p,opt} = 0.36 \left(\frac{L_{3-4}}{R_{eq}} \right)^{0.26} \quad (7.31)$$

B. Deceleration ratio: $c_{3q}/c_2 = 0.7$ to 0.85 (at $n_q < 12$ to 15 , c_{3q}/c_2 may be higher).

C. Ratio length to throat width: $2.5 < L_{3-4}/a_3 < 6$.

D. The ratio of inlet width b_3 to throat width a_3 should preferably be in the range of $0.8 < b_3/a_3 < 2$. Cross sections which are too flat are unfavorable but can sometimes not be avoided with very wide impellers and low specific speeds.

E. The outlet cross section A_4 of the diffusing channels of multistage pumps should be dimensioned so that the kinetic energy at the diffuser outlet is in the range of $c_4^2/(2 \times g \times H_{opt}) = 0.02$ to 0.04 in order to keep the losses in the overflow channels low. However, the outlet velocity c_4 must also be matched to the inlet velocity c_{1m} into the downstream impeller: $0.85 < c_4/c_{1m} < 1.25$.

F. The equivalent opening angle must be checked according to Eq. (1.45); it must not exceed the allowable opening angle given by Eq. (1.44) or a more stringent criterion based on test results should such be available. Exceeding the allowable diffuser angle would lead to premature stall the consequence of which could be increased pressure pulsations and dynamic stresses. This is particularly true for the last stage diffuser of a multistage pump or when the diffuser is followed by a volute

where the discharge pressure is non-uniform around the circumference, Chap. 10.7. Obviously this criterion gathers significance as the head per stage is increased.

11. Return vanes: The outlet velocity c_{6m} from the return vanes should be lower than the inlet velocity c_{1m} into the following impeller in order to smooth the impeller approach flow through a slight acceleration: $c_{6m} = (0.85 \text{ to } 0.9) \times c_{1m}$. Accordingly, the width b_6 is obtained from Eq. (T7.7.12).

The course of the flow areas between the return channel and the impeller inlet should be designed so that the flow is accelerated continuously. The width b_5 at the inlet of the return vanes can be selected equal to the outlet width (or different, depending on the configuration of the channel). The meridional component of the inlet velocity is obtained from Eq. (T7.7.13).

The velocity c_4 is obtained from continuity with dimensions a_4 and b_4 from the drawing. The circumferential component c_{4u} and the radial component of the vector c_4 are subsequently determined according to Fig. 7.47. The circumferential component of the velocity at the inlet to the return channels is calculated from the conservation of angular momentum, Eq. (T7.7.14). From c_{5u} and c_{5m} it is possible to determine the inlet angle of the return vanes as per Eq. (T7.7.15). The camber angle of the return vanes can be selected equal to this flow angle (plus blockage effect).

To obtain the desired flow angle at the outlet of the return channels, a slight angle exaggeration is required. For example, if an outflow angle of $\alpha_6 = 90^\circ$ is desired, the camber angle of the vane should be selected 4 to 6° larger, that is $\alpha_{6B} = 94^\circ$ to 96° .

If the diffuser is designed to give some pre-swirl to the subsequent impeller, the outflow angle can be estimated from $\alpha_6 \approx \arcsin a_6/t_6$.

Frequently return vane and diffuser vane numbers are selected identical. Sometimes fewer return vanes are selected (i.e. $z_R < z_{Le}$). In this case the solidity of the return vanes should be $L/t > 2$ so that the flow is deflected sufficiently.

7.9.2 Design and Shaping of Radial Diffusers

The design of radial diffusers according to Fig. 7.47 comprises the following steps:

1. Circles with r_3 and r_4 or r_{Le} are drawn in the plan view.
2. Based on the vane pitch the start of two vanes is constructed. The vane thickness at the inlet is $e_3 = (0.01 \text{ to } 0.015) \times d_2$. The camber angle α_{3B} is drawn and the profile defined at the inlet.
3. An arc of a circle with radius a_3 defines a point on the opposite vane surface which determines the throat area. The concave vane surface can now be provisionally drawn as a smooth curve (or a combination of circular arcs) up to the outer diameter d_4 . From the intersection of this curve with the diameter d_4 it is now possible to mark off a_4 as an arc of a circle. This produces a point on the

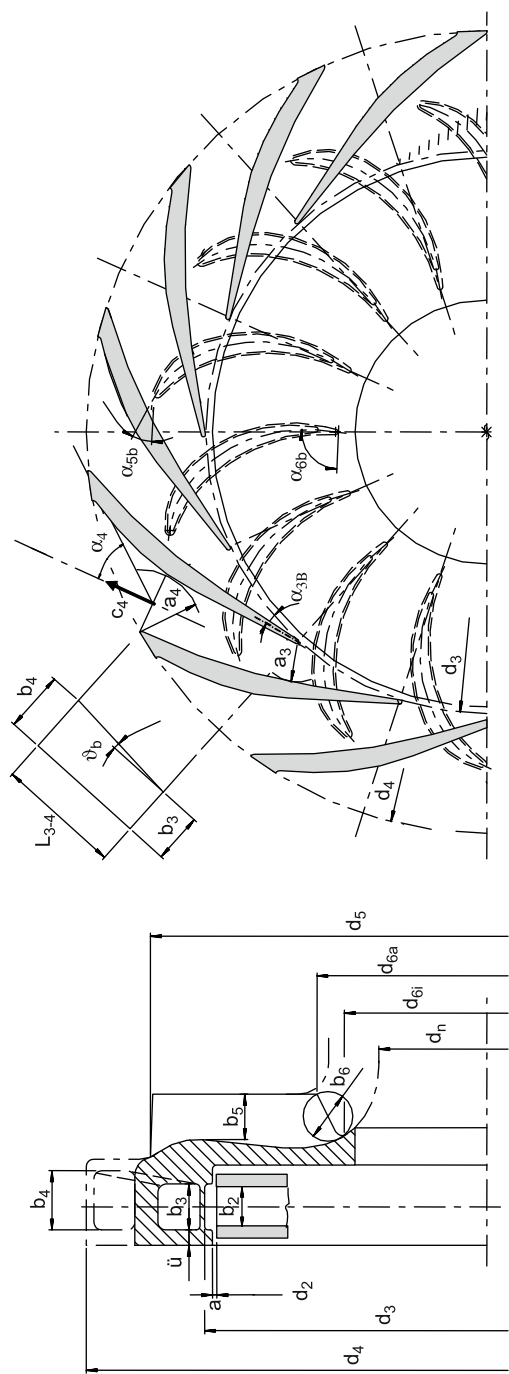


Fig. 7.47 Construction of a radial diffuser

convex vane surface which - together with the concave vane surface - supplies the vane thickness at the end of the channel. The resulting vane thickness must satisfy the requirements in terms of strength and castability. If this is not the case, the selected values of d_4 and a_4 cannot be maintained simultaneously but must be modified iteratively.

4. By changing the assumed curve of the concave vane surface (if adjusting d_4 is not desired) the channel geometry can be optimized so that the desired values for a_4 , L_{3-4} and the pressure recovery coefficient c_p are achieved as closely as possible. Flow areas and vane curvatures should vary smoothly in order to reduce the risk of premature stall. Alternatively, the first section of the vane can also be configured as a logarithmic spiral or with special profiles, such as were analyzed in [69].
5. The meridional section is drawn and, taking into account d_1 and d_n of the subsequent impeller, the inlet and outlet widths b_5 and b_6 of the return channels and the position of the leading and trailing edges of the return vanes on the diameters d_5 , d_{6a} and d_{6i} can be defined. At the same time, the section through the stage must be drafted.
6. The camber angle α_{5B} and d_5 is now drawn in the plan view. The camber line of the return vanes can, for instance, be constructed as a circular arc of the radius r_{sk} :

$$r_{sk} = \frac{r_5^2 - r_6^2}{2(r_5 \cos \alpha_{5B} - r_6 \cos \alpha_{6B})} \quad (7.32)$$

and subsequently given a thickness distribution and with leading and trailing edge profiles. This construction is performed in a way similar to Chap. 7.3.1 and Fig. 7.11. An elliptical profile at the inlet makes the vane less sensitive to incident flow. This is important since the approach flow direction is only a rough estimation in view of the complex geometry between diffuser outlet and return vane inlet. The return vanes should taper off to a thin trailing edge in order to reduce the blockage, but sufficient vane strength must be ensured.

7. With regard to position and effect of the trailing edges of the return vanes on the stability of the characteristics, reference is made to Chap. 5.6.

7.10 Semi-axial Diffusers

When designing a semi-axial diffuser, it is first necessary to determine the inlet and outlet diameters for the outer and inner streamline. Drafting therefore starts with the meridional section the shape of which has to be matched to the impeller (especially d_{2a} and d_{2i}). The diffuser outlet of single-stage pumps must be matched to the column pipe and the shaft protection pipe if applicable. In multistage pumps the

diffuser outlet is matched to the impeller inlet of the subsequent stage. Design procedure (see also Fig. 7.22):

1. The inlet and outlet diameters d_{3a} , d_{3i} , d_{4a} , d_{4i} are largely determined by aspects of the mechanical design. The distance between impeller blades and diffuser vanes can be selected according to Table 10.2 to avoid excessive pressure pulsations and hydraulic excitation forces. According to [B.18] it can also be selected as $\text{Gap B} = 0.3 \times b_2$.
2. The axial length of the diffuser must be sufficiently large to keep the losses due to flow deflection and deceleration within acceptable limits. Iteration is usually necessary since the allowable opening angle of the diffuser can only be verified when the length of the channel has been established from a drawing. A reference value for the minimum axial length L_{Le} can be determined from [B.18]:

$$\frac{L_{Le}}{D_{2m}} = 0.72 \left(\frac{n_q}{n_{q,Ref}} \right)^{0.19} \quad n_{q,Ref} = 200 \quad (7.33)$$

3. The diffuser inlet width is obtained with the impeller outlet width from $b_3 = (1.02 \text{ to } 1.05) \times b_2$. The smaller values apply to higher specific speeds.
4. After the main dimensions have been determined in this way, the course of the flow areas in the meridional section can be established so that a steady deceleration and an even curvature are obtained. The procedure is similar to Chap. 7.2.2.1.
5. When drawing the meridional section, various requirements of the mechanical design need to be taken into account, for example flange and stud dimensions, intermediate bearings, shaft protection sleeves and annular seal configurations. With high specific speeds the meridional section may be made of cylinders and conical sections, in particular when welded components are used. This type of design can yield very good efficiencies because flow turning can be minimized.
6. After the outer and inner streamlines have been drawn, the mean streamline can be constructed. Alternatively, the design can start from a mean streamline. The channel width from b_3 to b_4 is then defined by drawing circles which yield a smooth course of the flow area from inlet to outlet. Outer and inner streamlines are subsequently found from the envelopes to said circles. The procedure is similar to that described for impellers, Chap. 7.2.2.1.
7. The vane angles at the diffuser inlet are calculated for each streamline from Eqs. (T7.7.2 to 7.7.6).
8. The vane angles at the diffuser outlet are usually designed for swirl-free outflow. A slight angle exaggeration of 4 to 6° can be employed since the flow is not able to completely follow the vanes.
9. The number of diffuser vanes is usually selected greater than the number of impeller blades. The rules of Chap. 7.9.1 are applicable. In particular, diffuser and impeller should not have the same number of vanes nor vane number

combinations with common multipliers in order to avoid excessive pressure pulsations.

10. The diffuser throat area can be calculated according to Chap. 7.9.1 and Eq. (T7.7.10) which take into account the fact that the fluid in the vaneless space between impeller and diffuser flows in accordance with the conservation of angular momentum. The calculation therefore applies also to large distances between impeller and diffuser. The factor f_{a3} is close to 1.0; it has to be optimized in terms of hydraulic losses, stability and steepness of the Q-H-curve in the overload range. Generally applicable guidelines for controlling these three-dimensional effects are not available. CFD may be used for optimization.
11. The development of the vanes is drafted with the selected inlet and outlet angles α_{3B} and α_{4B} for various streamlines. The vane design procedure is similar to Chap. 7.2.2.2. In the inlet region of the vanes it is advisable to keep α_{3B} roughly constant in order to produce little flow deflection in the region with strong deceleration.
12. After the vanes have been drafted, the cross sections must be verified to ensure as steady as possible a flow deceleration. Because of the complex geometry this work is highly involved; it is best carried out using a 3D-CAD system.

7.11 Volute Combined with a Diffuser or Stay Vanes

In some types of pumps a vaned diffuser is arranged between the impeller and a single volute in order to reduce the radial thrust, Fig. 7.48. The diffuser and the volute are configured according to Chaps. 7.9 and 7.8, respectively.

At high specific speeds, stay vanes between impeller and volute are often necessary for very large pumps to reduce the deformation of the volute casing under the internal pressure and to reduce casing stresses. The length and thickness of the stay vanes must be determined on the basis of strength and requirements of mechanical design. The stay vanes are usually shaped in a way that they have little effect on the flow in the design point. They are not designed to serve as a diffuser but to allow the fluid to flow according to the conservation of angular momentum with the least impediment possible. To this end, the vane inlet angle is dimensioned according to Eqs. (T7.7.2 to 7.7.6) with zero incidence. The camber line is drafted as a logarithmic spiral with $b_3 = b_4$ (or approximately as an arc of a circle). Such types of stay vanes hardly reduce the radial thrust.

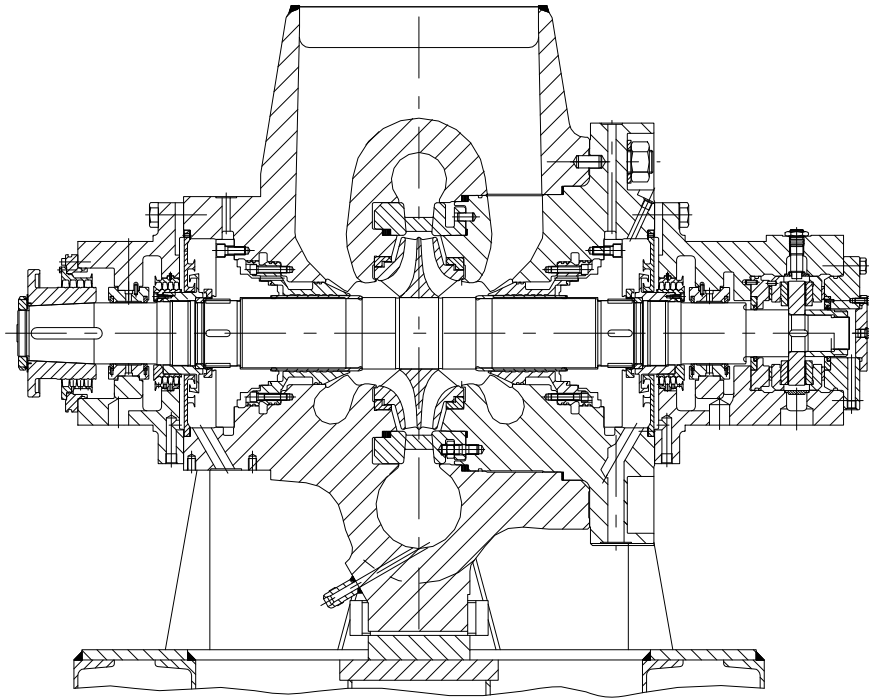


Fig. 7.48 Boiler feed pump with double-entry impeller, volute and diffuser, Sulzer Pumps

7.12 Annular Casings and Vaneless Diffusers

Because of their lower manufacturing costs, annular casings are sometimes preferred to volute casings for small pumps. Annular collectors also offer an alternative for pumps with low specific speeds since the efficiency impairment caused by the annular casing diminishes with falling n_q . For $n_q < 12$ even higher efficiencies are expected than with volutes according to [B.9]. The ratio of the pump efficiency expected with an annular collector η_{RR} to that with a volute η_{Sp} amounts to approximately $\eta_{RR}/\eta_{Sp} = (n_{q,Ref}/n_q)^{0.45}$ (with $n_{q,Ref} = 12$). This relation is valid for $n_q < 14$; it was derived from a diagram in [B.9].

To reduce the losses in the annular collector, the exchange of momentum between the impeller and the flow in the annulus should be intensified and smooth surfaces should be produced (see Chap. 7.3.2). The ratio b_3/b_2 must therefore not be selected too small; it amounts to approximately $b_3/b_2 = 1.2$ to 1.4 . A low radial extension of the annular collector is favorable to reduce friction losses and secondary flows. The ratio width B to height H is likely to be at an optimum in the range of $B/H = 1.5$ to 2.5 . The cross section $A_{RR} = B \times H$ of the annular collector must be considerably larger than corresponds to the circumferential component c_{2u} as stipulated by Eq. (7.34).

$$A_{RR} = (1.5 \text{ to } 2) \frac{Q_{opt}}{c_{2u}} \quad (7.34)$$

The fluid is discharged from the annular collector through a diffuser the throat area of which is calculated from Eq. (T7.7.9). At low specific speeds the diffuser throat area determines the best efficiency flow rate, as corroborated by Fig. 7.15. Optimization of the diffuser is carried out in accordance with Chap. 1.6 or Eqs. (7.30 and 7.31). The diffuser is arranged tangentially because high deflection losses would result at the diffuser inlet if it were configured radially.

It is also possible to combine an annular collector with a volute by designing an annular collector over (for example) 180° followed by a volute leading the fluid into the discharge nozzle/diffuser. This combination improves the efficiency since the transition to the discharge nozzle can be shaped much more favorably. In addition, the radial thrust at $Q = 0$ and low partload is reduced compared with the volute, Chap. 9.3.

Vaneless diffusers are rarely used in pumps. Their hydraulic characteristics have been discussed in Chap. 3.7. The pressure increase in the vaneless diffuser is calculated from Eq. (7.32). The radius ratio is determined on the basis of the desired pressure increase and design requirements. In general the inlet width b_3 should be selected not much larger than the impeller outlet width b_2 in order to reduce the tendency to flow separation.

7.13 Inlet Casings for Between-Bearing Pumps

For between-bearing pumps an inlet casing is required to deflect the fluid from the inlet nozzle - mainly arranged normally to the axis - into the axial direction. Generally a small inlet casing is desired with respect to costs. The most important hydraulic requirement is to design the inlet casing so as to achieve as uniform a velocity distribution as possible at the impeller inlet. Quantitative criteria for the quality of the approach flow are given in Chap. 8.8.2, item 17. These criteria could be applied when optimizing inlet casings by CFD or testing.

Different pump types require various types of inlet casings according to Fig. 7.49. With reference to a section through the pump axis, symmetrical inlet casings, such as used for barrel pumps, can be constructed as annular collectors with surfaces of revolution (Fig. 7.49a). Alternatively, symmetrical inlet casings can be designed similar to volutes with cross sections decreasing in circumferential direction (Fig. 7.49b). In order to reduce the tangential velocity components near the rib R1, a compromise between the angular shape and the volute-type area distribution can be sought.

Axial-split pump casings require asymmetric inlet casings where the inlet nozzle is located below the split flange. Asymmetric casings can either be designed so that the flow rate is divided “quasi-symmetrically” into two identical flows around the

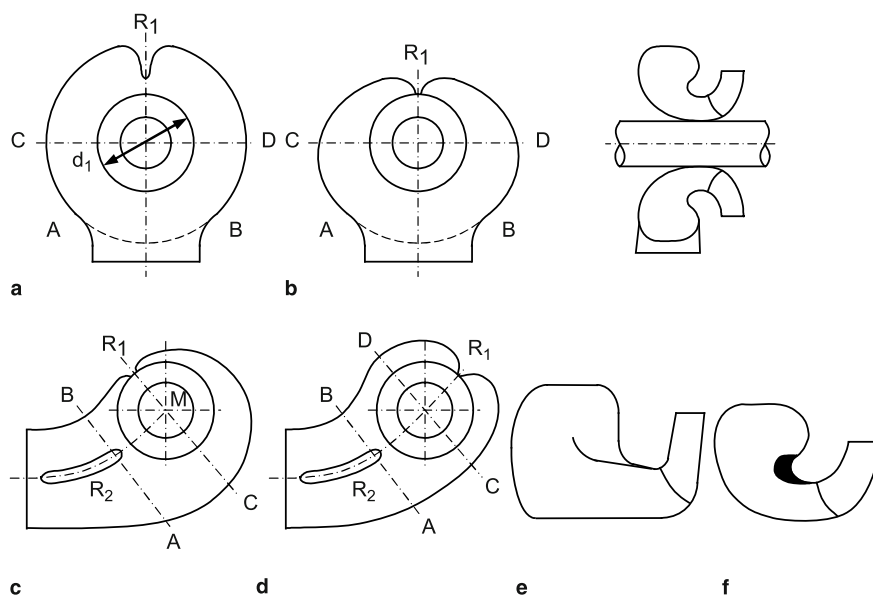


Fig. 7.49 Types of inlet casings. **a** Annulus, **b** spiral-shaped symmetrical, **c** asymmetric, **d** quasisymmetrical, **e** inlet cone, **f** inlet ring

hub (Fig. 7.49d) or that the lower half handles a significantly larger flow rate according to Fig. 7.49c.

All casing types have in common that the radial suction nozzle and the flow around the hub generate *circumferential* components in the velocity distribution at the impeller inlet. As a result, pre-rotation is generated over roughly one half of the impeller circumference and counter-rotation in the second half, Fig. 7.50. However, the flow through both halves is not uniform either since the liquid impinges on the rib which is absolutely indispensable in each and every inlet. As a consequence,

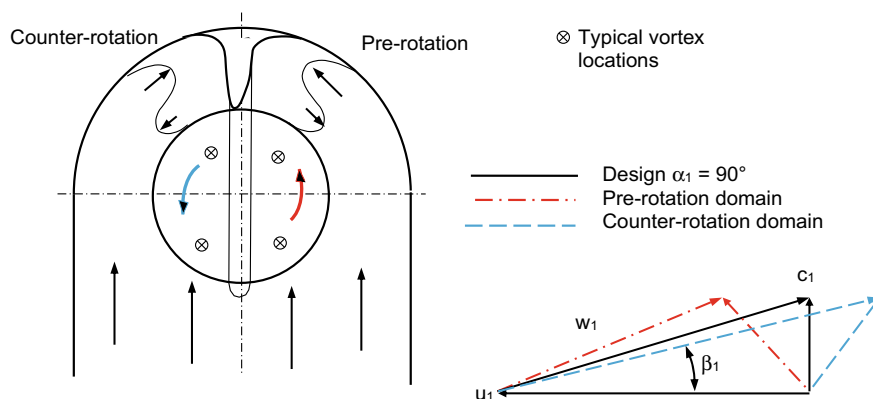


Fig. 7.50 Flow patterns in inlet casings

counter-currents near the hub occur in the region of the rib as indicated in Fig. 7.50. In the half with pre-rotation the approach flow angle increases compared to an approach flow with $\alpha_1 = 90^\circ$. Conversely, the flow angle decreases in the half with counter-rotation, as shown by the velocity triangles in Fig. 7.50 (refer also to Fig. 3.1).

The uneven velocity distribution at the impeller inlet has the following disadvantages:

- The angle of the approach flow to the impeller blades varies over the circumference and the radius, resulting in correspondingly different pressure distributions and cavity lengths on the blades.
- Noise and vibrations are excited.
- Loss in efficiency
- According to Chap. 9.3.7, Fig. 9.32, radial forces are generated.

Inflow disturbances of this kind always grow with the specific speed, with the circumferential velocity u_1 and (due to increasing inertia) with the flow rate. For a given pump, they are highest at overload $q^* > 1$.

The flow through inlet casings generates 3-dimensional boundary layers the separation of which can cause vortex ropes. The pressure in the vortex core is lower than in the surrounding fluid due to the centrifugal forces caused by the rotation. If the pressure in the core drops below the vapor pressure, the vortex cavitates as discussed in Chap. 6.8.2. If the liquid contains appreciable amounts of dissolved gas, a gaseous vortex core appears even at local pressures above vapor pressure when gas separates from the liquid.

The 3-dimensional flow in the inlet casing cannot be calculated by simple means, but some general rules for favorable flow conditions can be given:

1. The meridional component of the approach flow velocity profile upstream of the impeller inlet can be smoothed out by flow acceleration and an extended axial flow path upstream of the impeller. The acceleration should preferably take place in the region with the strongest flow turning (or downstream of it).
2. To foster a more uniform flow over the circumference of the impeller, the fluid has to be accelerated towards the impeller inlet. The cross section A-B in Fig. 7.49 must therefore be selected so that an acceleration ratio $c_{1m}/c_{A-B} = 1.5$ to 2.2 is obtained. The lower value applies to $n_q = 15$ and the upper value to $n_q > 60$. Half of the flow rate in cross section A-B flows through cross section C-D (or C-M in Fig. 7.49c). The cross sections C-D or C-M must be designed so that approximately the same velocity as in A-B prevails there or that the same acceleration ratio $c_{1m}/c_{C-D} = 1.5$ to 2.2 is achieved from C-D to the impeller inlet.
3. The circumferential component of the approach flow velocity can be decreased by reducing the blockage caused by the hub. If the inlet casing is very large, an inlet cone (or bellmouth) according to Fig. 7.49e ensures a favorable flow distribution upstream of the impeller. This type of bellmouth can be considered for a retrofit of an over-sized problem casing.

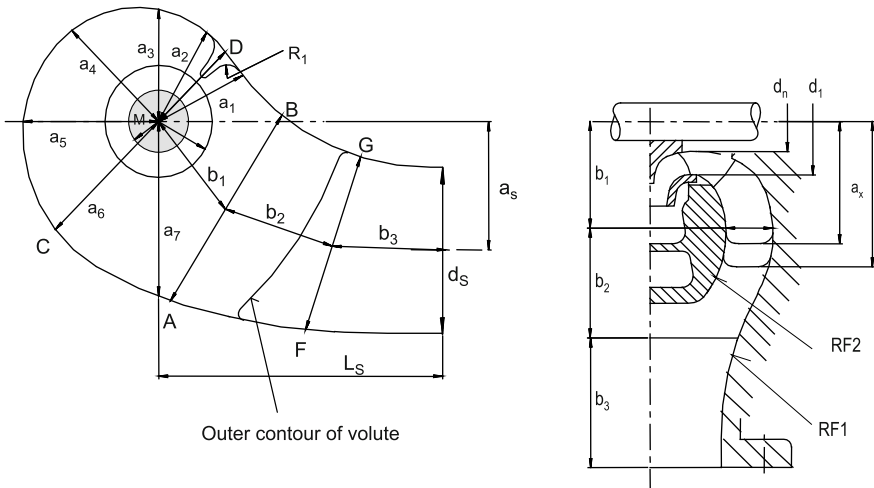


Fig. 7.51 Inlet casing design

4. If the inlet casing is over-sized, the flow can be smoothed out using a ring as shown in Fig. 7.49f. The effect of an *asymmetric* ring of this type is reported on in [42]. The asymmetric ring is better able to re-distribute the fluid to both halves of the inlet casing. Inlet rings according to Fig. 6.39 can also help. If the radial expansion of the inlet casing is large, it must be noted that the fluid tends to maintain its angular momentum according to $c_u \times r = \text{constant}$ when moving radially inward, which causes the circumferential velocities to be intensified and increases the risk of generating vortices. An inlet casing should therefore be expanded as little as possible in radial direction but rather axially.
5. Severe local decelerations from the suction nozzle to the inlet casing cause flow separation and vortices. Over-sized annular casing cross sections (which per se would favor the flow distribution over the circumference) should therefore be avoided.
6. Every inlet inevitably requires a rib R1 (Fig. 7.49) at the location where the two partial flows around the hub meet. If this rib is omitted, cavitation characteristics and work transfer (head coefficient and efficiency) are seriously affected. A periodic pre-rotation can also be induced which would result in unsteady operating.
7. The rib R2 is not absolutely necessary with symmetrical inlets but desirable for reasons of mechanical design in order to limit the casing deformation under the internal pressure (note pressure of hydrostatic test, if applicable). With asymmetric inlets a rib R2 can be expected to foster a more uniform flow.
8. All ribs must be rounded at the leading edge and profiled asymmetrically at the trailing edge according to Table 10.13 in order to reduce the formation of vortices and prevent vortex streets.

The step-wise procedure for drafting an inlet casing for a single-stage double-entry pump is explained below with reference to Fig. 7.51. This design of an asymmetric inlet casing is particularly complex because the inlet casing has to be developed around the volute casing. The design methodology essentially follows [B.2, B.9, B.21, B.22, 41]. The cross sections develop on the surfaces of revolution RF1 and RF2.

1. Known parameters are the inlet type, the inlet nozzle diameter d_s and the impeller inlet: d_1 , d_n and the impeller inlet area $A_1 = (\pi/4) \times (d_1^2 - d_n^2)$. The position of the inlet nozzle is given by the dimensions L_s and a_s .
2. Impeller inlet and hub diameter are drawn in the plan view. In an inlet according to Fig. 7.49c, the rib R1 is positioned approximately 45° over the partition plane of the axially split casing.
3. The outer contour from the rib to point C is defined by the radial dimensions a_1 to a_6 :

$$\begin{aligned} a_1 &= (1.5 \text{ to } 1.8) \times r_1 \\ a_6 &= (2.5 \text{ to } 2.8) \times r_1 \end{aligned} \quad (7.39)$$

The dimensions a_2 to a_5 increase with the circumferential angle ε according to:

$$\begin{aligned} a_i &= a_1 + (a_6 - a_1) \times \varepsilon^\circ / 180. \\ a_7 &= (3 \text{ to } 3.5) \times r_1 \\ (A - B) &= (1.7 \text{ to } 2.2) \times d_1 \end{aligned} \quad (7.40)$$

The distance A-B is positioned (approximately at a distance of $b_1 \approx d_1$ from the shaft axis) normally to the flow direction. With the details for the inlet nozzle (d_s , a_s , L_s) it is now possible to draw the inlet contour in the plan view as a connection of the checkpoints defined above. With regard to the split flange of axial-split pumps (near point B), a steep contour facilitates the positioning of the studs as closely as possible to the area subjected to pressure.

4. The outline of the volute is drawn. Another section F-G is positioned between the suction nozzle and A-B. The distances b_1 to b_3 between the centers of the sections A-B and F-G and the suction nozzle are defined.
5. The meridional section of the impeller and the volute casing are now drawn and the surfaces of revolution RF1 and RF2 are designed taking into account the hub contour which usually houses part of the shaft seal.
6. It is now possible to draft the flow areas ensuring a continuous velocity variation along the flow path. As an important element the cross section A-B must be selected so that the acceleration ratio c_{1m}/c_{A-B} defined under item (2) is achieved. The cross sections with the radial dimensions a_6 to a_1 are determined such that the velocity is approximately identical to that in cross section A-B. In doing so, it is assumed that the flow rate in the cross section M-C (or D-C) is

half that of the cross section A-B and that the flow rate subsequently decreases in proportion to the circumferential angle.

7. The areas of the cross sections between the inlet nozzle A_s and section A-B should vary in a continuous fashion. If the fluid is decelerated in this region, flow separation must be prevented under any circumstances (to be verified according to the rules for the diffuser design in Chap. 1.6). Any diffusion in the inlet casing should be designed with opening angles lower than those given by Eq. (1.44) or Fig. 1.20 in order to take into account possible flow distortions in the suction pipe.
8. The rib R2 in the inlet is arranged essentially along the mean streamline.

Vaned inlet casings: While all of the inlet casings depicted in Fig. 7.49 invariably generate a non-axisymmetric velocity distribution in front of the impeller blades, the tangential velocity components - and the resulting variation of incidence over the circumference - can be almost eliminated by a vaned inlet casing. An earlier version developed in the 1970s is presented in Fig. 7.51-1. A more compact design is shown in Figs. 7.51-2, 2.8 and 2.10 and discussed in [77, 78]. The tangential velocity components are virtually removed by several radial vanes with a length-to-pitch ratio of typically $L/t > 2$. This can be verified by visual observation

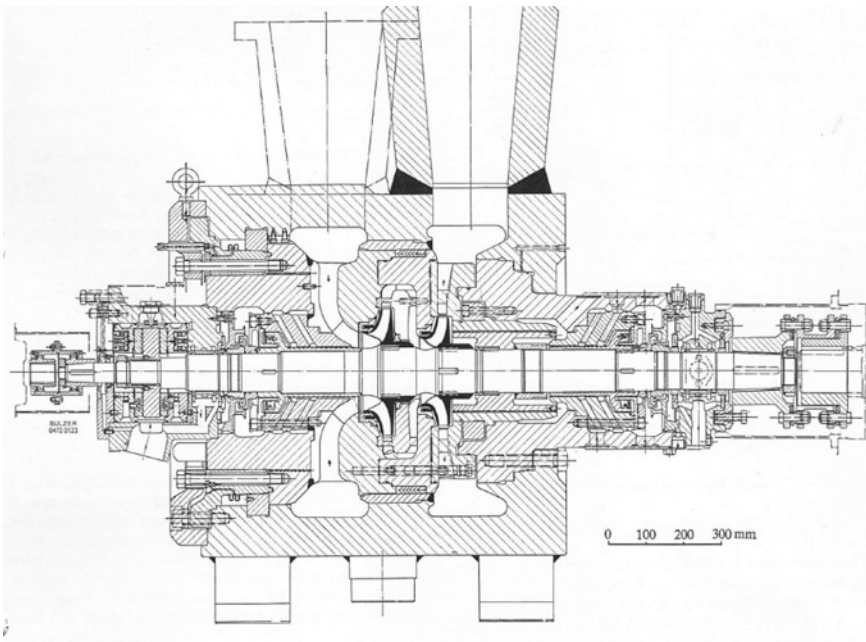


Fig. 7.50-1 “Advanced-class” boiler feed pump $n_q = 26$ with $H_{opt, st} = 1200$ m, [76]

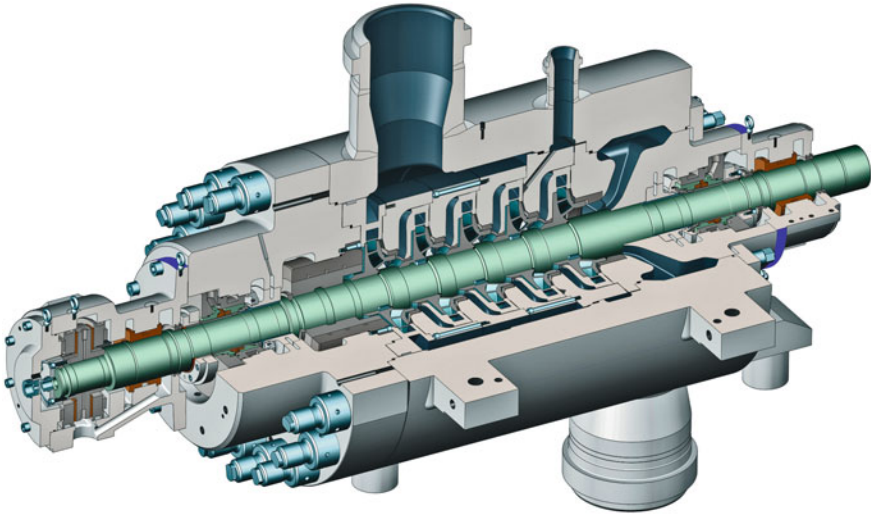


Fig. 7.50-2 5-stage boiler feed pump with $H_{\text{opt,st}} = 800$ m, [77]



Fig. 7.51-3 Model test for flow observation in a vaned inlet casing, [77]

of the cavity length (which is then uniform over the circumference) and by CFD analysis as shown in Figs. 7.51-3 and 7.51-4 respectively. Furthermore, the ribs provide an even force transmission from the stage casings to the bearing assembly which greatly benefits the mechanical design. Slight disadvantages of the vaned inlet casing are increased hydraulic losses caused by higher velocities and friction losses on the ribs. Therefore, this design is specific to high-pressure pumps where these losses and the associated increase in NPSH_R are insignificant.

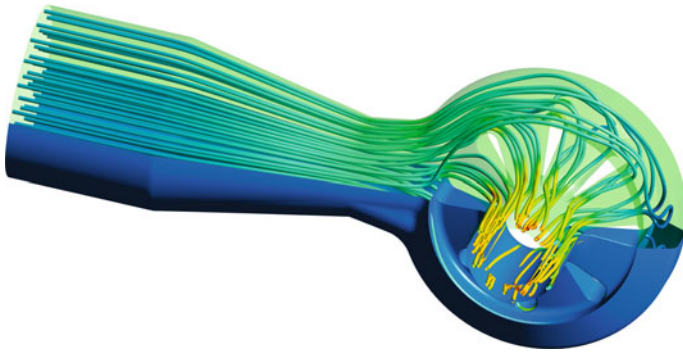


Fig. 7.51-4 CFD study of vaned inlet casing, [77]

7.14 Analytical Method for Impeller Design

7.14.1 Motivation, Scope and Objectives

For a specified performance given by n , Q_{opt} , H_{opt} and $\text{NPSH}_{R,\text{opt}}$ and for given boundary conditions, an impeller can be designed in quite different ways. For the following argument, let us assume that the pressure coefficient ψ_{opt} is selected. In this case, the impeller size is fixed. Still, the main parameters z_{La} , b_2 , d_{1a} , d_{1i} , β_{2B} , incidence, rake and others can be selected in a certain range. Even if the main parameters would be selected according to a design systematic, the meridional section of the impeller is usually drawn as free-style curves; formerly this was done by hand, nowadays Bezier coefficients are often used. The same applies to the blade development which determines the blade load distribution over the blade length.

The consequences are:

- For a given task every designer gets a different geometry since each designer makes his individual (arbitrary) choices: every impeller is a *unique* design. The complex shapes have an impact on performance, which cannot be quantified. There is a virtually infinite number of combinations of selected parameters and streamline and blade shapes. It is the *combination of these parameters and shapes* rather than any single parameter which determines the efficiency, the stability of the Q-H-curve, the hydraulic excitation forces and the cavitation characteristics $\text{NPSH}_R = f(Q)$.
- Since every designer produces a different geometry for a given set of specified conditions, each design will generate different velocity distributions and performance data. The differences in performance are admittedly often within the more or less accepted range of tolerances. However, the arbitrariness inherent to the generation of the impeller geometry is one of the reasons why the accuracy of the prediction of efficiency, head, Q-H-stability, NPSH_R and excitation forces is not always entirely satisfactory - and once in a while downright disappointing.

Since the flow undoubtedly depends on the geometry of the flow passages, it may be concluded that we must attempt to control the geometry in order to be able to control the flow patterns in the pump. In other words: consistent geometry is a *necessary* (but unfortunately *not sufficient*) condition for *consistent* performance.

In the following, design procedures are developed, which allow designing any range of pumps in a mathematically consistent way. The objectives and considerations are:

- Fully analytical description of impeller geometry: every designer produces exactly the same impeller geometry for a given set of specified data and boundary conditions.
- Develop and optimize a design system (“design systematic”) rather than individual hydraulics.
- Improve accuracy of performance prediction and reduce the costs for corrective action in case of performance deficits.
- Reduction of development effort: the impeller geometry can be created within minutes.
- A design systematic is equally useful for the design of: (1) standardized pump ranges, (2) engineered pumps, (3) pumps tailored to any client’s specifications.
- Even prior to developing a single pump it is advisable to establish a design systematic in order to have a basis for future pumps of the same type.
- The design systematic is not meant to restrict the freedom of the innovative designer in any way. If better rules are known or new ideas become available, these should be implemented in the systematic.
- The design systematic produces consistent designs with a minimum of effort. It attempts to achieve more consistent performance through a tight control of the geometry. It does not, as such, guarantee better performance. As with all other hydraulic designs this can be achieved only, if the design systematic has been tested with good results.

Limitations: (1) Due to secondary flows the fluid moves on a kind of spiraling streamlines through a given impeller. Hence, the velocity profiles at the diffuser/volute inlet depend on the specific speed. The velocity profile is critical for the losses in diffuser or volute and the stability of the Q-H-curve which is always very difficult to predict. (2) Mechanical design constraints which have an impact on the flow path but do not correlate with the specific speed. Depending on the priorities defined for a development project, a pump range can be designed in two ways:

1. Performance priority: The dimensions of the hydraulic components are given from a design systematic, previous projects or a model test. The mechanical design is made to integrate these components without modifications to the flow paths. If a tested hydraulic is used, the performance uncertainty is essentially determined by manufacturing tolerances only.
2. Cost priority: The mechanical design is largely influenced by considerations of product standardization and costs. The hydraulic components are designed to fit into the mechanical design. Scaling of hydraulic components with a given

specific speed to different pump sizes is no more possible. The mechanical constraints may not allow to optimize efficiency and required NPSH. Performance prediction is subject to higher uncertainties.

7.14.2 Meridional Section

Obviously, the shape of the outer streamline has a profound impact on the flow through the impeller *and* the diffuser - hence on efficiency, $NPSH_R$, Q-H-curve stability and hydraulic excitation forces. Second to this is the inner streamline. As a consequence, the key issue for developing a systematic impeller design procedure without any arbitrariness of the individual designer is a *fully analytical* formulation of the outer and inner streamlines of the impeller meridional section. This description must be independent of the specific speed, the suction specific speed (or impeller inlet diameter), the pump size and the hub diameter which is given by mechanical design constraints.

Over the years, quite a number of attempts have been made by the author to develop equations which would fulfill the above requirements. The difficulty was that the tangents to the streamline vary between zero at the inlet and infinite at the outlet of radial impellers (obviously, there is no such problem with semi-axial impellers). The only satisfactory general solution found so far is presented in the following.

Both, the outer and inner streamlines are calculated from “normalized streamlines” (NSL) by the procedure described in Table 7.8. The NSL are given by a number of discrete points in z^* , r^* -coordinates. The parameters used for developing the meridional section from normalized streamlines are shown in Fig. 7.52. The curved portion of the outer and inner streamlines can be followed by a straight section between r_{ga} (and r_{gi}) and r_2 . The correlations (T7.8.5 and T7.8.6) define these straight sections as a function of the specific speed. The constants in these equations could be modified to get different meridional shapes.

One possible set of streamlines is defined by the dimensionless coordinates $r^* = f(z^*)$ given in Table 7.9, but *any other shape* of streamline can be normalized by Eqs. (T7.8.1 and T7.8.2). The number of points might be increased by a Nurbs function or other forms of interpolation in order to get a more accurate fit of the curves (say 100 points per streamline).

The selected type of normalization ensures that the method can be used for:

- any specific speed
- shapes where ϵ_{DS} significantly increases with the specific speed
- the method is equally suitable for designing semi-axial impellers
- any impeller inlet diameter (e.g. large-eye suction impeller, small-eye series stages impeller)
- any number of different streamline definitions could be implemented

Table 7.8 Normalized stream line equationsOHH = overhung impellers; BB5 = impellers for multistage pumps; $n_{q,Ref} = 1.0$

Normalization			axial coordinate	$z^* = \frac{z}{z_{ga}}$	7.8.1		
			radial coordinate	$r^* = \frac{r - r_1}{r_{ga} - r_1}$	7.8.2		
Streamline The outer streamline starts at $z = z_{in}$			axial coordinate	$z = z_{in} + z_{ga} z^*$	7.8.3		
			radial coordinate	$r = r_1 + (r_{ga} - r_1) r^*$	7.8.4		
	OHH and sewage	BB5	r_{ga}/r_{1a}	$\frac{r_{ga}}{r_{1a}} = a \left(\frac{n_{q,Ref}}{n_q} \right)^x$	7.8.5		
a	2.75	2.0					
x	0.16	0.1	r_{gi}/r_{ga}	$\frac{r_{gi}}{r_{ga}} = 0.83 \left(\frac{n_q}{n_{q,Ref}} \right)^{0.021}$	7.8.6		
Straight exit, r is the independent variable			between r_g and r_2	$z = z_2 - (r_2 - r) \tan \epsilon_{DS}$	7.8.7		
Inlet length, relevant for annular seal length			OHH: $a = 0.12$, $d_n = 0$ BB5: $a = 0.25$	$z_{in} = a (d_{1a} - d_n)$	7.8.8		
Axial length of outer streamline (projection)				$z_E = \frac{d_{1a} - d_n}{2} \left(\frac{n_q}{n_{q,Ref}} \right)^x a$	7.8.9		
			a			0.75	0.33
			x			-0.05	0.227
Diameter of mid stream-line at leading edge				OHH	BB5	7.8.10	
			a	0.04	0.05		
			x	0.05	0.00		
			$d_{1m,LE} = d_{1m,z=0} + a \left(\frac{n_q}{n_{q,Ref}} \right)^x \left(\frac{r_{ga}^* + r_{gi}^*}{2} d_2 - d_{1m,z=0} \right)$				7.8.11
Diameter of mid stream-line at $z = 0$			$d_{1m,z=0} = \left(\frac{d_{1a}^2 + d_n^2}{2} \right)^{0.5}$			7.8.12	
Diameter of inner stream-line at leading edge (OHH)			if $d_n < 0.35d_{1a}$, $d_{1i} = 0.35 d_{1a}$; else $d_{1i} = 1.25 d_n$			7.8.13	
Start of inner streamline			$\frac{z_{iSL}}{z_E + b_2} = a + b \left(\frac{n_q}{n_{q,Ref}} \right)$ sewage: $a = 0.2$; $b = 0.002$; other: $a = b = 0$			7.8.14	

Table 7.9 Normalized stream line coordinates			
Outer streamline		Inner streamline	
Z_a^*	r_a^*	Z_i^*	r_i^*
1.0000	1.0000	1.0000	1.0000
0.9986	0.9335	0.9911	0.8068
0.9945	0.8692	0.9735	0.6959
0.9878	0.8072	0.9526	0.6195
0.9784	0.7475	0.9302	0.5610
0.9664	0.6901	0.9070	0.5134
0.9519	0.6351	0.8834	0.4729
0.9349	0.5826	0.8595	0.4374
0.9155	0.5325	0.8353	0.4056
0.8938	0.4849	0.8110	0.3767
0.8698	0.4397	0.7863	0.3500
0.8437	0.3971	0.7614	0.3253
0.8156	0.3569	0.7362	0.3021
0.7855	0.3192	0.7106	0.2803
0.7537	0.2839	0.6846	0.2597
0.7201	0.2511	0.6581	0.2401
0.6850	0.2206	0.6310	0.2214
0.6484	0.1925	0.6033	0.2036
0.6106	0.1667	0.5749	0.1865
0.5716	0.1431	0.5458	0.1701
0.5317	0.1218	0.5158	0.1543
0.4910	0.1025	0.4849	0.1392
0.4496	0.0852	0.4531	0.1246
0.4079	0.0700	0.4202	0.1106
0.3658	0.0565	0.3861	0.0972
0.3237	0.0449	0.3508	0.0843
0.2818	0.0348	0.3143	0.0719
0.2402	0.0264	0.2763	0.0602
0.1992	0.0193	0.2370	0.0489
0.1723	0.0153	0.2099	0.0418
0.1458	0.0119	0.1821	0.0349
0.1199	0.0089	0.1536	0.0283
0.0944	0.0064	0.1244	0.0220
0.0820	0.0053	0.1095	0.0190
0.0697	0.0043	0.0944	0.0160
0.0576	0.0034	0.0792	0.0131
0.0456	0.0026	0.0637	0.0103
0.0339	0.0018	0.0481	0.0076
0.0224	0.0012	0.0323	0.0050
0.0111	0.0005	0.0162	0.0024
0.0000	0.0000	0.0000	0.0000

Fig. 7.52 Meridional section with outer and inner streamline. The colored streamline sections are normalized

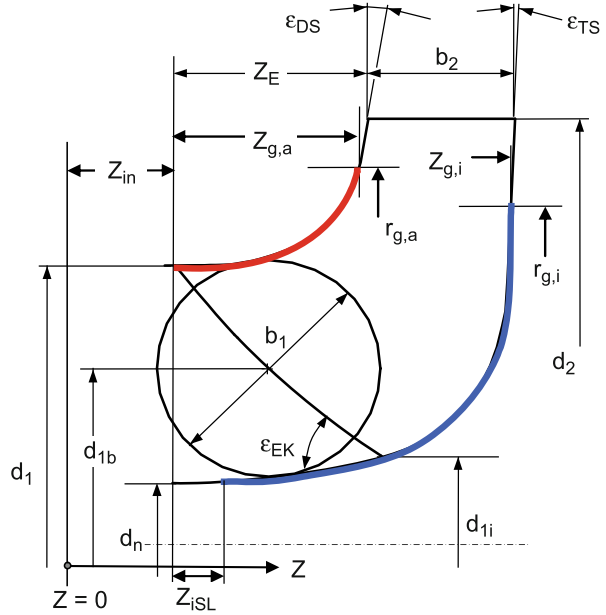
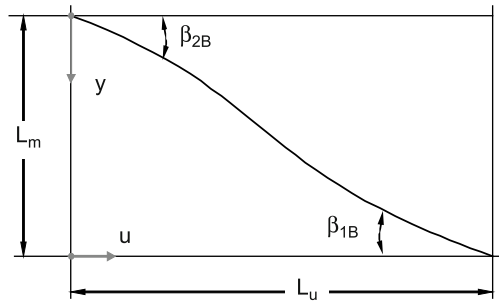


Fig. 7.53 Blade development



- Overhung impellers or between bearing impellers (single- or multi-stage pumps) with any hub diameter for high-pressure as well as low-pressure applications.

Once the outer and hub streamlines have been calculated, additional streamlines can be developed from geometric operations on the basis that the same amount of fluid flows between any two adjacent streamlines. For conventional designs the mid streamline can be calculated by averaging the dimensionless coordinates (z^* , r^*) of the outer and inner streamlines. The analogous procedure can be applied for developing additional streamlines between outer and mid as well as inner and mid streamlines.

The streamline defined in Table 7.9 yields a radial impeller exit. Some arguments might be advanced in favor of this design with $\epsilon_{DS} = \epsilon_{TS} = 0$:

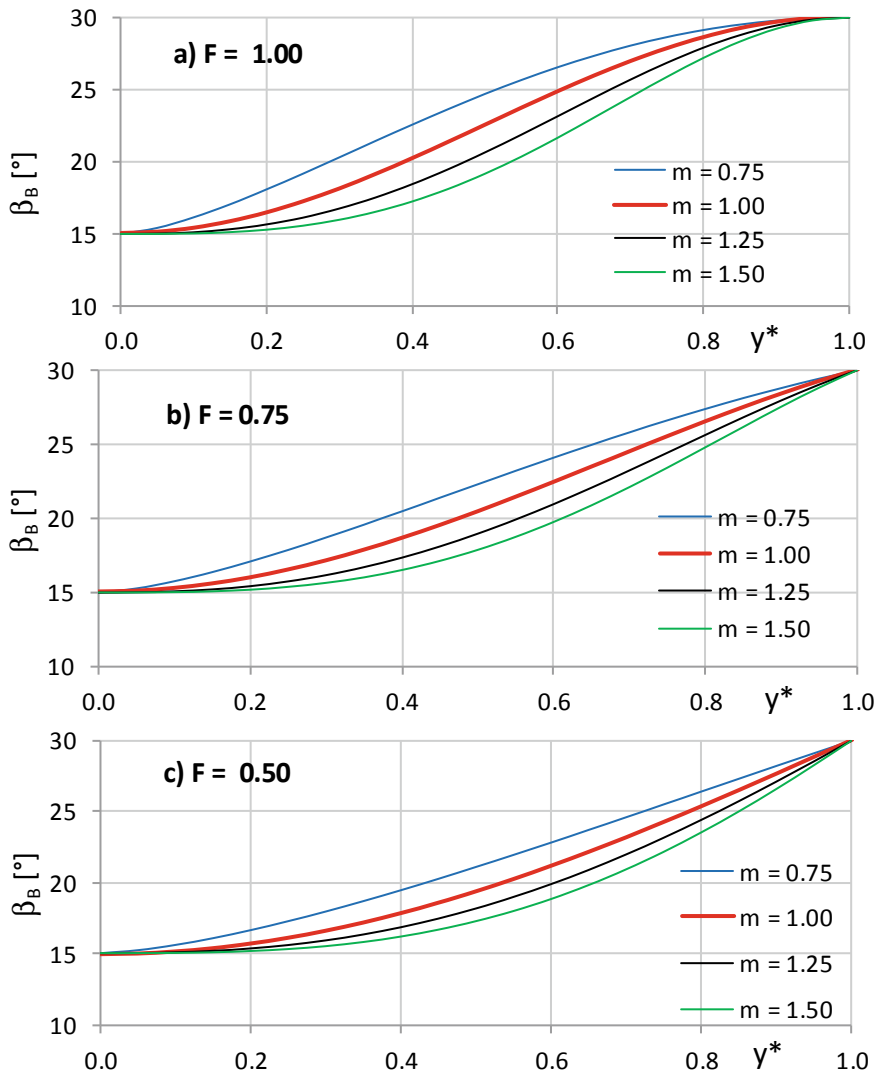


Fig. 7.53-1 Blade development according to cosine functions: effect of exponent m and factor F on blade loading distribution

- Smallest axial velocity component at the impeller outlet (the goal is $c_{2,ax} = 0$); hence less mixing losses in diffuser or casing
- Strongest decrease in width “ b ” in the region of strongest curvature (compare to Kaplan bend design)
- Radial forces are reduced since the projected area of the front shroud is smaller than in a design with $\varepsilon_{DS} > 0$. This because, according to Chap. 9.3.3, the contribution of the front shroud to the radial force is large if not dominating.

- Lower hydraulic impeller interaction improves rotor dynamic stability; this would be notably relevant for multistage pumps with a high head per stage, Chap. 10.6.3

Table 7.10 Blade design			
Blade development on 5 streamlines			
Developed length of streamline in meridional section	length element	$\Delta m = \sqrt{\Delta z^2 + \Delta r^2}$	7.10.1
	coordinate	$y_j = y_i + \Delta m$	7.10.2
	total length of streamline	$L_m = \sum \Delta m$	7.10.3
	normalized streamline developed in meridional section	$y^* = \frac{y}{L_m}$	7.10.4
Normalized blade angle Linear option: $a = 1.0$, $b = c = 0$ For single-blade impellers a 6 th order polynomial is used		$\beta_B^* \equiv \frac{\beta_B - \beta_{1B}}{\beta_{2B} - \beta_{1B}} = ay^* + by^{*2} + cy^{*3}$	7.10.5
Blade development according to cosine function		$\beta_B^* \equiv \frac{\beta_B - \beta_{1B}}{\beta_{2B} - \beta_{1B}} = 0.5 - 0.5 \cos(\pi F y^{*m})$ $\beta_B = \beta_{1B} + \frac{(\beta_{2B} - \beta_{1B})\beta_B^*}{0.5 - 0.5 \cos(\pi F)}$ exponent $m = 1.0$ and $F = 1.0$ yields true cosine function	7.10.6 7.10.6.a
Blade angle: <i>inlet to outlet</i> (modified)		$\beta_B = \beta_{1B} + (\beta_{2B} - \beta_{1B})\beta_B^*$	7.10.7
		$\Delta u = \frac{\Delta y}{\tan \beta_B} = \frac{\Delta m}{\tan \beta_B}$	7.10.8
		$y = y^* L_m$	7.10.9
		$u_j = u_i + \Delta u$	7.10.10
Length of blade element		$\Delta L = \sqrt{\Delta u^2 + \Delta y^2}$	7.10.11
Wrap angle	of one element r is the mean radius under the element Δu	$\Delta \varepsilon = \frac{\Delta u}{r}$ [radian]	7.10.12
	of entire blade on the individual streamline	$\varepsilon_{sch} = \sum \Delta \varepsilon$	7.10.13
Total blade length		$L_{sch} = \sum \Delta L$	7.10.14
Elliptical blade profile e_{nom} = nominal blade thickness, according to Table 14.3, Eq. (T14.3.1) L_E = length of leading edge profile		$\frac{y}{e_{nom}} = \sqrt{1 - \frac{x^2}{L_E^2} \left(1 - \frac{e_1^2}{e_{nom}^2} \right)}$	7.10.15

- A stronger shroud curvature might improve the Q-H-stability of diffuser pumps since it provides a defined location of flow separation at partload - thus reducing the risk of flow pattern switching, Chap. 5.6.5.
- Suction impellers can be designed with little increase in radius near the inlet for a given axial length z_E .

However, radial impellers with $\varepsilon_{DS} > 0$, typically 3 to 10°, are frequently used. Flow deceleration within the impeller channels, head coefficient and efficiency may be improved. A systematic experimental study should show which design is optimum.

7.14.3 Blade Design

The blades are developed on 5 streamlines. Starting points are the coordinates z, r at the impeller blade trailing edge of these streamlines in the meridional section. The parameters used are shown in Fig. 7.53. The design procedure has been discussed already in Chap. 7.2.2.2; the formulae used are given in Table 7.10.

During the layout process, the blade inlet and outlet angles β_{1B} and β_{2B} are calculated for each streamline in order to get the specified head and desired incidence at the impeller inlet. As discussed in Chap. 7.2.2.2 the blade angles between inlet and outlet can be determined by any functions which yield these values of β_{1B} and β_{2B} as boundary conditions at the leading and trailing edge. The normalized blade development given by Eqs. (T7.10.5 to 7) ensures just this. Therefore, a function can be chosen for defining $\beta_B^* = f(y^*)$, which allows to optimize the hydraulic features for any specific application. Three examples may be quoted: (1) a linear function for the series-stages of multistage pumps or low-pressure pumps with ample $NPSH_A$; (2) if cavitation criteria significant for the design, a cosine function as per Eq. (T7.10.6) or a 3rd order polynomial may be adequate for achieving a low blade loading near the impeller inlet; (3) a single-blade sewage impeller will need a 6th order polynomial.

The cosine function offers a simple, but very efficient means to control the blade loading. Cosine-blades are unloaded at the inlet and at the outlet. Unloading at the inlet is beneficial to reduce cavitation and incidence effects, while unloading at the outlet is expected to reduce the rotor/stator interaction, noise and vibrations. Unloading inlet and outlet implies that the blade angles increase more strongly in the mid-portion of the blades (roughly between A_{1q} and A_{2q}). As a consequence, more work is transferred in the domain where the blades overlap; the slip may decrease slightly in this way. In addition, the blade angle distribution according to Eq. (T7.10.6) can be modified: by applying an exponent m on y^* , and/or by truncating the range of β_B^* which is used: truncation factor F . If F is not equal to 1.0, Eq. (T7.10.6a) must be used for calculating the blade angles in order to obtain the specified blade outlet angle β_{2B} .

Figure 7.53.1a demonstrates how the blade loading can be controlled by the exponent m in Eq. (7.10.6). Increasing exponent to $m > 1$ unloads the blade at the inlet and puts more load towards the rear. In contrast, $m < 1$ unloads the blades at the outlet and reduces the rotor/stator interaction but rendering the impeller more susceptible to cavitation. Figure 7.53-1b, c show the effect of the truncation factor F . In summary, the cosine blade development provides much freedom to define an optimized blade loading in a fully analytical way.

7.14.4 Procedure for Developing a Design Systematic

As an integral part of a comprehensive product planning a hydraulic specification should be established the outset of the development of a pump range (or an individual pump). A sample format is given in Chap. 17.7; it may also serve as a check list.

From the hydraulic specification are given:

- Range of pumps: range chart, for each pump size: n , Q_{opt} , H_{opt}
- Required suction specific speed or $\text{NPSH}_{3,\text{opt}}$
- Boundary conditions for mechanical design

Procedure:

1. As far as possible and expedient all parameters and criteria are described by analytical functions of the specific speed, suction specific speed or geometric quantities.
2. Use smooth correlation for pressure coefficient, Eq. (3.26) with appropriate factor for f_T , impeller outlet width, Eq. (7.1), impeller inlet diameter as function of suction specific speed, Eqs. (T15.1.12 and T15.1.13) and other geometric parameters as required.
3. If necessary, adapt correlations for hydraulic shapes as per Table 7.8 and 7.10; notably Eqs. (T7.8.5 to T7.8.13) and Eq. (T7.10.5).
4. Adapt conditions and correlations for the mechanical design; e.g. shaft/hub diameter, impeller fixation, wall thicknesses, axial length.
5. Determine main dimensions of all pump sizes.

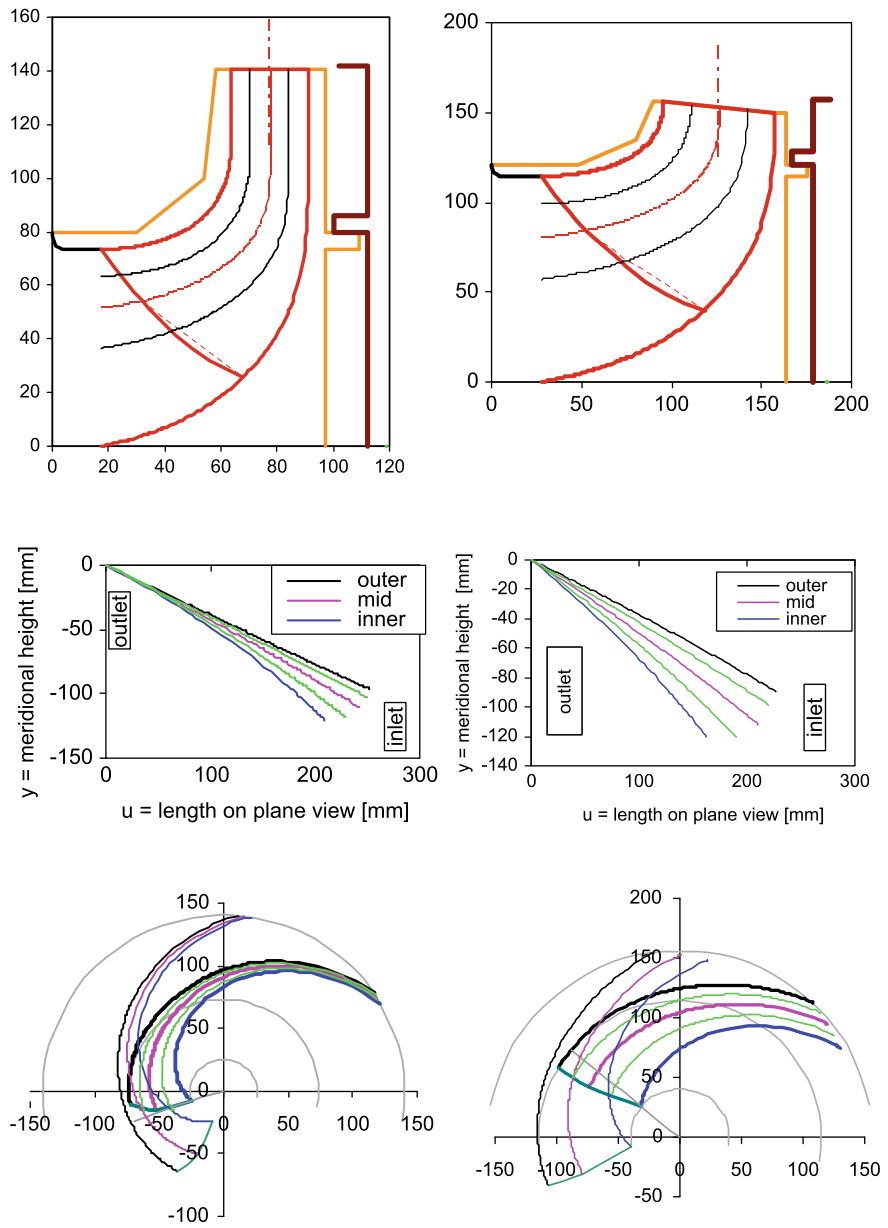


Fig. 7.54 Impellers with axial inflow from analytical design procedure; *left:* $n_q = 36$, *right:* $n_q = 100$

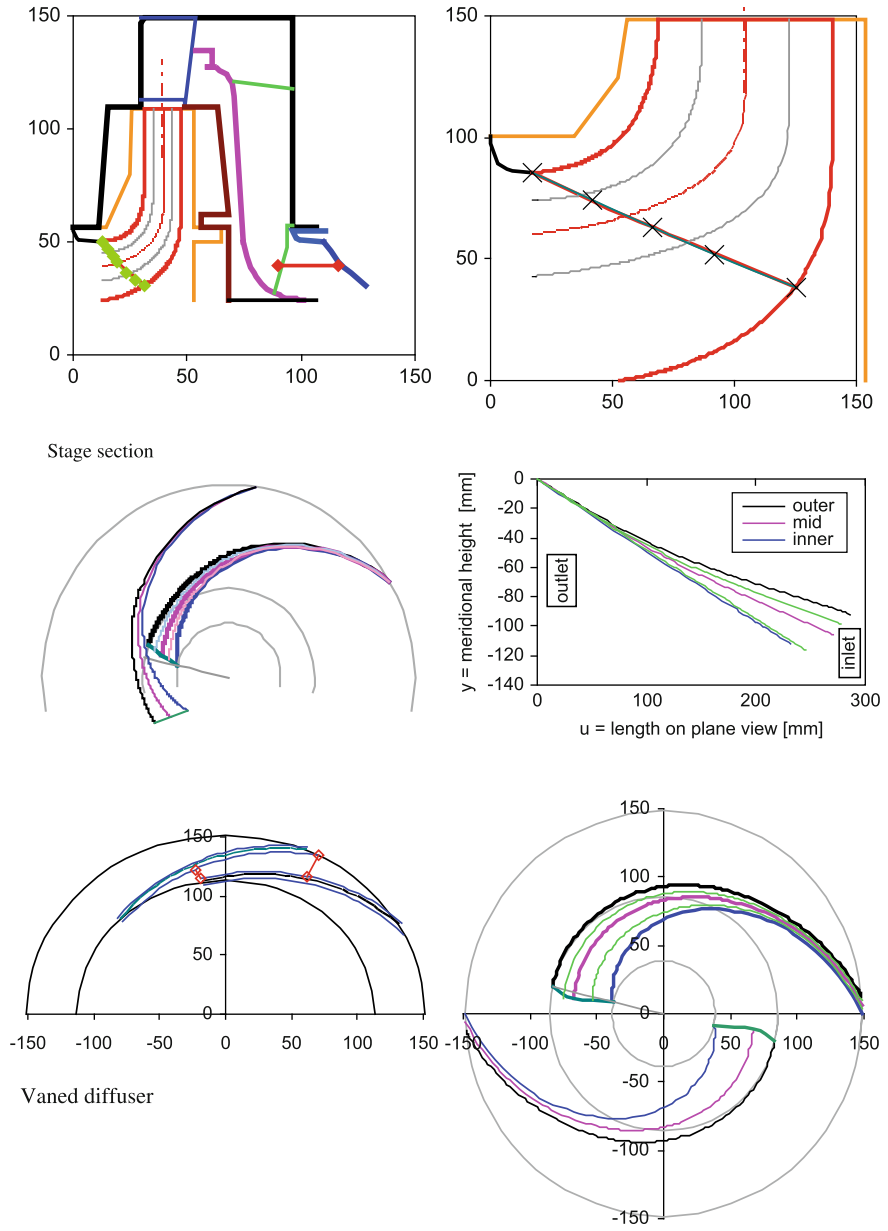


Fig. 7.55 Components from analytical design procedure; *left*: multistage pump $n_q = 24$, *right*: sewage pump with 2 blades, $n_q = 46$

Table D7.5 Sample calculations

Case		Single-stage pump with overhung impeller		Multistage pump	Sewage pump
n_q		36	100	24	46
n	rpm	1450	1450	2950	1450
Q_{opt}	m ³ /h	200	954	105	290
H_{opt}	m	20	15	58	18
d_{2a}/d_{2i}	mm	281	312/300	217	297
d_l	mm	147	229	101	171
d_n	mm	0	0	48	0
n_{ss}		220	220	180	–

6. Check the meridional section of impeller and diffuser (volute), the stage section, the blade angle development and blade plane view.

* Adapt dimensions in stage section as required

* Adapt length of streamlines as required to get the desired shape of the leading edge.

7. Step 6 is to be done for (say) 3 to 5 specific speeds which cover the entire range of n_q required for the project. The correlations can then be adapted to get smooth geometry variations throughout the range. This step should ensure that all remaining specific speeds can well be interpolated.
8. After completing step 7, all parameters are fixed in a way which ensures that the impeller geometries vary smoothly over the entire n_q -range required. Once this is achieved, *no more changes are allowed to individual impellers*. Otherwise the systematic approach will be lost: geometry, hence performance, are no longer under control.
9. Subsequently, all impellers can be designed in a very short time.
10. The coordinates of the meridional section and of the blade camber surfaces are transferred to a program which creates the 3D-model of the impeller. The blade thickness with proper profiling is applied in this program. The leading edge profile can be calculated from Eq. (T7.10.15).
11. Check impeller throat area to achieve the desired deceleration at the BEP:
- axial approach flow: $w_{1q}/w_{1m} = 0.75$ to 0.85 at Q_{opt}
 - radial approach flow: $w_{1q}/w_{1m} = 0.65$ to 0.75 at Q_{opt}
- Note: at specified maximum flow keep $w_{1q}/w_{1m} < 1.0$
 w_{1m} is determined at the geometric mean diameter of the inlet $d_{1m} = \{(d_{1a}^2 + d_n^2)/2\}^{0.5}$
12. Check smoothness of blade surface in the 3D-model.
13. CFD calculations at BEP with one impeller channel and axial approach flow with the objective to check:

* theoretical head

- * pressure distribution at impeller inlet at BEP and maximum flow ($q^* = 1.3$ to 1.5) in order to assess cavitation inception. Aim: get flat pressure distribution at the inlet of suction impellers
 - * quality of flow, non-uniformity factors (span-wise and pitch-wise) according to Table 8.2.
14. If the results of steps 11 to 13 reveal a potential for optimizing the geometry, the design systematic is to be adapted accordingly.

7.14.5 Some Results

In the following some examples of impellers or stages designed according to the above procedures are presented. Figure 7.54 shows impellers designed for a range of single-stage pumps with overhung impellers and axial approach flow; depicted are the designs for $n_q = 36$ and $n_q = 100$. Figure 7.55 shows the impeller and diffuser of a low-pressure multistage pump and the stage section in the left column. The right column shows a sewage pump impeller with two blades for $n_q = 46$.

All of these designs are examples from ranges of pump sizes which roughly cover $n_q = 7$ to 100 for the OHH pumps, $n_q = 10$ to 50 for the multistage pumps and $n_q = 20$ to 100 for the sewage pumps with 2 or 3 blades. The layout and design strictly followed the rules given in this book. The main data of the sample designs are listed in Table D7.5.

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